Group Project

AI Help Analyze Data: A case study of COVID-19 prediction

Course: Artificial Intelligence and Deep Learning (1001)

Tutor: Dr. Lily Bei LUO

Member

Ziheng LYU	2330036105
Zeming CHEN	2330036010
Jiaming LIU	2330036089
Xin WU	2330036159
Ang MA	2330036106

Contents

1 Background **3**

1.1 Data Resources . . . 3

1.2 Data Preparation . . . 3

1.2.1 CSV File . . . 3

1.2.2 CT Images . . . 4

2 Machine Learning Method **4**

2.1 Overlapping Content . . . 4

2.1.1 Evaluation Metrics . . . 4

2.1.2 ROC Curve (Receiver Operating Characteristic) . . . 4

2.1.3 AUC (Area Under the Curve) . . . 5

2.1.4 Using K-fold to Train-Test Spilt . . . 5

2.1.5 Parameter Optimization . . . 5

2.2 Naive Bayes . . . 5

2.2.1 Objective . . . 5

2.2.2 Theoretical Basis . . . 5

2.2.3 Gaussian Naive Bayes . . . 6

2.2.4 Bernoulli Naive Bayes . . . 6

2.2.5 Basic Model Code . . . 6

2.2.6 Parameter Optimization . . . 6

2.3 Logistic Regression . . . 7

2.3.1 Maximum Likelihood Estimates . . . 7

2.3.2 Activation: Sigmoid(σ) Function . . . 8

2.3.3 Gradient Descent/Ascent . . . 8

2.3.4 Parameters Optimization . . . 9

2.3.5 Evaluation . . . 9

2.4 Decision Tree . . . 10

2.4.1 Objective . . . 10

2.4.2 Entropy . . . 10

2.4.3 Information Gain . . . 10

2.4.4 Feature Selection and Recursion . . . 10

2.4.5 Parameter Optimization . . . 11

2.4.6 Evaluation . . . 11

2.5 Support Vector Machine . . . 11

2.5.1 Vector Modeling . . . 11

2.5.2 Hard-Margin SVM . . . 12

2.5.3 Soft-Margin SVM . . . 12

2.5.4 kernel Trick . . . 13
2.5.5 Evaluation Metrics . . . 13
2.5.6 Parameter Optimization . . . 14
2.6 Multi-stage Decision System . . . 14
2.6.1 Background . . . 14
2.6.2 MPGA Modeling . . . 14
2.6.3 Desicion Weight Matrix . . . 15
2.6.4 Model Weight Calculation . . . 16
2.6.5 Verification . . . 16
2.6.6 Setting Thresholds and New Model Evaluation . . . 16

3 Deep Learning Method **16**
3.1 Introduction of Convolutional Neural Network (CNN) . . . 17
3.2 Data Prepare . . . 17
3.3 Lenet-5 . . . 17
3.4 Convolution . . . 18
3.5 Pooling . . . 18
3.6 Fully-connected . . . 18

4 Conclusion **18**

A Code **19**

Abstract

The report utilizes machine learning models, decision support model and deep learning model to analyze and deal with classification issue which related to COVID-19 diagnosis. Initially, we fit the dataset and promote **Naive Bayes**, **Decision Tree**, **Logistic Regression** and **Support Vector Machine** using K-Fold cross validation method. Then, **Analytic Hierarchy Process** is applied in order to create a new model MPGA combining previous 4 models in our multi-stage decision system. Meanwhile, MPGA will assign priority for further **Convolutional Neural Network(CNN)** detection to individual who has high probability of infecting. This model performs high accuracy in predicting the COVID-19 by analyzing lung CT-images.

Key-words: Naive Bayes, Decision Tree, Logistic Regression, Support Vector Machine, Analytic Hierarchy Process, Convolutional Neural Network (CNN)

1 Background

In 2020 Feb epidemic, the shortage of medical resources caused a great problem of Misdiagnosis and delayed treatment. Therefore, we've decided to create a solution to handle this issue for future occurrence.

1.1 Data Resources

1. For machine learning section, we utilize CSV file from Kaggle, contain the symptoms during 2020 Feb [1].

2. For deep learning section, we utilize CT Images from Kaggle during 2020 Feb [2].

1.2 Data Preparation

1.2.1 CSV File

CSV file contains 20 categorical features (Breathing Problem, Fever, Dry Cough, Sore throat, Running Nose, Asthma, Chronic Lung Disease, Headache, Heart Disease, Diabetes, Hyper Tension, Fatigue, Gastrointestinal, Abroad travel, Contact with COVID Patient, Attended Large Gathering, Visited Public Exposed Places, Family working in Public Exposed Places, Wearing Masks, Sanitization from Market) and 1 class COVID-19. We drop `N/A` value get first 5 rows of dataset.

Then we transfer `Yes` or `No` to 1 or 0 with label encoding.

Ultimately, we define prediction Y is COVID-19 column and feature X is data columns except COVID-19.

[1]This CSV file could be downloaded at https://www.kaggle.com/datasets/hemanthhari/symptoms-and-covid-presence

[2]This Image Dataset could be downloaded at https://www.kaggle.com/datasets/prashant268/chest-xray-covid19-pneumonia

1.2.2 CT Images

The official file has already given divided images include information on whether individuals have COVID-19, as well as photos for both the test and training datasets in the compressed file.

2 Machine Learning Method

2.1 Overlapping Content

2.1.1 Evaluation Metrics

In our model evaluation, we initially utilize evaluation metrics which describe model performance roundly. Evaluation metrics is based on confusion matrix which is made up of:

True Positives (TP) The model correctly identified TP "infection" cases.

True Negatives (TN) The model correctly identified TN "no infection" cases.

False Positives (FP) The model incorrectly predicted FP "no infection" cases as "infection".

False Negatives (FN) The model incorrectly predicted FN "infection" cases as "no infection"

And metrics is defined by

- $$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \tag{1}$$
  identifies how many correctly classified samples from the total sample.

- $$\text{Precision} = \frac{TP}{TP+FP} \tag{2}$$
  states that what proportion of exact positives from the predicted positives.

- $$\text{Recall} = \frac{TP}{TP+FN} \tag{3}$$
  identifies how many samples are predicted to positive from the exact positive samples.

- $$\text{F1} - \text{Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$
  which importantly test the performance of precision and recall.

2.1.2 ROC Curve (Receiver Operating Characteristic)

ROC Curve is derived by plotting the True Positive Rate (TPR, also known as Recall) against the False Positive Rate (FPR, where $FPR = FP/(FP+TN)$) under various thresholds. It represents the performance of a classification model across different thresholds, showing the relationship between TPR and FPR.

2.1.3 AUC (Area Under the Curve)

AUC is the area under the ROC curve, used to evaluate the overall performance of a classification model. The area under the ROC curve is a metric to measure the overall effectiveness of the classification model, with values closer to 1 indicating better performance.

2.1.4 Using K-fold to Train-Test Spilt

Instead of using `train_test_split()`, we utilize K-fold to split train and test dataset avoiding overfitting if the data distribution is imbalanced or if certain samples dominate the training set and introducing variability due to randomness or "luck" in the split.

2.1.5 Parameter Optimization

We decide use `GridSearchCV()` to find the optimal parameter in each machine learning model by systematically searching through a predefined grid of parameter combinations because it uses cross-validation

2.2 Naive Bayes

2.2.1 Objective

The objective of this study is to utilize the provided dataset, which contains 20 feature columns, to predict the target variable indicating whether an individual has a specific disease ("yes" or "no"). To achieve this, we aim to implement a Naive Bayes model, which will classify the observations into the two target categories based on the probabilities derived from the Bayes theorem, assuming independence among the features.

2.2.2 Theoretical Basis

For a class Y and features $X_1 \ldots, X_n$, the posterior probability is as follows

$$P(Y = y_k | X_1, \ldots, X_n) = \frac{P(Y = y_k) P(X_1, \ldots, X_n | Y = y_k)}{\sum_j P\left(Y = y_j\right) P\left(X_1, \ldots, X_n | Y = y_j\right)} \tag{5}$$

Meanwhile, Naive Bayes algorithms assumes that condition independence for all features

$$P(X_1, \ldots, X_n | Y = y_k) = \prod P(X_i | Y = y_k) \tag{6}$$

Therefore, we have the ultimate form of Bayes rule

$$P(Y = y_k | X_1, \ldots, X_n) = \frac{P(Y = y_k) \prod_i P(X_i | Y = y_k)}{\sum_j P\left(Y = y_j\right) \prod_i \left(X_i | Y = y_j\right)} \tag{7}$$

To classify a new sample $X^{\text{new}} = \left\{X_1^{\text{new}}, \ldots, X_n^{\text{new}}\right\}$, Naive Bayes pick the most probably Y

$$Y^{\text{new}} = \arg\max_{y_k} P(Y = y_k) \prod_i P(X_i^{\text{new}} | Y = y_k) \tag{8}$$

2.2.3 Gaussian Naive Bayes

Gaussian Naive Bayes is designed for continuous features, assuming that each feature follows a Gaussian (normal) distribution. The likelihood of a feature is

$$P(X_i|Y=y_k) = \frac{1}{\sqrt{2\pi\sigma_{i,y_k}^2}} \exp\left[-\frac{(X_i-\mu_{i,y_k})}{2\sigma_{i,y_k}^2}\right]^2 \tag{9}$$

where μ_{i,y_k} and σ_{i,y_k}^2 are the mean and variance of feature X_i for class y_k.

Then, we compute the posterior probability and found the highest one which is a class we intend to identify.

2.2.4 Bernoulli Naive Bayes

Bernoulli Naive Bayes is designed for binary features, assuming that each feature represents the presence or absence of a particular characteristic, modeled as a Bernoulli distribution (Swathi Priyadarshini & Abdul Hameed, 2024). The likelihood of a feature is

$$P(X_i|Y=y_k)^{X_i}\left[1-P(X_i|Y=y_k)\right]^{1-X_i} \tag{10}$$

Then, we compute posterior probabilities and select the class with the highest posterior probability

2.2.5 Basic Model Code

Initially, we utilize Gaussian Naive Bayes to predict with codes using K-fold method. However, the model furnish a low accuracy. Hence, with the features X_i and class Y all are binary (1/0), we decide to use Bernoulli Naive Bayes which conforms the requirement greatly (see Appendix).

The code (see Appendix) refers to a Bernoulli Naive Bayes model prediction under K-fold method. In this way, evaluation metrics give a fabulous model performance which average values are all higher than 0.9.

2.2.6 Parameter Optimization

For Bernoulli Naive Bayes model, Smoothing Parameter α and Feature Binarization Threshold are only two hyper-parameters we can alter.

Smoothing Parameter α The α parameter is used for Laplace smoothing, which prevents probabilities from becoming zero when a feature is absent in a specific class. The default α value is 1 which is a widely utilization. Meanwhile, in order to remain robustness, we decide to apply the default value in the model.

Feature Binarization Threshold The `binarize` parameter is used to convert continuous features into binary values (0 or 1). We have already done this task, thus we can ignore this parameter.

In conclusion, all hyperparameters are adjusted and inputed in the Bernoulli Naive Bayes model.

2.3 Logistic Regression

Logistic regression is a kind of supervised machine learning algorithm in order to tackle binary classification tasks. It estimates the parameters of a logistic model. This model also requires an specific activation function to transform a linear vector of input variables/features into a probability between 0 (False) and 1 (True), i.e., this model can predict two predefined outcomes. By calculating the probability, input features will be grouped by the importance, so that provides a precise decision-making according to the experimental(training) datasets.

2.3.1 Maximum Likelihood Estimates

The Maximum Likelihood Estimates (MLE) is a statistical method used to estimate the weights of a machine learning model, and MLE chooses the weights that make the observed data most probable in logistic regression process. The purpose of MLE is to find parameters that maximize the likelihood of prediction, which expresses the probability of the observed data according to the weights of the model.

MLE function is constructed by the conditional probability of observation. The main manipulation is to suppose that there are n samples, every samples have multiple features $\mathbf{X}^{(i)}$, and one objective variable $y^{(i)}$. Because our data output is a Binary Distribution, so the Bernoulli Function can be used to illustrate the total probability density

$$\begin{aligned}\mathscr{L}_n(\mathbf{X},\mathbf{y};\omega) &= \prod_{i=1}^{n} f_i^{\text{Bernoulli}}\left(\mathbf{X}^{(i)},y^{(i)};\omega\right) \\ &= \prod_{i=1}^{n}\left[P_\sigma\left(y^{(i)}\middle|\mathbf{X}^{(i)},\omega\right)\right]^{y^{(i)}}\left[1-P_\sigma\left(y^{(i)}\middle|\mathbf{X}^{(i)},\omega\right)\right]^{1-y^{(i)}}\end{aligned} \tag{11}$$

where ω is a vector contains parameters governing the joint distribution $\omega=[\omega_0,\omega_1,\ldots,\omega_d]^\mathsf{T}$, $\mathbf{y}$ is the observed output sample $\mathbf{y}=\left(y^{(1)},y^{(2)},\ldots,y^{(n)}\right)$, and P_σ is the activated probability function (introduce at Section 2.3.2).

The meaning of MLE is to assign the values to model parameters that maximize the likelihood function over the parameter space

$$\hat{\omega}=\underset{\omega\in\Omega}{\arg\max}\,\mathscr{L}_n(\mathbf{X},\mathbf{y};\omega)=\underset{\omega}{\arg\max}\prod_{j=1}^{n} f_i^{\text{Bernoulli}}\left(\mathbf{X}^{(j)},y^{(j)}\middle|\,\omega\right) \tag{12}$$

where Ω is the set that contains all possible parameter ω, n is the dimension of input vector $\mathbf{X}$, $\mathbf{X}^{(j)}$ and $\mathbf{y}^{(j)}$ are input features and output vectors from the j-th sample (observation).

2.3.2 Activation: Sigmoid(σ) Function

The Sigmoid Function (σ), particularly the logistic function, is a bounded, differentiable, and monotonic function that always have positive value and non-negative derivative which is bell shaped in order to classify the output to 0 and 1

$$\sigma(z) = \frac{1}{1+\exp(-z)} = \frac{\exp(z)}{1+\exp(z)} = 1-\sigma(-z) \tag{13}$$

So if we take the sigmoid function (σ) as activation function, then the MLE could be expressed as

$$\begin{aligned}\mathscr{L}_n(\mathbf{X},\mathbf{y};\omega) &= \prod_{i=1}^{n}\left[P_\sigma\left(y^{(i)}=1\left|\mathbf{X}^{(i)},\omega\right.\right)\right]^{y^{(i)}}\left[P_\sigma\left(y^{(i)}=0\left|\mathbf{X}^{(i)},\omega\right.\right)\right]^{1-y^{(i)}} \\ &= \prod_{i=1}^{n}\left[\frac{\exp(z)}{1+\exp(z)}\right]^{y^{(j)}}\left[\frac{1}{1+\exp(z)}\right]^{1-y^{(j)}}\end{aligned} \tag{14}$$

where z is the polynomial to sum up the product of weights and features if we choose to activate the observation to probability output $z = \omega_0 + \sum_{i=1}^{d}\omega_i X_i^{(j)}$, and d is the dimension of the features $\mathbf{X}$ from j-th observation.

And if we want to get the $\arg\max_\omega \mathscr{L}_n(\mathbf{X},\mathbf{y};\omega)$, this expression could be simplified by taking the logarithm

$$\begin{aligned}\mathscr{L}_n(\mathbf{X},\mathbf{y};\omega) &\equiv \ln\prod_{i=1}^{n}\left[P_\sigma\left(y^{(i)}=1\left|\mathbf{X}^{(i)},\omega\right.\right)\right]^{y^{(i)}}\left[P_\sigma\left(y^{(i)}=0\left|\mathbf{X}^{(i)},\omega\right.\right)\right]^{1-y^{(i)}} \\ &= \sum_{j=1}^{n}\left\{y^{(j)}\left(\omega_0+\sum_{i=1}^{d}\omega_i X_i^{(j)}\right)-\ln\left[1+\exp\left(\omega_0+\sum_{i=1}^{d}\omega_i X_i^{(j)}\right)\right]\right\}\end{aligned} \tag{15}$$

2.3.3 Gradient Descent/Ascent

Gradient descent is a iterative algorithm in order to find mathematical optimum by minimizing the derivative of specific point within the gradient. This method is preferred for its particularly effectiveness in finding global optimum by minimizing the cost or loss in machine learning in order to find local optimum when the gradient is strictly concave or convex

$$f(tx_1+(1-t)x_2) > tf(x_1)+(1-t)f(x_2) \quad \forall\, x_1,x_2 \in D \tag{16}$$

although it has no mathematical closed-form solution to maximize $\mathscr{L}(\mathbf{X},\mathbf{y};\omega)$. In this situation, gradient descent could help find out the optimal solution of MLE by adjusting weight ω_i.

Since the Formula 15 is concave in $\mathbf{w}$, the local optimum is the global optimum, i.e., the maximum of $\mathscr{L}_n(\mathbf{X},\mathbf{y};\omega)$ can be reached when the gradient of likelihood function

$$\nabla_{\mathbf{w}}\mathscr{L}(\mathbf{w}) = \left[\frac{\partial\mathscr{L}(\mathbf{w})}{\partial\omega_0},\frac{\partial\mathscr{L}(\mathbf{w})}{\partial\omega_1},\ldots,\frac{\partial\mathscr{L}(\mathbf{w})}{\partial\omega_n}\right] = \mathbf{0} \tag{17}$$

And the core idea is to calculate the gradient of loss function, differentiating and updating the parameters along the gradient direction

$$\omega_i^{(t+1)} = \omega_i^{(t)} + \eta_t \left. \frac{\partial \mathscr{L}(\omega)}{\partial \omega_i} \right|_t \tag{18}$$

where η_t is the t-th step size (learning rate) which is predefined based on the scale of features. And the possible stopping criteria of gradient descent is $||\nabla f(x_t)|| < \varepsilon$ for some $\varepsilon > 0$ because gradient descent could only find feasible solution, not theoretical solution. And here is the iteration of gradient descent

$$\begin{aligned} \omega_0^{(t+1)} &= \omega_0^{(t)} + \eta \sum_{j=1}^{n} \left[y^{(j)} - \hat{P}_\sigma \left(y^{(j)} = 1 \middle| \mathbf{X}^{(j)}, \omega^{(t)} \right) \right] \\ \omega_i^{(t+1)} &= \omega_i^{(t)} + \eta \sum_{j=1}^{n} x_i^{(j)} \left[y^{(j)} - \hat{P} \left(y^{(j)} = 1 \middle| \mathbf{X}^{(j)}, \omega^{(t)} \right) \right] \end{aligned} \tag{19}$$

2.3.4 Parameters Optimization

C is the inverse of regularization strength λ to prevent model from over-fitting. The larger the C is, the weaker the regularization is.

Penalty is used to add in the loss function to penalty weights to improve the smoothness of models (Josephus et al., 2021). L_1 uses the absolute values of weights to encourage sparsity (make some weights equal to 0). L_2 is used as the square value of weights to shrink the weights towards but not equal to 0.

$$\begin{aligned} L_1 &: R(\omega) = \lambda \sum |\omega| \\ L_2 &: R(\omega) = \lambda \sum \omega^2 \end{aligned} \tag{20}$$

Solver lbfgs (Limited-memory Broyden-Fletcher-Goldfarb-Shanno) is a quasi-Newton method used for solving optimization problems, especially in cases where we want to minimize a loss function by logistic regression. It has high memory efficiency, fast convergence and strong robustness, which are suitable for our wide COVID-19 dataset.

2.3.5 Evaluation

Four Scores Average Accuracy: 0.9687, Average Precision: 0.9760, Average Recall: 0.9855, Average F1-Score: 0.9807 (3 of 4 scores are higher than before).

ROC, AOC Comprehensive view of a binary classification models' performance, illustrating the trade-off between true positive and the false positive rate, 0.99 highly conclude the excellent performance of our logistic regression model.

2.4 Decision Tree

2.4.1 Objective

The objective of this study is to utilize the provided dataset, which contains 20 feature columns, to predict the target variable indicating whether an individual has a specific disease ("yes" or "no"). To achieve this, we aim to implement a Decision Tree Model, which will classify the observations into the two target categories by sequentially splitting the dataset based on the feature values that maximize information gain (or minimize impurity) at each node (Manzella et al., 2023).

2.4.2 Entropy

If the probability distribution of a random variable is

$$P(X = x_i) = p_i, \quad i = 1, 2, \ldots, n \tag{21}$$

The Entropy is

$$H(X) = -\sum_{i=1}^{n} p_i \log p_i, \quad i = 1, 2, \ldots, n \tag{22}$$

Especially, for binary random variable

$$E(p_{\oplus}, p_{\ominus}) = H(S) \equiv -p_{\oplus} \log_2 p_{\oplus} - p_{\ominus} \log_2 p_{\ominus} \tag{23}$$

where $p_{\oplus}$ is the proportion of positive examples in S and $p_{\ominus}$ is the proportion of negative example in S.

2.4.3 Information Gain

Information Gain of A is the expected reduction in entropy of target variable Y for data sample S, due to sorting on variable A

$$\text{Gain}(S, A) = \text{Entropy}(S) - \sum_{v \in A} \frac{|S_v|}{|S|} \text{Entropy}(S_v) \tag{24}$$

where S_v is the subset of S for which feature A has value v.

2.4.4 Feature Selection and Recursion

For each feature A, calculate the information gain $\text{Gain}(S, A)$. Choose the feature with the highest gain to split the dataset. Then, Applying the same process to each subset until one of the stopping conditions is met:

1. All samples in a subset belong to the same class
2. No remaining features to split

2.4.5 Parameter Optimization

Although the performance metrics from the initial model are promising, we cannot be certain if they reflect the optimal solution. To improve the model's performance, we apply GridSearchCV to systematically explore various combinations of hyper-parameters for the decision tree model. The primary parameters we focus on include:

Criterion We explore both `'gini'` and `'entropy'` criteria.

Max Depth We test depths ranging from 3 to 20 to find the optimal tree size.

Min Samples Split We test values from 2 to 10 to avoid over-fitting.

Min Samples Leaf We experiment with values from 1 to 5. To identify the optimal hyper-parameters, we perform a grid search over all parameter combinations. We use a 5-fold cross-validation scheme, with F-1 score as the evaluation metric. This method ensures that the selected parameters provide the best balance between precision and recall.

2.4.6 Evaluation

Our attempt to optimize the algorithm is unsuccessful, the accuracy score dropped by 0.09 instead of rising. We have tried multiple times and this is the best result we got. The reasons this is happening, we believe it might be because the data and the algorithm is already so matched there is no better parameters.

2.5 Support Vector Machine

2.5.1 Vector Modeling

Firstly, a hyperplane in $\mathbb{R}^d$ can be defined as

$$f(x) = \mathbf{w}^x + b \tag{25}$$

where $\mathbf{w} \in \mathbb{R}^d$ is the weight vector, and $b \in \mathbb{R}$ is the bias term. With the sigmoid function, the decision function should be changed to

$$f(\mathbf{x}) = \sigma\left(\mathbf{w}^\mathsf{T}\mathbf{x} + b\right) \tag{26}$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid function which predict class 1 if $f(x) > 0.5$ and class 0 if $f(x) \leq 0.5$.

Therefore, the margin is given by

$$\text{Margin} = \frac{2}{||\omega||} \tag{27}$$

If we want to maximize the margin, $||\omega||^2$ should be minimize.

2.5.2 Hard-Margin SVM

The constraint ensuring that all data points are correctly classified. Hence , the optimization problem can be expressed as

$$\min_{\mathbf{w},b} \frac{1}{2}\|\mathbf{w}\|^2 \quad \text{subject to} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1, \quad \forall i. \tag{28}$$

Use Lagrange multipliers and transforming the optimization problem into the dual form

$$\mathscr{L}(\mathbf{w}, b, \alpha) = \frac{1}{2}\|\mathbf{w}\|^2 - \sum_{i=1}^{n} \alpha_i \Big[y_i(\mathbf{w}^\top \mathbf{x}_i + b) - 1 \Big] \tag{29}$$

Then, the dual optimization problem is

$$\begin{aligned} \max_{\alpha} \quad & \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^\mathsf{T} \mathbf{x}_j, \\ \text{subject to} \quad & \sum_{i=1}^{n} \alpha_i y_i = 0 \quad \text{and} \quad \alpha_i \geq 0 \ \forall i. \end{aligned} \tag{30}$$

The solution of dual problem provides the optimal α_i, from which the weight vector ω is computed as:

$$\omega = \sum_{i=1}^{n} \alpha_i y_i \mathbf{x}_i \tag{31}$$

Once ω and b are determined, the decision function can be expressed as:

$$f(\mathbf{x}) = \frac{1}{1 + e^{-(\omega^\mathsf{T} \mathbf{x} + b)}} \tag{32}$$

2.5.3 Soft-Margin SVM

To account for instances where some points may not be perfectly classified by using Hard-Margin SVM, slack variables ξ_i are introduced. These variables impose a penalty for such violations to allow few mis-classifying points exist within the margin. The optimization problem is written as

$$\begin{aligned} \min_{\mathbf{w},b,\xi} \quad & \frac{1}{2}\|\mathbf{w}\|^2 + C \sum_{i=1}^{n} \xi_i, \\ \text{subject to} \quad & y_i(\mathbf{w}^\mathsf{T} \mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad \forall i. \end{aligned} \tag{33}$$

Using Lagrange multipliers and transforming the optimization problem into the dual form:

$$\begin{aligned} \max_{\alpha} \quad & \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^\mathsf{T} \mathbf{x}_j, \\ \text{subject to} \quad & \sum_{i=1}^{n} \alpha_i y_i = 0, \quad 0 \leq \alpha_i \leq C, \quad \forall i \end{aligned} \tag{34}$$

After solving the optimal α_i, the decision function is computed as:

$$\begin{aligned} \omega &= \sum_{i=1}^{n} \alpha_i y_i \mathbf{x}_i \\ f(\mathbf{x}) &= \frac{1}{1 + e^{-(\omega^\top \mathbf{x} + b)}} \end{aligned} \tag{35}$$

2.5.4 kernel Trick

In order to perfectly separated data, we use a nonlinear mapping $\phi(\mathbf{x})$ to map data into a higher dimensional space. Because, in a higher dimensional space, the dataset may become linearly separable (Akbar et al., 2024). SVM uses the kernel function $K\left(\mathbf{x}_i, \mathbf{x}_j\right)$ to compute the inner product in the feature space:

$$K\left(\mathbf{X}_i, \mathbf{X}_j\right) = \phi\left(\mathbf{X}_i\right)^{\mathsf{T}} \phi\left(\mathbf{X}_j\right) \tag{36}$$

Using Lagrange multipliers and transforming the primal problem into dual optimization problem

$$\begin{aligned} \max_{\alpha} \quad & \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j K(\mathbf{x}_i, \mathbf{x}_j), \\ \text{subject to} \quad & \sum_{i=1}^{n} \alpha_i y_i = 0, \quad 0 \leq \alpha_i \leq C, \quad \forall i. \end{aligned} \tag{37}$$

These kernel functions are commonly used

Linear kernel

$$K\left(\mathbf{x}_i, \mathbf{x}_j\right) = \mathbf{x}_i^{\mathsf{T}} \mathbf{x}_{j'} \tag{38}$$

Polynomial kernel

$$K\left(\mathbf{x}_i, \mathbf{x}_j\right) = \left(\mathbf{x}_i^{\mathsf{T}} \mathbf{x}_j + c\right)^p \tag{39}$$

Radial Basis Funtion (RBF) kernel

$$K\left(\mathbf{x}_i, \mathbf{x}_j\right) = \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2) \tag{40}$$

After gaining the optimal α_i, the decision function is computed as

$$f(\mathbf{x}) = \operatorname{sign}\left(\sum_{i=1}^{n} \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b\right). \tag{41}$$

At the very beginning, we are uncertain whether the problem is a linearly separable or not, hence, we initially assumed it to be a linearly separable problem and chose `kernel = "linear"` to address it.

2.5.5 Evaluation Metrics

After determining the model, we aim to assess its capability to correctly classify the categories given the 20 features columns. We apply K-fold cross-validation to ignore potential errors from randomly splitting the data, hence, obtain the average scores from a more reliable evaluation.

2.5.6 Parameter Optimization

Although the scores are high, we are uncertain whether it can represent the optimal solution. Therefore, we use GridSearchCV function to achieve the best possible model performance on our dataset. Typically, we focus on these following parameters:

C **(Regularization Parameter)** The parameter *C* controls the tradeoff between maximizing the margin and minimizing classification errors. A smaller *C* allows more margin violations, leading to a more smoother decision boundary, while a larger *C* penalizes errors more heavily, leading to a more complex model.

γ **(Kernel Coefficient)** The parameter γ defines the influence of each individual training sample. A high γ leads the model becomes sensitive to individual data points, but may overfit if too high. A lower γ increases the model's ability to generalize, but may underfit if too small.

Kernel Type The kernel determines the type of decision boundary the SVM will use. Linear: Suitable when the data is linearly separable which uses a straight-line decision; RBF (Radial Basis Function): Suitable for non-linear decision boundaries which uses a more flexible, curved boundary.

Hence, we go through all the number from 0.1 to 10.0 with the step of 0.1 to find the optimal value of *C*, select three classical number (0.01, 0.1, 1) for γ and two kernels of Linear and RBF which can cover most scenarios

2.6 Multi-stage Decision System

After enhancing all previous 4 models into the best performance, we intend to aggregate them together for achieving better prediction on COVID-19 infection.

2.6.1 Background

To tackle the limited medical resources issue during early disease detection mentioned before, we create a multi-stage decision system for reducing the occurrence of the chaos mentioned earlier.

Therefore, we initially create a new model called MPGA with Analytic Hierarchy Process(AHP) which combine previous 4 models, and we assign priority for CT scans based on the predictions (0/1) from this model.

2.6.2 MPGA Modeling

Initially, we obtain advantages of each model and then assign each model a important weight listing as follows:

Bernoulli Naive Bayes

Role Performs well in simple scenarios (e.g., quick screening) but may struggle with complex data.

Weight Lower than other models but still significant.

Decision Tree

Role Suitable for medium-sized datasets, useful for analyzing key influencing factors (e.g., age, health conditions).

Weight Higher than Naïve Bayes, due to its interpretability and ability to provide concrete decision-making criteria.

Logistic Regression

Role Ideal for identifying key risk factors (e.g., high risk in elderly or patients with chronic diseases).

Weight Slightly lower than SVM but still holds an important position.

Support Vector Machine

Role Excels in precise predictions for multi-dimensional features (e.g., medical resource distribution, infection trends).

Weight Highest, owing to its superior performance with complex data.

Then, we start to create decision weight matrix.

2.6.3 Desicion Weight Matrix

Generally, decision matrix can be demonsrtated by

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \tag{42}$$

And each item in the matrix have the requirement:

$$a_{ij} > 0, \quad a_{ij} = \frac{1}{a_{ji}}, \quad a_{ii} = 1,$$

where a_{ij} represents the relative importance of criterion j compared to criterion i . Hence, we utilize saaty table as follows to evaluate the relative importance of each pair of model, then we can fill the scale into the matrix and obtain the matrix.

2.6.4 Model Weight Calculation

Initially, we compute n dimension vector in a row

$$\bar{W}_i = \sqrt{\prod_{j=1}^{n} a_{ij}} \quad i, j = 1, 2, \cdots, n \tag{43}$$

Then, we can obtain each model's weight vector by normalization

$$W_i = \frac{\bar{W}_t}{\sum_{i=1}^{n} W_j} \tag{44}$$

2.6.5 Verification

Nevertheless, we should verify whether the weight meets the quality requirements, thus we utilize Consistency Ratio Test to check.

$$\mathrm{CR} = \mathrm{CI}/\mathrm{RI} \tag{45}$$

where RI can be found in The Random Consistency Index. And CI can be calculated by this formula:

$$\mathrm{CI} = \frac{\lambda_{\max} - n}{n - 1} \tag{46}$$

Where $\lambda_{\max}$ can be computed by decision matrix $\mathbf{A}$ and weight vectors W_i

$$\lambda_{\max} = \frac{1}{n} \sum_{i=1}^{n} \frac{(\mathbf{A}w)_i}{w_i} \tag{47}$$

With calculation, we have $\mathrm{CR} = 0.04$ where less than 0, thus we can conclude that the weights are acceptable. Meanwhile, we can fill weight into following formula:

$$\begin{aligned} \mathrm{MPGA} &= w_{m_1} m_1 + w_{m_2} m_2 + w_{m_3} m_3 + w_{m_4} m_4 \\ 0.055 m_1 + 0.1178 m_2 + 0.2634 m_3 + 0.5638 m_4 \end{aligned} \tag{48}$$

2.6.6 Setting Thresholds and New Model Evaluation

Then, we should find thresholds for identifying whether the individual got COVID-19. In this way, we utilize `f1-score` which focus on recall and precision to iterate and then found the best thresholds is 0.31. The python code is as in Appendix. Also, the evaluation metrics of MPGA is greatly high. The python code and result is as in Appendix. Also, the evaluation metrics of MPGA is greatly high. The python code and result are in Appendix

3 Deep Learning Method

The previous Section (2) introduces the models and algorithms of Machine Learning to predict whether a person has COVID-19 or not by analyzing the symptoms of the tested person. However, relying solely on empirical existing features to determine whether a person is affected does not meet the medical system's criteria for judging the person being tested.

So in this Section, our team is committed to finding a more direct and scientific approach, which is to design a specific algorithm to analyze the lung CT images of the person being tested, so as to more objectively predict whether the person being tested has COVID-19. Here our group design a model using Convolutional Neural Network (CNN)

3.1 Introduction of Convolutional Neural Network (CNN)

A Convolutional Neural Network (CNN) is a feed-forward neural network that learns features by itself via filter (or kernel) optimization. This is a kind of shift invariant or space invariant artificial neural networks which are fully connected networks with each neuron in one layer is connected to all neuron in the next layer. This model is also based on the shared-weight architecture of the convolution kernels to provide translation-equivariant responses–feature maps.

3.2 Data Prepare

The file we downloaded has automatically segmented the COVID-19 or non-COVID-19 data for us, but we don't need the training set and test set segmentation it prepared for us, because we want a more universal model (Pham et al., 2025). So in the data preparation part, we made the following adjustments:

Resize the image The reason why we resize the images is that photos of different sizes or resolutions will extract sample information of different sizes during convolution, which will lead to different importance of observed samples at the data level. So the purpose of unifying them into one size is to prevent inequality of the samples themselves.

Enhance the image We enhance the image by increasing contrast, reducing gray-scale, and increasing sharpness in order to make the information provided by each observation more obvious.

Convert it into a tensor The purpose of converting to tensor is to perform matrix operations and speed up the efficiency of our parallel computing.

Store it in the GPU memory We use the GPU's video memory to process our images faster, and using the GPU can speed up the construction of deep learning models.

3.3 Lenet-5

LeNet 5 is a series of CNN, which is the one of the most significant and powerful CNN Architecture which contains 2 Convolutional Layers, 2 Pooling Layers and 2 Full Connection Layer.

3.4 Convolution

We are actually doing matrix multiplication in the convolutional layer. We have an input tensor and an operation tensor. After passing through a convolutional layer, the image becomes abstracted to a feature map, also called an activation map

$$(\mathbf{x} * \mathbf{f})(i,j,c) = \varphi\left(\sum_{k=-K}^{K}\sum_{l=-L}^{L}\sum_{c'} f^{c}_{k,l,c'} \cdot x_{i+k,j+l,c'} + b_c\right) \tag{49}$$

where the i, j, c is the length, width, dimensions of inputs tensors, k, l, c' is the length, width, dimensions of operation tensors, and φ is the activation function which we would like to use rectifier (ReLU) function

$$\text{ReLU} = \max(0, x) \tag{50}$$

where x is the input to a neuron. In fact, we do this kind of multiplication is to get the feature or useful information about the boundary shape and the lightness distribution of the image in order to construct hidden layers to recognize the images.

3.5 Pooling

Convolutional networks often incorporate both local and global pooling layers alongside standard convolutional layers. These pooling layers minimize data dimensions by aggregating the outputs of neuron clusters from one layer into a single neuron in the subsequent layer

$$\text{MaxPool}(x|f,s)_{i,j,c} = \max\left(x_{is_H:is_H+f_H-1,js_W:js_W+f_W-1,c}\right) \tag{51}$$

where is_H, js_W is the starting index, f_H, f_W is the ending index. This layer does operations to find the maximum value of this sub-matrix in order to find the most significant information and remove redundant data in order to simplify the following calculations.

3.6 Fully-connected

Fully connected layers link every neuron in one layer to every neuron in the next, functioning similarly to a traditional multilayer perceptron (MLP). In image classification, the flattened matrix passes through a fully connected layer to perform the classification task. We do this in order to integrate the existing results and balance the output of each node, and finally return the expected result.

4 Conclusion

In 2020 Feb epidemic, the shortage of medical resources caused a great problem of Misdiagnosis and delayed treatment. Through using the MPGA model that we construct, we can make an initial determination of individuals whose test results are ”Yes” when the evaluation value

of MPGA is greater than 0.31. Then, these individuals can be prioritized for CT scans to confirm whether they are truly positive cases. Therefore, our model contributes to enhance the efficiency and effectiveness of the system of medical services, and minimize resource wastage to the greatest extent possible.

References

Akbar, S., Azam, H., Almutairi, S. S., Alqahtani, O., Shah, H., & Aleryani, A. (2024). Contemporary study for detection of covid-19 using machine learning with explainable ai. *Computers, Materials amp; Continua*, *80*(1), 1075–1104. https://doi.org/10.32604/cmc.2024.050913

Josephus, B. O., Nawir, A. H., Wijaya, E., Moniaga, J. V., & Ohyver, M. (2021). Predict mortality in patients infected with covid-19 virus based on observed characteristics of the patient using logistic regression. *Procedia Computer Science*, *179*, 871–877. https://doi.org/10.1016/j.procs.2021.01.076

Manzella, F., Pagliarini, G., Sciavicco, G., & Stan, I. (2023). The voice of covid-19: Breath and cough recording classification with temporal decision trees and random forests. *Artificial Intelligence in Medicine*, *137*, 102486. https://doi.org/10.1016/j.artmed.2022.102486

Pham, N. T., Ko, J., Shah, M., Rakkiyappan, R., Woo, H. G., & Manavalan, B. (2025). Leveraging deep transfer learning and explainable ai for accurate covid-19 diagnosis: Insights from a multi-national chest ct scan study. *Computers in Biology and Medicine*, *185*, 109461. https://doi.org/10.1016/j.compbiomed.2024.109461

Swathi Priyadarshini, T., & Abdul Hameed, M. (2024). Developing heart stroke prediction model using deep learning with combination of fixed row initial centroid method with navie bayes, decision tree, and artificial neural network. *Measurement: Sensors*, *34*, 101237. https://doi.org/10.1016/j.measen.2024.101237

A Code

NavieBayes(done)

December 7, 2024

1 Naive Bayes

[1]:
```
import numpy as np
import pandas as pd

from sklearn.naive_bayes import GaussianNB, BernoulliNB
from sklearn.model_selection import KFold,GridSearchCV
from sklearn.metrics import accuracy_score, precision_score, recall_score,␣
↪f1_score, roc_curve, auc

import matplotlib.pyplot as plt
```

[2]:
```
df = pd.read_csv("Covid Dataset.csv")
df1 = df.dropna()
df1.head()
```

[2]:

	Breathing Problem	Fever	Dry Cough	Sore throat	Running Nose	Asthma	\
0	Yes	Yes	Yes	Yes	Yes	No	
1	Yes	Yes	Yes	Yes	No	Yes	
2	Yes	Yes	Yes	Yes	Yes	Yes	
3	Yes	Yes	Yes	No	No	Yes	
4	Yes	Yes	Yes	Yes	Yes	No	

	Chronic Lung Disease	Headache	Heart Disease	Diabetes	...	Fatigue	\
0	No	No	No	Yes	...	Yes	
1	Yes	Yes	No	No	...	Yes	
2	Yes	Yes	No	Yes	...	Yes	
3	No	No	Yes	Yes	...	No	
4	Yes	Yes	Yes	Yes	...	No	

	Gastrointestinal	Abroad travel	Contact with COVID Patient	\
0	Yes	No	Yes	
1	No	No	No	
2	Yes	Yes	No	
3	No	Yes	No	
4	Yes	No	Yes	

```
  Attended Large Gathering Visited Public Exposed Places  \
0                        No                           Yes
1                       Yes                           Yes
2                        No                            No
3                       Yes                           Yes
4                        No                           Yes

  Family working in Public Exposed Places Wearing Masks  \
0                                     Yes            No
1                                      No            No
2                                      No            No
3                                      No            No
4                                      No            No

  Sanitization from Market COVID-19
0                       No      Yes
1                       No      Yes
2                       No      Yes
3                       No      Yes
4                       No      Yes

[5 rows x 21 columns]
```

[3]:
```
for i in df1.columns:
    df1[i] = df1[i].apply(lambda x: 1 if x == "Yes" else 0)

df1.head()
```

[3]:

	Breathing Problem	Fever	Dry Cough	Sore throat	Running Nose	Asthma	\
0	1	1	1	1	1	0	
1	1	1	1	1	0	1	
2	1	1	1	1	1	1	
3	1	1	1	0	0	1	
4	1	1	1	1	1	0	

	Chronic Lung Disease	Headache	Heart Disease	Diabetes	...	Fatigue	\
0	0	0	0	1	...	1	
1	1	1	0	0	...	1	
2	1	1	0	1	...	1	
3	0	0	1	1	...	0	
4	1	1	1	1	...	0	

	Gastrointestinal	Abroad travel	Contact with COVID Patient	\
0	1	0	1	
1	0	0	0	
2	1	1	0	
3	0	1	0	

```
4                   1              0                             1

   Attended Large Gathering  Visited Public Exposed Places  \
0                         0                              1
1                         1                              1
2                         0                              0
3                         1                              1
4                         0                              1

   Family working in Public Exposed Places  Wearing Masks  \
0                                        1              0
1                                        0              0
2                                        0              0
3                                        0              0
4                                        0              0

   Sanitization from Market  COVID-19
0                         0         1
1                         0         1
2                         0         1
3                         0         1
4                         0         1

[5 rows x 21 columns]
```

[4]:
```
Y = df1["COVID-19"]
X = df1.drop("COVID-19", axis=1)
```

1.1 Gaussian Naive Bayes (Directly using KFold)

[5]:
```
model = GaussianNB()

# Use KFold for cross-validation
kf = KFold(n_splits=5, shuffle=True, random_state=42)

accuracy_scores = []

for train_index, test_index in kf.split(X):
    # Split data into training and testing sets
    X_train, X_test = X.iloc[train_index], X.iloc[test_index]
    Y_train, Y_test = Y.iloc[train_index], Y.iloc[test_index]

    # Train the model
    model.fit(X_train, Y_train)

    # Test the model
    predictions = model.predict(X_test)
```

```
    acc = round(accuracy_score(Y_test, predictions),4)
    accuracy_scores.append(acc)


print(f"Accuracy per fold: {accuracy_scores}")

print("\nAverage accuracy:", np.mean(accuracy_scores))
```

```
Accuracy per fold: [0.7663, 0.7341, 0.7489, 0.7884, 0.7643]

Average accuracy: 0.7604
```

1.2 Bernoulli Naive Bayes

[6]:
```
model1 = BernoulliNB()

kf = KFold(n_splits=5, shuffle=True, random_state=42)

accuracy_scores = []
precision_scores = []
recall_scores = []
f1_scores = []

for train_index, test_index in kf.split(X):

    X_train1, X_test1 = X.iloc[train_index], X.iloc[test_index]
    Y_train1, Y_test1 = Y.iloc[train_index], Y.iloc[test_index]

    model1.fit(X_train1, Y_train1)
    predictions = model1.predict(X_test1)

    acc = round(accuracy_score(Y_test1, predictions),4)
    precision = round(precision_score(Y_test1, predictions),4)
    recall = round(recall_score(Y_test1, predictions),4)
    f1 = round(f1_score(Y_test1, predictions),4)

    accuracy_scores.append(acc)
    precision_scores.append(precision)
    recall_scores.append(recall)
    f1_scores.append(f1)

print("Accuracy per fold:", accuracy_scores)
print("Precision per fold:", precision_scores)
print("Recall per fold:", recall_scores)
print("F1-Score per fold:", f1_scores)
```

```
print("\nAverage Accuracy: {:.4f}".format(np.mean(accuracy_scores)))
print("Average Precision: {:.4f}".format(np.mean(precision_scores)))
print("Average Recall: {:.4f}".format(np.mean(recall_scores)))
print("Average F1-Score: {:.4f}".format(np.mean(f1_scores)))
```

```
Accuracy per fold: [0.9706, 0.9586, 0.9733, 0.9706, 0.9549]
Precision per fold: [np.float64(0.9733), np.float64(0.9568), np.float64(0.9849),
np.float64(0.9757), np.float64(0.966)]
Recall per fold: [np.float64(0.991), np.float64(0.9931), np.float64(0.9815),
np.float64(0.9888), np.float64(0.9782)]
F1-Score per fold: [np.float64(0.9821), np.float64(0.9746), np.float64(0.9832),
np.float64(0.9822), np.float64(0.9721)]

Average Accuracy: 0.9656
Average Precision: 0.9713
Average Recall: 0.9865
Average F1-Score: 0.9788
```

1.2.1 Evaluation

ROCAUC

[7]:
```
kf = KFold(n_splits=5, shuffle=True, random_state=42)

# Initialize lists for FPR and TPR
tprs = []
mean_fpr = np.linspace(0, 1, 100)
roc_auc_list = []

for train_index, test_index in kf.split(X):

    X_train3, X_test3 = X.iloc[train_index], X.iloc[test_index]
    y_train3, y_test3 = Y.iloc[train_index], Y.iloc[test_index]

    model1.fit(X_train3, y_train3)

    # Predict probabilities for ROC curve
    y_test_proba = model1.predict_proba(X_test3)[:, 1]

    # Compute ROC curve and AUC
    fpr, tpr, thresholds = roc_curve(y_test3, y_test_proba)
    roc_auc = auc(fpr, tpr)
    roc_auc_list.append(roc_auc)

    # Interpolate TPR values for a common FPR range
    tprs.append(np.interp(mean_fpr, fpr, tpr))
    tprs[-1][0] = 0.0  # Ensure the curve starts at (0,0)

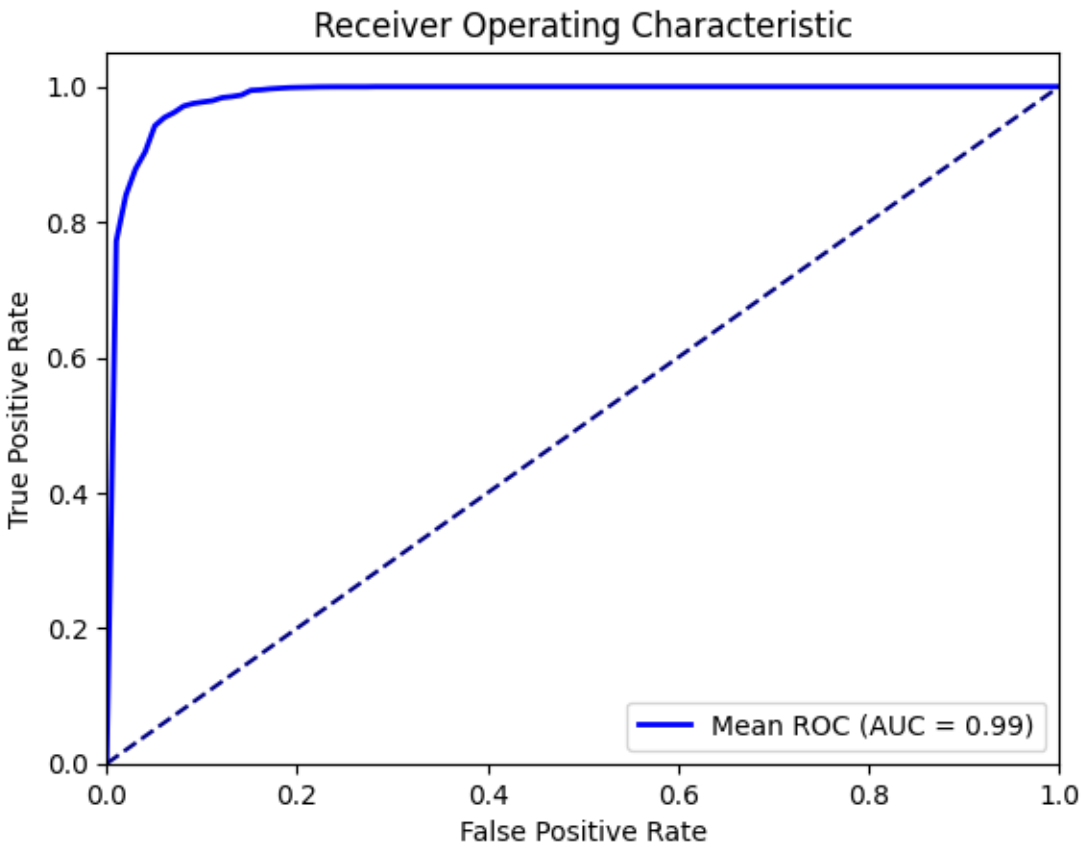
# Calculate the mean TPR and its standard deviation
```

```
mean_tpr = np.mean(tprs, axis=0)
mean_tpr[-1] = 1.0  # Ensure the curve ends at (1,1)
std_tpr = np.std(tprs, axis=0)

# Compute mean ROC AUC
mean_auc = np.mean(roc_auc_list)
std_auc = np.std(roc_auc_list)


# Plot the mean ROC curve
plt.figure()
plt.plot(mean_fpr, mean_tpr, color='b',label=r'Mean ROC (AUC = %0.2f)' % mean_auc
, lw=2)
plt.plot([0, 1], [0, 1], color='navy', linestyle='--')

#plot
plt.xlim([0.0, 1.0])
plt.ylim([0.0, 1.05])
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('Receiver Operating Characteristic')
plt.legend(loc='lower right')
plt.show()
```


Receiver Operating Characteristic
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Mean ROC (AUC = 0.99)

LogisticRegression(done)

December 7, 2024

[1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import GridSearchCV,KFold
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, precision_score, recall_score,␣
↪f1_score, roc_curve, auc
```

[2]:
```
df = pd.read_csv("Covid Dataset.csv")
df1 = df.dropna()
df1.head()
```

[2]:

	Breathing Problem	Fever	Dry Cough	Sore throat	Running Nose	Asthma	\
0	Yes	Yes	Yes	Yes	Yes	No	
1	Yes	Yes	Yes	Yes	No	Yes	
2	Yes	Yes	Yes	Yes	Yes	Yes	
3	Yes	Yes	Yes	No	No	Yes	
4	Yes	Yes	Yes	Yes	Yes	No	

	Chronic Lung Disease	Headache	Heart Disease	Diabetes	...	Fatigue	\
0	No	No	No	Yes	...	Yes	
1	Yes	Yes	No	No	...	Yes	
2	Yes	Yes	No	Yes	...	Yes	
3	No	No	Yes	Yes	...	No	
4	Yes	Yes	Yes	Yes	...	No	

	Gastrointestinal	Abroad travel	Contact with COVID Patient	\
0	Yes	No	Yes	
1	No	No	No	
2	Yes	Yes	No	
3	No	Yes	No	
4	Yes	No	Yes	

	Attended Large Gathering	Visited Public Exposed Places	\
0	No	Yes	
1	Yes	Yes	

```
2                        No                              No
3                       Yes                             Yes
4                        No                             Yes

  Family working in Public Exposed Places Wearing Masks  \
0                                       Yes            No
1                                        No            No
2                                        No            No
3                                        No            No
4                                        No            No

  Sanitization from Market COVID-19
0                        No      Yes
1                        No      Yes
2                        No      Yes
3                        No      Yes
4                        No      Yes

[5 rows x 21 columns]
```

[3]:
```
for i in df1.columns:
    df1[i] = df1[i].apply(lambda x: 1 if x == "Yes" else 0)
```

[4]:
```
Y = df1["COVID-19"]
X = df1.drop("COVID-19", axis=1)
```

[5]:
```
model = LogisticRegression()

kf = KFold(n_splits=5, shuffle=True, random_state=42)

accuracy_scores = []

for train_index, test_index in kf.split(X):
    # Split data into training and testing sets
    X_train, X_test = X.iloc[train_index], X.iloc[test_index]
    Y_train, Y_test = Y.iloc[train_index], Y.iloc[test_index]

    # Train the model
    model.fit(X_train, Y_train)

    # Test the model
    predictions = model.predict(X_test)

    acc = accuracy_score(Y_test, predictions)
    accuracy_scores.append(acc)
```

```
print("Accuracy per fold:", accuracy_scores)

print("\nAverage accuracy:", np.mean(accuracy_scores))
```

```
Accuracy per fold: [0.9724011039558418, 0.9604415823367065, 0.9724011039558418,
0.9705611775528978, 0.9640883977900553]

Average accuracy: 0.9679786731182686
```

[6]:
```
model1 = LogisticRegression()

kf = KFold(n_splits=5, shuffle=True, random_state=42)

accuracy_scores = []
precision_scores = []
recall_scores = []
f1_scores = []

for train_index, test_index in kf.split(X):

    X_train1, X_test1 = X.iloc[train_index], X.iloc[test_index]
    Y_train1, Y_test1 = Y.iloc[train_index], Y.iloc[test_index]

    model1.fit(X_train1, Y_train1)
    predictions = model1.predict(X_test1)

    acc = accuracy_score(Y_test1, predictions)
    precision = precision_score(Y_test1, predictions)
    recall = recall_score(Y_test1, predictions)
    f1 = f1_score(Y_test1, predictions)

    accuracy_scores.append(acc)
    precision_scores.append(precision)
    recall_scores.append(recall)
    f1_scores.append(f1)

print("Accuracy per fold:", accuracy_scores)
print("Precision per fold:", precision_scores)
print("Recall per fold:", recall_scores)
print("F1-Score per fold:", f1_scores)

print("\nAverage Accuracy: {:.4f}".format(np.mean(accuracy_scores)))
print("Average Precision: {:.4f}".format(np.mean(precision_scores)))
print("Average Recall: {:.4f}".format(np.mean(recall_scores)))
print("Average F1-Score: {:.4f}".format(np.mean(f1_scores)))
```

```
Accuracy per fold: [0.9724011039558418, 0.9604415823367065, 0.9724011039558418,
0.9705611775528978, 0.9640883977900553]
```

```
Precision per fold: [np.float64(0.9755011135857461), np.float64(0.9609375),
np.float64(0.9826589595375722), np.float64(0.9778024417314095),
np.float64(0.9695603156708005)]
Recall per fold: [np.float64(0.9909502262443439),
np.float64(0.9907940161104718), np.float64(0.9826589595375722),
np.float64(0.986562150055991), np.float64(0.9862385321100917)]
F1-Score per fold: [np.float64(0.9831649831649831),
np.float64(0.9756373937677054), np.float64(0.9826589595375722),
np.float64(0.9821627647714605), np.float64(0.9778283115406481)]

Average Accuracy: 0.9680
Average Precision: 0.9733
Average Recall: 0.9874
Average F1-Score: 0.9803
```

[7]:
```python
# Initialize Logistic Regression model with default parameters
model2 = LogisticRegression(random_state=42, max_iter=2000)


param_grid = {
    'C': np.arange(0.7001, 1.0001, 0.0001),
    'penalty': ['l2'],
    'solver': ['lbfgs']  }

grid_search = GridSearchCV(model, param_grid, verbose=2, cv=kf)


grid_search.fit(X_train, Y_train)


best_params = grid_search.best_params_
print(f"Best Parameters: {best_params}")

best_model = grid_search.best_estimator_
```

```
Fitting 5 folds for each of 3001 candidates, totalling 15005 fits
[CV] END ...C=0.7001, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7001, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7001, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7001, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7001, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7001999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7001999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7001999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7001999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7001999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7002999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7002999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7002999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7002999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7002999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7003999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7003999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7003999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7003999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7003999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7004999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7004999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7004999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7004999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7004999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7005999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7005999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7005999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7005999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7005999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7006999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7006999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7006999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7006999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7006999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7007999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7007999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7007999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7007999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7007999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7008999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7008999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7008999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7008999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7009999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7009999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7009999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7009999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7009999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7010999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7010999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7010999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7010999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7010999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7011999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7011999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7011999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7011999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7012999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7012999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7012999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7012999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7012999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7013999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7013999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7013999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7013999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7013999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7014999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7014999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7014999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7014999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7014999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7015999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7015999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7015999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7015999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7015999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7016999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7016999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7016999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7016999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7016999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7017999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7017999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7017999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7017999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7017999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7018999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7018999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7018999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7018999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7018999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7019999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7019999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7019999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7019999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7019999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7020999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7020999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7020999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7020999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7020999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7021999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7021999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7021999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7021999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7021999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7022999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7022999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7022999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7022999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7022999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7023999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7023999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7023999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7023999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7023999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7024999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7024999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7024999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7024999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7024999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7025999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7025999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7025999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7025999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7025999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7026999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7026999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7026999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7026999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7026999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7027999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7027999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7027999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7027999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7027999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7028999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7028999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7028999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7028999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7028999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7029999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7029999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7029999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7029999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7029999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7030999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7030999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7030999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7030999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7030999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7031999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7031999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7031999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7031999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7031999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7032999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7032999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7032999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7032999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7032999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7033999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7033999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7033999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7033999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7033999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7034999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7034999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7034999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7034999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7034999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7035999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7035999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7035999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7035999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7035999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7036999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7036999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7036999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7036999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7036999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7037999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7037999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7037999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7037999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7037999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7038999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7038999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7038999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7038999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7038999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7039999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7039999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7039999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7039999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7039999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7040999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7040999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7040999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7040999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7040999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7041999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7041999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7041999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7041999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7041999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7042999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7042999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7042999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7042999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7042999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7043999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7043999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7043999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7043999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7043999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7044999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7044999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7044999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7044999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7044999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7045999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7045999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7045999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7045999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7045999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7046999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7046999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7046999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7046999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7046999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7047999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7047999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7047999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7047999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7047999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7048999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7048999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7048999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7048999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7048999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7049999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7049999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7049999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7049999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7049999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7050999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7050999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7050999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7050999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7050999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7051999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7051999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7051999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7051999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7051999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7052999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7052999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7052999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7052999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7052999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7053999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7053999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7053999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7053999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7053999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7054999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7054999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7054999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7054999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7054999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7055999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7055999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7055999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7055999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7055999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7056999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7056999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7056999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7056999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7056999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7057999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7057999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7057999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7057999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7057999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7058999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7058999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7058999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7058999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7058999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7059999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7059999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7059999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7059999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7060999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7060999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7060999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7060999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7060999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7061999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7061999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7061999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7061999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7061999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7062999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7062999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7062999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7062999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7062999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7063999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7063999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7063999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7063999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7063999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7064999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7064999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7064999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7064999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7064999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7065999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7065999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7065999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7065999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7065999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7066999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7066999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7066999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7066999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7066999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7067999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7067999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7067999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7067999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7067999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7068999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7068999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7068999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7068999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7068999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7069999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7069999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7069999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7069999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7069999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7070999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7070999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7070999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7070999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7070999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7071999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7071999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7071999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7071999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7071999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7072999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7072999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7072999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7072999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7072999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7073999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7073999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7073999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7073999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7073999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7074999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7074999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7074999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7074999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7074999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7075999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7075999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7075999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7075999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7075999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7076999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7076999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7076999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7076999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7076999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7077999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7077999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7077999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7077999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7077999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7078999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7078999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7078999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7078999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7078999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7079999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7079999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7079999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7079999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7079999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7080999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7080999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7080999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7080999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7080999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708199999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708199999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708199999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708199999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708199999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708299999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708299999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708299999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708299999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708299999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708399999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708399999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708399999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708399999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708399999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708499999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708499999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708499999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708499999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708499999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708599999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708599999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708599999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708599999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708599999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708699999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708699999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708699999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708699999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708699999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708799999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708799999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708799999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708799999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708799999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708899999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708899999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708899999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708899999999999, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.708899999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.708999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.709099999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.709099999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.709099999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.709099999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.709099999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7091999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7091999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7091999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7091999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7091999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7092999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7092999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7092999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7092999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7092999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7093999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7093999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7093999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7093999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7093999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7094999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7094999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7094999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7094999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7094999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7095999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7095999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7095999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7095999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7095999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7096999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7096999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7096999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7096999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7096999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7097999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7097999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7097999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7097999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7097999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7098999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7098999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7098999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7098999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7098999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7099999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7099999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7099999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7099999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7099999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7100999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7100999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7100999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7100999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7100999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7101999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7101999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7101999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7101999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7101999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7102999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7102999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7102999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7102999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7102999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7103999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7103999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7103999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7103999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7103999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7104999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7104999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7104999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7104999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7104999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7105999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7105999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7105999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7105999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7105999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7106999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7106999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7106999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7106999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7106999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7107999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7107999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7107999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7107999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7108999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7108999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7108999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7108999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7108999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7109999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7109999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7109999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7109999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7109999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7110999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7110999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7110999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7110999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7110999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7111999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7111999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7111999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7111999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7111999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7112999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7112999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7112999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7112999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7112999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7113999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7113999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7113999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7113999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7113999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7114999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7114999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7114999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7114999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7114999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7115999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7115999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7115999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7115999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7115999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7116999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7116999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7116999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7116999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7116999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7117999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7117999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7117999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7117999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7117999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7118999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7118999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7118999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7118999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7118999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7119999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7119999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7119999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7119999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7119999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7120999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7120999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7120999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7120999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7120999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7121999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7121999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7121999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7121999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7121999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7122999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7122999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7122999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7122999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7122999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7123999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7123999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7123999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7123999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7123999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7124999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7124999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7124999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7124999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7124999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7125999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7125999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7125999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7125999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7125999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7126999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7126999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7126999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7126999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7126999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7127999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7127999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7127999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7127999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7127999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7128999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7128999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7128999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7128999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7128999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7129999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7129999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7129999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7129999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7129999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7130999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7130999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7130999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7130999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7130999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7131999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7131999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7131999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7131999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7131999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7132999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7132999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7132999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7132999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7132999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7133999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7133999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7133999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7133999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7133999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7134999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7134999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7134999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7134999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7134999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7135999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7135999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7135999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7135999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7135999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7136999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7136999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7136999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7136999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7136999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7137999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7137999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7137999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7137999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7137999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7138999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7138999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7138999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7138999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7138999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7139999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7139999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7139999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7139999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7139999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7140999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7140999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7140999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7140999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7140999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7141999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7141999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7141999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7141999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7141999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7142999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7142999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7142999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7142999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7142999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7143999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7143999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7143999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7143999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7143999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7144999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7144999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7144999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7144999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7144999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7145999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7145999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7145999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7145999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7145999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7146999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7146999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7146999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7146999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7146999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7147999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7147999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7147999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7147999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7147999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7148999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7148999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7148999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7148999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7148999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7149999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7149999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7149999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7149999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7149999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7150999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7150999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7150999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7150999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7150999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7151999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7151999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7151999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7151999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7151999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7152999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7152999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7152999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7152999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7152999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7153999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7153999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7153999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7153999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7153999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7154999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7154999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7154999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7154999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7154999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7155999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7155999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7155999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7155999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7156999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7156999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7156999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7156999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7156999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7157999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7157999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7157999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7157999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7157999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7158999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7158999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7158999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7158999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7158999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7159999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7159999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7159999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7159999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7159999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7160999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7160999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7160999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7160999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7160999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7161999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7161999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7161999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7161999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7161999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7162999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7162999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7162999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7162999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7162999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7163999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7163999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7163999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7163999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7163999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7164999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7164999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7164999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7164999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7164999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7165999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7165999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7165999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7165999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7165999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7166999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7166999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7166999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7166999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7166999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7167999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7167999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7167999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7167999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7167999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7168999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7168999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7168999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7168999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7168999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7169999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7169999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7169999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7169999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7169999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7170999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7170999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7170999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7170999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7170999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7171999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7171999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7171999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7171999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7171999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717299999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717299999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717299999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717299999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717299999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717399999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717399999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717399999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717399999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717399999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717499999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717499999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717499999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717499999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717499999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717599999999998, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.717599999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717599999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717599999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717599999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717699999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717699999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717699999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717699999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717699999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717799999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717799999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717799999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717799999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717799999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717899999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717899999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717899999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717899999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717899999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.717999999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.718099999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.718099999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.718099999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.718099999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.718099999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.718199999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.718199999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.718199999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.718199999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7182999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7182999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7182999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7182999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7182999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7183999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7183999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7183999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7183999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7183999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7184999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7184999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7184999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7184999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7184999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7185999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7185999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7185999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7185999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7185999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7186999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7186999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7186999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7186999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7186999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7187999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7187999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7187999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7187999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7187999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7188999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7188999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7188999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7188999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7188999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7189999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7189999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7189999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7189999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7189999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7190999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7190999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7190999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7190999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7190999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7191999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7191999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7191999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7191999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7191999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7192999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7192999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7192999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7192999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7192999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7193999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7193999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7193999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7193999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7193999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7194999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7194999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7194999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7194999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7194999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7195999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7195999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7195999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7195999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7195999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7196999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7196999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7196999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7196999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7196999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7197999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7197999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7197999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7197999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7197999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7198999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7198999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7198999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7198999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7198999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7199999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7199999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7199999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7199999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7199999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7200999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7200999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7200999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7200999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7200999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7201999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7201999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7201999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7201999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7201999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7202999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7202999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7202999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7202999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7202999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7203999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7203999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7203999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7203999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7204999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7204999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7204999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7204999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7204999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7205999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7205999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7205999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7205999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7205999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7206999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7206999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7206999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7206999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7206999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7207999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7207999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7207999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7207999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7207999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7208999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7208999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7208999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7208999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7208999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7209999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7209999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7209999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7209999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7209999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7210999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7210999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7210999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7210999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7210999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7211999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7211999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7211999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7211999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7211999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7212999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7212999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7212999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7212999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7212999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7213999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7213999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7213999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7213999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7213999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7214999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7214999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7214999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7214999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7214999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7215999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7215999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7215999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7215999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7215999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7216999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7216999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7216999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7216999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7216999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7217999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7217999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7217999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7217999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7217999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7218999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7218999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7218999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7218999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7218999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7219999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7219999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7219999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7219999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7219999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7220999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7220999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7220999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7220999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7220999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7221999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7221999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7221999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7221999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7221999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7222999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7222999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7222999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7222999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7222999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7223999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7223999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7223999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7223999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7223999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7224999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7224999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7224999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7224999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7224999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7225999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7225999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7225999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7225999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7225999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7226999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7226999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7226999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7226999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7226999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7227999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7227999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7227999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7227999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7227999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7228999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7228999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7228999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7228999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7228999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7229999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7229999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7229999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7229999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7230999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7230999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7230999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7230999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7230999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7231999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7231999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7231999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7231999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7231999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7232999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7232999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7232999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7232999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7232999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7233999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7233999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7233999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7233999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7233999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7234999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7234999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7234999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7234999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7234999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7235999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7235999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7235999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7235999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7235999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7236999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7236999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7236999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7236999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7236999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7237999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7237999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7237999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7237999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7237999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7238999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7238999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7238999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7238999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7238999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7239999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7239999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7239999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7239999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7239999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7240999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7240999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7240999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7240999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7240999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7241999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7241999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7241999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7241999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7241999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7242999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7242999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7242999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7242999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7242999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7243999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7243999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7243999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7243999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7243999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7244999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7244999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7244999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7244999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7244999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7245999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7245999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7245999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7245999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7245999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7246999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7246999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7246999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7246999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7246999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7247999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7247999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7247999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7247999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7247999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7248999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7248999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7248999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7248999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7248999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7249999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7249999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7249999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7249999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7250999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7250999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7250999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7250999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7250999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7251999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7251999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7251999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7251999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7252999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7252999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7252999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7252999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7252999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7253999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7253999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7253999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7253999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7253999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7254999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7254999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7254999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7254999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7254999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7255999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7255999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7255999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7255999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7255999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7256999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7256999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7256999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7256999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7256999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7257999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7257999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7257999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7257999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7257999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7258999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7258999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7258999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7258999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7258999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7259999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7259999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7259999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7259999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7259999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7260999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7260999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7260999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7260999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7260999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7261999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7261999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7261999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7261999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7261999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7262999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7262999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7262999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7262999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7262999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726399999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726399999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726399999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726399999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726399999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726499999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726499999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726499999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726499999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726499999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726599999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726599999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726599999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726599999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726599999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726699999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726699999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726699999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726699999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726699999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726799999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726799999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726799999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726799999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726799999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726899999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726899999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726899999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726899999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726899999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.726999999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.727099999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.727099999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.727099999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.727099999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.727099999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.727199999999997, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.727199999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.727199999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.727199999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.727199999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.727299999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.727299999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.727299999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.727299999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.727299999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7273999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7273999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7273999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7273999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7273999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7274999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7274999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7274999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7274999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7274999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7275999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7275999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7275999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7275999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7275999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7276999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7276999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7276999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7276999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7276999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7277999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7277999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7277999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7277999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7277999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7278999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7278999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7278999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7278999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7278999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7279999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7279999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7279999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7279999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7279999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7280999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7280999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7280999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7280999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7280999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7281999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7281999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7281999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7281999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7281999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7282999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7282999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7282999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7282999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7282999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7283999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7283999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7283999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7283999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7283999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7284999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7284999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7284999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7284999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7284999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7285999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7285999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7285999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7285999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7285999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7286999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7286999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7286999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7286999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7287999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7287999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7287999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7287999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7287999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7288999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7288999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7288999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7288999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7288999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7289999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7289999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7289999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7289999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7289999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7290999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7290999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7290999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7290999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7290999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7291999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7291999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7291999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7291999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7291999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7292999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7292999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7292999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7292999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7292999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7293999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7293999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7293999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7293999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7293999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7294999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7294999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7294999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7294999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7294999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7295999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7295999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7295999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7295999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7295999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7296999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7296999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7296999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7296999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7296999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7297999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7297999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7297999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7297999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7297999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7298999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7298999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7298999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7298999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7298999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7299999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7299999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7299999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7299999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7300999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7300999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7300999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7300999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7300999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7301999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7301999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7301999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7301999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7301999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7302999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7302999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7302999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7302999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7302999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7303999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7303999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7303999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7303999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7303999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7304999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7304999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7304999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7304999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7304999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7305999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7305999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7305999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7305999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7305999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7306999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7306999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7306999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7306999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7306999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7307999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7307999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7307999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7307999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7307999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7308999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7308999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7308999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7308999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7308999999999966, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7309999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7309999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7309999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7309999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7309999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7310999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7310999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7310999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7310999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7310999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7311999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7311999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7311999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7311999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7311999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7312999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7312999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7312999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7312999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7312999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7313999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7313999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7313999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7313999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7313999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7314999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7314999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7314999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7314999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7314999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7315999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7315999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7315999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7315999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7315999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7316999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7316999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7316999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7316999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7316999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7317999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7317999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7317999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7317999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7317999999999965, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7318999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7318999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7318999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7318999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7318999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7319999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7319999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7319999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7319999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7319999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7320999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7320999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7320999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7320999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7320999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7321999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7321999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7321999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7321999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7321999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7322999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7322999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7322999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7322999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7322999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7323999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7323999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7323999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7323999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7323999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7324999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7324999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7324999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7324999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7324999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7325999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7325999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7325999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7325999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7325999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7326999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7326999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7326999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7326999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7326999999999964, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7327999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7327999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7327999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7327999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7327999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7328999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7328999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7328999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7328999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7328999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7329999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7329999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7329999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7329999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7329999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7330999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7330999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7330999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7330999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7330999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7331999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7331999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7331999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7331999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7331999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7332999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7332999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7332999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7332999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7332999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7333999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7333999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7333999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7333999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7333999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7334999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7334999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7334999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7334999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7334999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7335999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7335999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7335999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7335999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7335999999999963, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7336999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7336999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7336999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7336999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7336999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7337999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7337999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7337999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7337999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7337999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7338999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7338999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7338999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7338999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7338999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7339999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7339999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7339999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7339999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7339999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7340999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7340999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7340999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7340999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7340999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7341999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7341999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7341999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7341999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7341999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7342999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7342999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7342999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7342999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7342999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7343999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7343999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7343999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7343999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7343999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7344999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7344999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7344999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7344999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7344999999999962, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7345999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7345999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7345999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7345999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7345999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7346999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7346999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7346999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7346999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7346999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7347999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7347999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7347999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7347999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7348999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7348999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7348999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7348999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7348999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7349999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7349999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7349999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7349999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7349999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7350999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7350999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7350999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7350999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7350999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7351999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7351999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7351999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7351999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7351999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7352999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7352999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7352999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7352999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7352999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7353999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7353999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7353999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7353999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7353999999999961, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735499999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735499999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735499999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735499999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735499999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735599999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735599999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735599999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735599999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735599999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735699999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735699999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735699999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735699999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735699999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735799999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735799999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735799999999996, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.735799999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735799999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735899999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735899999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735899999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735899999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735899999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.735999999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.736099999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.736099999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.736099999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.736099999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.736099999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.736199999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.736199999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.736199999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.736199999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.736199999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.736299999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.736299999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.736299999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.736299999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.736299999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.736399999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.736399999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.736399999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.736399999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.736399999999996, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7364999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7364999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7364999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7364999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7364999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7365999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7365999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7365999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7365999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7365999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7366999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7366999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7366999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7366999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7366999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7367999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7367999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7367999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7367999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7367999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7368999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7368999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7368999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7368999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7368999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7369999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7369999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7369999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7369999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7369999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7370999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7370999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7370999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7370999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7370999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7371999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7371999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7371999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7371999999999959, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7372999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7372999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7372999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7372999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7372999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7373999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7373999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7373999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7373999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7374999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7374999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7374999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7374999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7374999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7375999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7375999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7375999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7375999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7375999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7376999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7376999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7376999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7376999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7376999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7377999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7377999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7377999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7377999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7377999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7378999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7378999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7378999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7378999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7378999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7379999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7379999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7379999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7379999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7379999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7380999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7380999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7380999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7380999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7380999999999958, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7381999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7381999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7381999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7381999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7381999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7382999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7382999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7382999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7382999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7382999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7383999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7383999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7383999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7383999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7383999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7384999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7384999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7384999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7384999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7384999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7385999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7385999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7385999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7385999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7385999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7386999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7386999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7386999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7386999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7386999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7387999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7387999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7387999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7387999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7387999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7388999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7388999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7388999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7388999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7388999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7389999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7389999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7389999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7389999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7389999999999957, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7390999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7390999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7390999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7390999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7390999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7391999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7391999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7391999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7391999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7391999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7392999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7392999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7392999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7392999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7392999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7393999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7393999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7393999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7393999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7393999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7394999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7394999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7394999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7394999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7394999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7395999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7395999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7395999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7395999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7396999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7396999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7396999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7396999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7396999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7397999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7397999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7397999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7397999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7397999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7398999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7398999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7398999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7398999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7398999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7399999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7399999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7399999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7399999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7399999999999956, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7400999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7400999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7400999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7400999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7400999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7401999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7401999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7401999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7401999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7401999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7402999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7402999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7402999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7402999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7402999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7403999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7403999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7403999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7403999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7403999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7404999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7404999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7404999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7404999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7404999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7405999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7405999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7405999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7405999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7405999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7406999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7406999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7406999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7406999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7406999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7407999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7407999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7407999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7407999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7407999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7408999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7408999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7408999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7408999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7408999999999955, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7409999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7409999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7409999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7409999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7409999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7410999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7410999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7410999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7410999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7410999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7411999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7411999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7411999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7411999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7411999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7412999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7412999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7412999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7412999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7412999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7413999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7413999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7413999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7413999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7413999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7414999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7414999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7414999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7414999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7414999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7415999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7415999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7415999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7415999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7415999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7416999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7416999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7416999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7416999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7416999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7417999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7417999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7417999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7417999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7417999999999954, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7418999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7418999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7418999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7418999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7418999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7419999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7419999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7419999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7419999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7419999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7420999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7420999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7420999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7420999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7420999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7421999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7421999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7421999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7421999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7421999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7422999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7422999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7422999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7422999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7422999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7423999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7423999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7423999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7423999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7423999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7424999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7424999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7424999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7424999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7424999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7425999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7425999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7425999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7425999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7425999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7426999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7426999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7426999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7426999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7426999999999953, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7427999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7427999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7427999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7427999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7427999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7428999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7428999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7428999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7428999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7428999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7429999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7429999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7429999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7429999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7429999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7430999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7430999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7430999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7430999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7430999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7431999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7431999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7431999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7431999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7431999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7432999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7432999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7432999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7432999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7432999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7433999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7433999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7433999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7433999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7433999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7434999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7434999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7434999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7434999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7434999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7435999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7435999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7435999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7435999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7435999999999952, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7436999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7436999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7436999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7436999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7436999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7437999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7437999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7437999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7437999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7437999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7438999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7438999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7438999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7438999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7438999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7439999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7439999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7439999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7439999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7439999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7440999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7440999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7440999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7440999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7440999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7441999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7441999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7441999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7441999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7441999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7442999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7442999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7442999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7442999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7442999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7443999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7443999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7443999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7443999999999951, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.744499999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744499999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744499999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744499999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744499999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744599999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744599999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744599999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744599999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744599999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744699999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744699999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744699999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744699999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744699999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744799999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744799999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744799999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744799999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744799999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744899999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744899999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744899999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744899999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744899999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.744999999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.745099999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.745099999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.745099999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.745099999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.745099999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.745199999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.745199999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.745199999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.745199999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.745199999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.745299999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.745299999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.745299999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.745299999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.745299999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.745399999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.745399999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.745399999999995, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.745399999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.745399999999995, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7454999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7454999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7454999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7454999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7454999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7455999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7455999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7455999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7455999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7455999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7456999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7456999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7456999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7456999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7456999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7457999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7457999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7457999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7457999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7457999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7458999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7458999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7458999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7458999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7458999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7459999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7459999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7459999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7459999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7459999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7460999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7460999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7460999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7460999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7460999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7461999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7461999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7461999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7461999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7461999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7462999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7462999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7462999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7462999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7462999999999949, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7463999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7463999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7463999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7463999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7463999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7464999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7464999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7464999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7464999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7464999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7465999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7465999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7465999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7465999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7465999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7466999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7466999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7466999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7466999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7466999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7467999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7467999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7467999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7467999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7467999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7468999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7468999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7468999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7468999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7468999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7469999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7469999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7469999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7469999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7470999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7470999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7470999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7470999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7470999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7471999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7471999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7471999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7471999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7471999999999948, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7472999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7472999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7472999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7472999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7472999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7473999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7473999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7473999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7473999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7473999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7474999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7474999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7474999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7474999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7474999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7475999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7475999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7475999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7475999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7475999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7476999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7476999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7476999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7476999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7476999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7477999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7477999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7477999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7477999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7477999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7478999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7478999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7478999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7478999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7478999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7479999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7479999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7479999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7479999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7479999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7480999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7480999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7480999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7480999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7480999999999947, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7481999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7481999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7481999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7481999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7481999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7482999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7482999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7482999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7482999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7482999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7483999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7483999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7483999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7483999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7483999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7484999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7484999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7484999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7484999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7484999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7485999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7485999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7485999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7485999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7485999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7486999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7486999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7486999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7486999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7486999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7487999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7487999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7487999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7487999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7487999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7488999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7488999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7488999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7488999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7488999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7489999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7489999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7489999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7489999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7489999999999946, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7490999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7490999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7490999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7490999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7490999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7491999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7491999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7491999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7491999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7492999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7492999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7492999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7492999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7492999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7493999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7493999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7493999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7493999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7493999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7494999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7494999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7494999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7494999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7494999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7495999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7495999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7495999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7495999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7495999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7496999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7496999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7496999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7496999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7496999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7497999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7497999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7497999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7497999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7497999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7498999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7498999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7498999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7498999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7498999999999945, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7499999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7499999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7499999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7499999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7499999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7500999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7500999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7500999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7500999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7500999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7501999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7501999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7501999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7501999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7501999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7502999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7502999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7502999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7502999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7502999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7503999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7503999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7503999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7503999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7503999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7504999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7504999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7504999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7504999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7504999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7505999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7505999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7505999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7505999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7505999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7506999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7506999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7506999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7506999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7506999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7507999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7507999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7507999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7507999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7507999999999944, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7508999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7508999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7508999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7508999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7508999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7509999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7509999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7509999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7509999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7509999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7510999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7510999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7510999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7510999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7510999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7511999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7511999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7511999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7511999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7511999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7512999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7512999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7512999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7512999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7512999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7513999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7513999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7513999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7513999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7513999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7514999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7514999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7514999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7514999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7514999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7515999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7515999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7515999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7515999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7515999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7516999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7516999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7516999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7516999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7516999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7517999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7517999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7517999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7517999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7517999999999943, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7518999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7518999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7518999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7518999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7518999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7519999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7519999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7519999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7519999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7519999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7520999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7520999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7520999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7520999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7520999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7521999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7521999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7521999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7521999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7521999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7522999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7522999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7522999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7522999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7522999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7523999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7523999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7523999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7523999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7523999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7524999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7524999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7524999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7524999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7524999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7525999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7525999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7525999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7525999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7525999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7526999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7526999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7526999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7526999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7526999999999942, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7527999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7527999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7527999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7527999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7527999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7528999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7528999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7528999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7528999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7528999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7529999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7529999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7529999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7529999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7529999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7530999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7530999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7530999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7530999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7530999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7531999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7531999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7531999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7531999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7531999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7532999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7532999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7532999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7532999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7532999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7533999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7533999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7533999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7533999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7533999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7534999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7534999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7534999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7534999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7534999999999941, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.753599999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.753599999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.753599999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.753599999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.753599999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.753699999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.753699999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.753699999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.753699999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.753699999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.753799999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.753799999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.753799999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.753799999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.753799999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.753899999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.753899999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.753899999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.753899999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.753899999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.753999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.753999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.753999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.753999999999994, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.754099999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.754099999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.754099999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.754099999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.754099999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.754199999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.754199999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.754199999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.754199999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.754199999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.754299999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.754299999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.754299999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.754299999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.754299999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.754399999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.754399999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.754399999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.754399999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.754399999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.754499999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.754499999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.754499999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.754499999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.754499999999994, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7545999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7545999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7545999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7545999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7545999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7546999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7546999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7546999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7546999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7546999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7547999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7547999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7547999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7547999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7547999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7548999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7548999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7548999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7548999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7548999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7549999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7549999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7549999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7549999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7549999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7550999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7550999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7550999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7550999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7550999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7551999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7551999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7551999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7551999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7551999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7552999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7552999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7552999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7552999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7552999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7553999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7553999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7553999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7553999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7553999999999939, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7554999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7554999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7554999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7554999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7554999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7555999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7555999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7555999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7555999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7555999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7556999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7556999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7556999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7556999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7556999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7557999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7557999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7557999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7557999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7557999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7558999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7558999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7558999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7558999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7558999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7559999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7559999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7559999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7559999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7559999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7560999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7560999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7560999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7560999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7560999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7561999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7561999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7561999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7561999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7561999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7562999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7562999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7562999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7562999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7562999999999938, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7563999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7563999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7563999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7563999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7563999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7564999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7564999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7564999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7564999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7564999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7565999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7565999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7565999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7565999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7565999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7566999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7566999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7566999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7566999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7566999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7567999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7567999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7567999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7567999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7567999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7568999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7568999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7568999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7568999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7568999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7569999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7569999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7569999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7569999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7569999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7570999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7570999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7570999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7570999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7570999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7571999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7571999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7571999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7571999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7571999999999937, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7572999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7572999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7572999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7572999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7572999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7573999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7573999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7573999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7573999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7573999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7574999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7574999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7574999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7574999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7574999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7575999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7575999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7575999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7575999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7575999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7576999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7576999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7576999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7576999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7576999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7577999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7577999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7577999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7577999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7577999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7578999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7578999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7578999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7578999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7578999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7579999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7579999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7579999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7579999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7579999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7580999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7580999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7580999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7580999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7580999999999936, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7581999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7581999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7581999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7581999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7581999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7582999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7582999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7582999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7582999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7582999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7583999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7583999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7583999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7583999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7583999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7584999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7584999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7584999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7584999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7584999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7585999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7585999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7585999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7585999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7585999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7586999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7586999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7586999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7586999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7586999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7587999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7587999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7587999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7587999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7588999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7588999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7588999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7588999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7588999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7589999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7589999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7589999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7589999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7589999999999935, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7590999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7590999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7590999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7590999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7590999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7591999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7591999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7591999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7591999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7591999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7592999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7592999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7592999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7592999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7592999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7593999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7593999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7593999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7593999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7593999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7594999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7594999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7594999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7594999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7594999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7595999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7595999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7595999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7595999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7595999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7596999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7596999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7596999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7596999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7596999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7597999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7597999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7597999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7597999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7597999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7598999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7598999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7598999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7598999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7598999999999934, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7599999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7599999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7599999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7599999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7599999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7600999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7600999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7600999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7600999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7600999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7601999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7601999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7601999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7601999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7601999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7602999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7602999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7602999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7602999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7602999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7603999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7603999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7603999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7603999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7603999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7604999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7604999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7604999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7604999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7604999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7605999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7605999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7605999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7605999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7605999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7606999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7606999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7606999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7606999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7606999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7607999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7607999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7607999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7607999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7607999999999933, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7608999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7608999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7608999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7608999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7608999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7609999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7609999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7609999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7609999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7609999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7610999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7610999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7610999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7610999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7610999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7611999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7611999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7611999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7611999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7612999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7612999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7612999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7612999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7612999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7613999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7613999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7613999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7613999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7614999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7614999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7614999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7614999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7614999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7615999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7615999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7615999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7615999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7615999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7616999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7616999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7616999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7616999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7616999999999932, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7617999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7617999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7617999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7617999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7617999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7618999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7618999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7618999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7618999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7618999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7619999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7619999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7619999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7619999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7619999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7620999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7620999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7620999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7620999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7620999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7621999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7621999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7621999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7621999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7621999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7622999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7622999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7622999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7622999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7622999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7623999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7623999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7623999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7623999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7623999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7624999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7624999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7624999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7624999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7624999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7625999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7625999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7625999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7625999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7625999999999931, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.762699999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.762699999999993, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.762699999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.762699999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.762699999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.762799999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.762799999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.762799999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.762799999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.762799999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.762899999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.762899999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.762899999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.762899999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.762899999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.762999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.762999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.762999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.762999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.762999999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763099999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763099999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763099999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763099999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763099999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763199999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763199999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763199999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763199999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763199999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763299999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763299999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763299999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763299999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763299999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763399999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763399999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763399999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763399999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763399999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763499999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763499999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763499999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763499999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763499999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763599999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763599999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763599999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763599999999993, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.763599999999993, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7636999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7636999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7636999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7636999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7636999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7637999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7637999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7637999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7637999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7637999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7638999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7638999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7638999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7638999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7638999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7639999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7639999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7639999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7639999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7639999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7640999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7640999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7640999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7640999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7640999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7641999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7641999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7641999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7641999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7641999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7642999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7642999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7642999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7642999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7642999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7643999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7643999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7643999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7643999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7643999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7644999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7644999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7644999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7644999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7644999999999929, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7645999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7645999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7645999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7645999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7645999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7646999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7646999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7646999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7646999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7646999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7647999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7647999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7647999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7647999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7647999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7648999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7648999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7648999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7648999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7648999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7649999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7649999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7649999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7649999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7649999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7650999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7650999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7650999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7650999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7650999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7651999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7651999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7651999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7651999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7651999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7652999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7652999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7652999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7652999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7652999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7653999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7653999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7653999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7653999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7653999999999928, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7654999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7654999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7654999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7654999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7654999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7655999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7655999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7655999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7655999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7655999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7656999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7656999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7656999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7656999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7656999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7657999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7657999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7657999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7657999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7657999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7658999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7658999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7658999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7658999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7658999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7659999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7659999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7659999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7659999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7659999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7660999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7660999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7660999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7660999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7660999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7661999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7661999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7661999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7661999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7661999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7662999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7662999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7662999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7662999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7662999999999927, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7663999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7663999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7663999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7663999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7663999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7664999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7664999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7664999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7664999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7664999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7665999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7665999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7665999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7665999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7665999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7666999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7666999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7666999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7666999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7666999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7667999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7667999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7667999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7667999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7667999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7668999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7668999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7668999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7668999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7668999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7669999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7669999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7669999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7669999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7669999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7670999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7670999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7670999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7670999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7670999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7671999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7671999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7671999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7671999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7671999999999926, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7672999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7672999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7672999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7672999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7672999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7673999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7673999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7673999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7673999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7673999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7674999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7674999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7674999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7674999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7674999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7675999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7675999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7675999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7675999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7675999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7676999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7676999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7676999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7676999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7676999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7677999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7677999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7677999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7677999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7677999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7678999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7678999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7678999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7678999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7678999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7679999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7679999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7679999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7679999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7679999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7680999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7680999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7680999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7680999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7680999999999925, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7681999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7681999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7681999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7681999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7681999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7682999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7682999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7682999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7682999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7682999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7683999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7683999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7683999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7683999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7684999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7684999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7684999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7684999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7684999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7685999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7685999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7685999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7685999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7685999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7686999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7686999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7686999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7686999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7686999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7687999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7687999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7687999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7687999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7687999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7688999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7688999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7688999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7688999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7688999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7689999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7689999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7689999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7689999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7689999999999924, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7690999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7690999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7690999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7690999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7690999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7691999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7691999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7691999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7691999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7691999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7692999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7692999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7692999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7692999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7692999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7693999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7693999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7693999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7693999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7693999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7694999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7694999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7694999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7694999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7694999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7695999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7695999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7695999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7695999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7695999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7696999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7696999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7696999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7696999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7696999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7697999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7697999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7697999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7697999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7697999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7698999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7698999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7698999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7698999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7698999999999923, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7699999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7699999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7699999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7699999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7699999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7700999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7700999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7700999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7700999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7700999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7701999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7701999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7701999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7701999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7701999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7702999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7702999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7702999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7702999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7702999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7703999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7703999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7703999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7703999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7703999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7704999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7704999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7704999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7704999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7704999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7705999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7705999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7705999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7705999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7705999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7706999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7706999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7706999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7706999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7706999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7707999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7707999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7707999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7707999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7707999999999922, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7708999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7708999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7708999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7708999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7708999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7709999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7709999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7709999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7709999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7709999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7710999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7710999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7710999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7710999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7710999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7711999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7711999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7711999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7711999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7711999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7712999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7712999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7712999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7712999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7712999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7713999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7713999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7713999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7713999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7713999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7714999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7714999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7714999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7714999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7714999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7715999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7715999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7715999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7715999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7715999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7716999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7716999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7716999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7716999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7716999999999921, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.771799999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.771799999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.771799999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.771799999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.771799999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.771899999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.771899999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.771899999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.771899999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.771899999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.771999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.771999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.771999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.771999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.771999999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772099999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772099999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772099999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772099999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772099999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772199999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772199999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772199999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772199999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772199999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772299999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772299999999992, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.772299999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772299999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772299999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772399999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772399999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772399999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772399999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772399999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772499999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772499999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772499999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772499999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772499999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772599999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772599999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772599999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772599999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772599999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772699999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772699999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772699999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772699999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.772699999999992, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7727999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7727999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7727999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7727999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7727999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7728999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7728999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7728999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7728999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7728999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7729999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7729999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7729999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7729999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7729999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7730999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7730999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7730999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7730999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7730999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7731999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7731999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7731999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7731999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7732999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7732999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7732999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7732999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7732999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7733999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7733999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7733999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7733999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7733999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7734999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7734999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7734999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7734999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7734999999999919, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7735999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7735999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7735999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7735999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7735999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7736999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7736999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7736999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7736999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7736999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7737999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7737999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7737999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7737999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7737999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7738999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7738999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7738999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7738999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7738999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7739999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7739999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7739999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7739999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7739999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7740999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7740999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7740999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7740999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7740999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7741999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7741999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7741999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7741999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7741999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7742999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7742999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7742999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7742999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7742999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7743999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7743999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7743999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7743999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7743999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7744999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7744999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7744999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7744999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7744999999999918, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7745999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7745999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7745999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7745999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7745999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7746999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7746999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7746999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7746999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7746999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7747999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7747999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7747999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7747999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7747999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7748999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7748999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7748999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7748999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7748999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7749999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7749999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7749999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7749999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7749999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7750999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7750999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7750999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7750999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7750999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7751999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7751999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7751999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7751999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7751999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7752999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7752999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7752999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7752999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7752999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7753999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7753999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7753999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7753999999999917, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7754999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7754999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7754999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7754999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7754999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7755999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7755999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7755999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7755999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7756999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7756999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7756999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7756999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7756999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7757999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7757999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7757999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7757999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7758999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7758999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7758999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7758999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7758999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7759999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7759999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7759999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7759999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7760999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7760999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7760999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7760999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7760999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7761999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7761999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7761999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7761999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7761999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7762999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7762999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7762999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7762999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7762999999999916, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7763999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7763999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7763999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7763999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7763999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7764999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7764999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7764999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7764999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7764999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7765999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7765999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7765999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7765999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7765999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7766999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7766999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7766999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7766999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7766999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7767999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7767999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7767999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7767999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7767999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7768999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7768999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7768999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7768999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7768999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7769999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7769999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7769999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7769999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7769999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7770999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7770999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7770999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7770999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7770999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7771999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7771999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7771999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7771999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7771999999999915, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7772999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7772999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7772999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7772999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7772999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7773999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7773999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7773999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7773999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7773999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7774999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7774999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7774999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7774999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7774999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7775999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7775999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7775999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7775999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7775999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7776999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7776999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7776999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7776999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7776999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7777999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7777999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7777999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7777999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7777999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7778999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7778999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7778999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7778999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7778999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7779999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7779999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7779999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7779999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7780999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7780999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7780999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7780999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7780999999999914, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7781999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7781999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7781999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7781999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7781999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7782999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7782999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7782999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7782999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7782999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7783999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7783999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7783999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7783999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7783999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7784999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7784999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7784999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7784999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7784999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7785999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7785999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7785999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7785999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7785999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7786999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7786999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7786999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7786999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7786999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7787999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7787999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7787999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7787999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7787999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7788999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7788999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7788999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7788999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7788999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7789999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7789999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7789999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7789999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7789999999999913, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7790999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7790999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7790999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7790999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7790999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7791999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7791999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7791999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7791999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7791999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7792999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7792999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7792999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7792999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7792999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7793999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7793999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7793999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7793999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7793999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7794999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7794999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7794999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7794999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7795999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7795999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7795999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7795999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7795999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7796999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7796999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7796999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7796999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7796999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7797999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7797999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7797999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7797999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7797999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7798999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7798999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7798999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7798999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7798999999999912, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7799999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7799999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7799999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7799999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7799999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7800999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7800999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7800999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7800999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7800999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7801999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7801999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7801999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7801999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7801999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7802999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7802999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7802999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7802999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7802999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7803999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7803999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7803999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7803999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7803999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7804999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7804999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7804999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7804999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7804999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7805999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7805999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7805999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7805999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7805999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7806999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7806999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7806999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7806999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7806999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7807999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7807999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7807999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7807999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7807999999999911, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.780899999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.780899999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.780899999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.780899999999991, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.780899999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.780999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.780999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.780999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.780999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.780999999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781099999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781099999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781099999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781099999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781099999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781199999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781199999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781199999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781199999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781199999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781299999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781299999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781299999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781299999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781299999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781399999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781399999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781399999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781399999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781399999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781499999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781499999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781499999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781499999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781499999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781599999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781599999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781599999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781599999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781599999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781699999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781699999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781699999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781699999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781699999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781799999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781799999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781799999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781799999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.781799999999991, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7818999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7818999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7818999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7818999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7818999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7819999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7819999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7819999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7819999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7819999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7820999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7820999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7820999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7820999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7820999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7821999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7821999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7821999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7821999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7821999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7822999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7822999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7822999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7822999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7822999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7823999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7823999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7823999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7823999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7823999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7824999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7824999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7824999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7824999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7824999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7825999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7825999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7825999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7825999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7825999999999909, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7826999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7826999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7826999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7826999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7826999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7827999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7827999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7827999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7827999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7828999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7828999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7828999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7828999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7828999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7829999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7829999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7829999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7829999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7829999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7830999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7830999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7830999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7830999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7830999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7831999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7831999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7831999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7831999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7831999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7832999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7832999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7832999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7832999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7832999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7833999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7833999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7833999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7833999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7833999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7834999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7834999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7834999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7834999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7834999999999908, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7835999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7835999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7835999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7835999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7835999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7836999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7836999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7836999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7836999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7836999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7837999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7837999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7837999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7837999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7837999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7838999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7838999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7838999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7838999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7838999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7839999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7839999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7839999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7839999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7839999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7840999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7840999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7840999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7840999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7840999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7841999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7841999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7841999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7841999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7841999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7842999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7842999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7842999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7842999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7842999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7843999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7843999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7843999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7843999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7843999999999907, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7844999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7844999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7844999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7844999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7844999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7845999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7845999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7845999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7845999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7845999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7846999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7846999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7846999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7846999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7846999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7847999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7847999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7847999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7847999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7847999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7848999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7848999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7848999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7848999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7848999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7849999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7849999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7849999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7849999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7849999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7850999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7850999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7850999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7850999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7850999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7851999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7851999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7851999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7851999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7852999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7852999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7852999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7852999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7852999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7853999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7853999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7853999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7853999999999906, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7854999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7854999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7854999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7854999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7854999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7855999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7855999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7855999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7855999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7856999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7856999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7856999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7856999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7856999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7857999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7857999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7857999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7857999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7857999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7858999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7858999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7858999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7858999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7858999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7859999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7859999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7859999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7859999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7859999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7860999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7860999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7860999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7860999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7860999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7861999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7861999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7861999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7861999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7861999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7862999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7862999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7862999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7862999999999905, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7863999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7863999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7863999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7863999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7863999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7864999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7864999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7864999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7864999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7864999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7865999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7865999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7865999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7865999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7865999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7866999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7866999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7866999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7866999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7866999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7867999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7867999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7867999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7867999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7867999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7868999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7868999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7868999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7868999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7868999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7869999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7869999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7869999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7869999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7869999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7870999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7870999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7870999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7870999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7870999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7871999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7871999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7871999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7871999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7871999999999904, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7872999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7872999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7872999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7872999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7872999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7873999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7873999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7873999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7873999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7873999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7874999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7874999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7874999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7874999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7874999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7875999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7875999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7875999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7875999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7876999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7876999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7876999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7876999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7876999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7877999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7877999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7877999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7877999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7877999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7878999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7878999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7878999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7878999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7878999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7879999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7879999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7879999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7879999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7879999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7880999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7880999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7880999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7880999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7880999999999903, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7881999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7881999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7881999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7881999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7881999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7882999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7882999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7882999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7882999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7882999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7883999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7883999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7883999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7883999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7883999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7884999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7884999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7884999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7884999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7884999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7885999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7885999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7885999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7885999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7885999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7886999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7886999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7886999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7886999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7886999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7887999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7887999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7887999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7887999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7887999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7888999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7888999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7888999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7888999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7888999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7889999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7889999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7889999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7889999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7889999999999902, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7890999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7890999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7890999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7890999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7890999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7891999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7891999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7891999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7891999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7891999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7892999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7892999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7892999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7892999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7892999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7893999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7893999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7893999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7893999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7893999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7894999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7894999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7894999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7894999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7894999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7895999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7895999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7895999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7895999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7895999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7896999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7896999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7896999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7896999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7896999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7897999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7897999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7897999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7897999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7897999999999901, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.78989999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.78989999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.78989999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.78989999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.78989999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.78999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.78999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.78999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.78999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.78999999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79009999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79009999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79009999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79009999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79009999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79019999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79019999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79019999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79019999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79019999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79029999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79029999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79029999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79029999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79029999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79039999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79039999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79039999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79039999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79039999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79049999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79049999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79049999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79049999999999, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.79049999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79059999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79059999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79059999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79059999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79059999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79069999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79069999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79069999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79069999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79069999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79079999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79079999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79079999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79079999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.79079999999999, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7908999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7908999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7908999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7908999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7908999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7909999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7909999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7909999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7909999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7909999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7910999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7910999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7910999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7910999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7910999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7911999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7911999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7911999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7911999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7911999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7912999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7912999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7912999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7912999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7912999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7913999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7913999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7913999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7913999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7914999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7914999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7914999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7914999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7914999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7915999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7915999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7915999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7915999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7915999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7916999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7916999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7916999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7916999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7916999999999899, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7917999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7917999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7917999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7917999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7917999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7918999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7918999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7918999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7918999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7918999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7919999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7919999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7919999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7919999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7919999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7920999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7920999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7920999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7920999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7920999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7921999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7921999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7921999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7921999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7921999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7922999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7922999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7922999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7922999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7922999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7923999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7923999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7923999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7923999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7924999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7924999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7924999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7924999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7924999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7925999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7925999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7925999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7925999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7925999999999898, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7926999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7926999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7926999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7926999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7926999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7927999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7927999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7927999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7927999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7927999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7928999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7928999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7928999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7928999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7928999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7929999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7929999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7929999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7929999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7929999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7930999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7930999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7930999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7930999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7930999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7931999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7931999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7931999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7931999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7931999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7932999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7932999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7932999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7932999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7932999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7933999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7933999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7933999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7933999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7933999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7934999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7934999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7934999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7934999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7934999999999897, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7935999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7935999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7935999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7935999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7935999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7936999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7936999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7936999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7936999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7936999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7937999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7937999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7937999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7937999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7937999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7938999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7938999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7938999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7938999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7938999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7939999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7939999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7939999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7939999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7939999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7940999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7940999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7940999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7940999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7940999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7941999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7941999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7941999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7941999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7941999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7942999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7942999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7942999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7942999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7942999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7943999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7943999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7943999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7943999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7943999999999896, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7944999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7944999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7944999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7944999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7944999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7945999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7945999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7945999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7945999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7945999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7946999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7946999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7946999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7946999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7946999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7947999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7947999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7947999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7947999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7947999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7948999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7948999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7948999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7948999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7948999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7949999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7949999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7949999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7949999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7950999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7950999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7950999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7950999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7950999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7951999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7951999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7951999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7951999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7951999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7952999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7952999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7952999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7952999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7952999999999895, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7953999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7953999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7953999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7953999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7953999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7954999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7954999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7954999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7954999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7954999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7955999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7955999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7955999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7955999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7955999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7956999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7956999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7956999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7956999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7956999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7957999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7957999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7957999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7957999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7957999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7958999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7958999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7958999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7958999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7958999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7959999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7959999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7959999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7959999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7959999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7960999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7960999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7960999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7960999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7960999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7961999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7961999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7961999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7961999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7961999999999894, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7962999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7962999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7962999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7962999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7962999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7963999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7963999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7963999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7963999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7963999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7964999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7964999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7964999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7964999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7964999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7965999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7965999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7965999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7965999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7965999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7966999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7966999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7966999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7966999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7966999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7967999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7967999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7967999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7967999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7967999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7968999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7968999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7968999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7968999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7968999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7969999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7969999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7969999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7969999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7969999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7970999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7970999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7970999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7970999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7970999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7971999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7971999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7971999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7971999999999893, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7972999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7972999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7972999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7972999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7972999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7973999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7973999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7973999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7973999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7973999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7974999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7974999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7974999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7974999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7974999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7975999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7975999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7975999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7975999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7975999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7976999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7976999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7976999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7976999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7976999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7977999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7977999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7977999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7977999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7977999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7978999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7978999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7978999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7978999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7978999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7979999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7979999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7979999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7979999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7979999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7980999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7980999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7980999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7980999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7980999999999892, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7981999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7981999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7981999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.7981999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7981999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7982999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7982999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7982999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7982999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7982999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7983999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7983999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7983999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7983999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7983999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7984999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7984999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7984999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7984999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7984999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7985999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7985999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7985999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7985999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7985999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7986999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7986999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7986999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7986999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7986999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7987999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7987999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7987999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7987999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7987999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7988999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7988999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7988999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7988999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7988999999999891, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.798999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.798999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.798999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.798999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.798999999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799099999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799099999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799099999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799099999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799099999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799199999999989, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.799199999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799199999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799199999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799199999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799299999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799299999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799299999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799299999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799299999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799399999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799399999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799399999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799399999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799399999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799499999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799499999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799499999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799499999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799499999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799599999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799599999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799599999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799599999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799599999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799699999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799699999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799699999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799699999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799699999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799799999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799799999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799799999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799799999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799899999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799899999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799899999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799899999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.799899999999989, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7999999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7999999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7999999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7999999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.7999999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8000999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8000999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8000999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8000999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8000999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8001999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8001999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8001999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8001999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8001999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8002999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8002999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8002999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8002999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8002999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8003999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8003999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8003999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8003999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8003999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8004999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8004999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8004999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8004999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8004999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8005999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8005999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8005999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8005999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8005999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8006999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8006999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8006999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8006999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8006999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8007999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8007999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8007999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8007999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8007999999999889, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8008999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8008999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8008999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8008999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8008999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8009999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8009999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8009999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8009999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8009999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8010999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8010999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8010999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8010999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8010999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8011999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8011999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8011999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8011999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8011999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8012999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8012999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8012999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8012999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8012999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8013999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8013999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8013999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8013999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8013999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8014999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8014999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8014999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8014999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8014999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8015999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8015999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8015999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8015999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8015999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8016999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8016999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8016999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8016999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8016999999999888, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8017999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8017999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8017999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8017999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8017999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8018999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8018999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8018999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8018999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8018999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8019999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8019999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8019999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8019999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8020999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8020999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8020999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8020999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8020999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8021999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8021999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8021999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8021999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8021999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8022999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8022999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8022999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8022999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8022999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8023999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8023999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8023999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8023999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8023999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8024999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8024999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8024999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8024999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8024999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8025999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8025999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8025999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8025999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8025999999999887, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8026999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8026999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8026999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8026999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8026999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8027999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8027999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8027999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8027999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8027999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8028999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8028999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8028999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8028999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8028999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8029999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8029999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8029999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8029999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8029999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8030999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8030999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8030999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8030999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8030999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8031999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8031999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8031999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8031999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8031999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8032999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8032999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8032999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8032999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8032999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8033999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8033999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8033999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8033999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8033999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8034999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8034999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8034999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8034999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8034999999999886, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8035999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8035999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8035999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8035999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8035999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8036999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8036999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8036999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8036999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8036999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8037999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8037999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8037999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8037999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8037999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8038999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8038999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8038999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8038999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8038999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8039999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8039999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8039999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8039999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8039999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8040999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8040999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8040999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8040999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8040999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8041999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8041999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8041999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8041999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8041999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8042999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8042999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8042999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8042999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8042999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8043999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8043999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8043999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8043999999999885, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8044999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8044999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8044999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8044999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8044999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8045999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8045999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8045999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8045999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8045999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8046999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8046999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8046999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8046999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8046999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8047999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8047999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8047999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8047999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8047999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8048999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8048999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8048999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8048999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8048999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8049999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8049999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8049999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8049999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8049999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8050999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8050999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8050999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8050999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8050999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8051999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8051999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8051999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8051999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8051999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8052999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8052999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8052999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8052999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8052999999999884, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8053999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8053999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8053999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8053999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8053999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8054999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8054999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8054999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8054999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8054999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8055999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8055999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8055999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8055999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8055999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8056999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8056999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8056999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8056999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8056999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8057999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8057999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8057999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8057999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8057999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8058999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8058999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8058999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8058999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8058999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8059999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8059999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8059999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8059999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8059999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8060999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8060999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8060999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8060999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8060999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8061999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8061999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8061999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8061999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8061999999999883, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8062999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8062999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8062999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8062999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8062999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8063999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8063999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8063999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8063999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8063999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8064999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8064999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8064999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8064999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8064999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8065999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8065999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8065999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8065999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8065999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8066999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8066999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8066999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8066999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8066999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8066999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8067999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8067999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8067999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8067999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8068999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8068999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8068999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8068999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8068999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8069999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8069999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8069999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8069999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8069999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8070999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8070999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8070999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8070999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8070999999999882, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8071999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8071999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8071999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8071999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8071999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8072999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8072999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8072999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8072999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8072999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8073999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8073999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8073999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8073999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8073999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8074999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8074999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8074999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8074999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8074999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8075999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8075999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8075999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8075999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8075999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8076999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8076999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8076999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8076999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8076999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8077999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8077999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8077999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8077999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8077999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8078999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8078999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8078999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8078999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8078999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8079999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8079999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8079999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8079999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8079999999999881, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808099999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808099999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808099999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808099999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808099999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808199999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808199999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808199999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808199999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808199999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808299999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808299999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808299999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808299999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808299999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808399999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808399999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808399999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808399999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808399999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808499999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808499999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808499999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808499999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808499999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808599999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808599999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808599999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808599999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808599999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808699999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808699999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808699999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808699999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808699999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.808799999999988, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8087999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8087999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8087999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8087999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8088999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8088999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8088999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8088999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8088999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8089999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8089999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8089999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8089999999999988, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8090999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8090999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8090999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8090999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8090999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8091999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8091999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8091999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8091999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8091999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8092999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8092999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8092999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8092999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8092999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8093999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8093999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8093999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8093999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8093999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8094999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8094999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8094999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8094999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8094999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8095999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8095999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8095999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8095999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8095999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8096999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8096999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8096999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8096999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8096999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8097999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8097999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8097999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8097999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8097999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8098999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8098999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8098999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8098999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8098999999999879, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8099999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8099999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8099999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8099999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8099999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8100999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8100999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8100999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8100999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8100999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8101999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8101999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8101999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8101999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8101999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8102999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8102999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8102999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8102999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8102999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8103999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8103999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8103999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8103999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8103999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8104999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8104999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8104999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8104999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8104999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8105999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8105999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8105999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8105999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8105999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8106999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8106999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8106999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8106999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8106999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8107999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8107999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8107999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8107999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8107999999999878, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8108999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8108999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8108999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8108999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8108999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8109999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8109999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8109999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8109999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8109999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8110999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8110999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8110999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8110999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8110999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8111999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8111999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8111999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8111999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8111999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8112999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8112999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8112999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8112999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8112999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8113999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8113999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8113999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8113999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8113999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8114999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8114999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8114999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8114999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8114999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8115999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8115999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8115999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8115999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8116999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8116999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8116999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8116999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8116999999999877, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8117999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8117999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8117999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8117999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8117999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8118999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8118999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8118999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8118999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8118999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8119999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8119999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8119999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8119999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8119999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8120999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8120999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8120999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8120999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8120999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8121999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8121999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8121999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8121999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8121999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8122999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8122999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8122999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8122999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8122999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8123999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8123999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8123999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8123999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8123999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8124999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8124999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8124999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8124999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8124999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8125999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8125999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8125999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8125999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8125999999999876, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8126999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8126999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8126999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8126999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8126999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8127999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8127999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8127999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8127999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8127999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8128999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8128999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8128999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8128999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8128999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8129999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8129999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8129999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8129999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8129999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8130999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8130999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8130999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8130999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8130999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8131999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8131999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8131999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8131999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8131999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8132999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8132999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8132999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8132999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8132999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8133999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8133999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8133999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8133999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8133999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8134999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8134999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8134999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8134999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8134999999999875, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8135999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8135999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8135999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8135999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8135999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8136999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8136999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8136999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8136999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8136999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8137999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8137999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8137999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8137999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8137999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8138999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8138999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8138999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8138999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8138999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8139999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8139999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8139999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8139999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8139999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8140999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8140999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8140999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8140999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8140999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8141999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8141999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8141999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8141999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8141999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8142999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8142999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8142999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8142999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8142999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8143999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8143999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8143999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8143999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8143999999999874, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8144999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8144999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8144999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8144999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8144999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8145999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8145999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8145999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8145999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8145999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8146999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8146999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8146999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8146999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8146999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8147999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8147999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8147999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8147999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8147999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8148999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8148999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8148999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8148999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8148999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8149999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8149999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8149999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8149999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8149999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8150999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8150999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8150999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8150999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8150999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8151999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8151999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8151999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8151999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8151999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8152999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8152999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8152999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8152999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8152999999999873, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8153999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8153999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8153999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8153999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8153999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8154999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8154999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8154999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8154999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8154999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8155999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8155999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8155999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8155999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8155999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8156999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8156999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8156999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8156999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8156999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8157999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8157999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8157999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8157999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8157999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8158999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8158999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8158999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8158999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8158999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8159999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8159999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8159999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8159999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8159999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8160999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8160999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8160999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8160999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8160999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8161999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8161999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8161999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8161999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8161999999999872, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8162999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8162999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8162999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8162999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8162999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8163999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8163999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8163999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8163999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8164999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8164999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8164999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8164999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8164999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8165999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8165999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8165999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8165999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8165999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8166999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8166999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8166999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8166999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8166999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8167999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8167999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8167999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8167999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8167999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8168999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8168999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8168999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8168999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8168999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8169999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8169999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8169999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8169999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8169999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8170999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8170999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8170999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8170999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8170999999999871, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817199999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817199999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817199999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817199999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817199999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817299999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817299999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817299999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817299999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817299999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817399999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817399999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817399999999987, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.817399999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817399999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817499999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817499999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817499999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817499999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817499999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817599999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817599999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817599999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817599999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817599999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817699999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817699999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817699999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817699999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817699999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817799999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817799999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817799999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817799999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817799999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817899999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817899999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817899999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817899999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817899999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.817999999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.818099999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.818099999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.818099999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.818099999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.818099999999987, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8181999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8181999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8181999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8181999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8181999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8182999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8182999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8182999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8182999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8182999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8183999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8183999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8183999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8183999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8183999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8184999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8184999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8184999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8184999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8184999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8185999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8185999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8185999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8185999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8185999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8186999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8186999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8186999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8186999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8186999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8187999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8187999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8187999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8187999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8187999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8188999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8188999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8188999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8188999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8188999999999869, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8189999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8189999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8189999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8189999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8189999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8190999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8190999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8190999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8190999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8190999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8191999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8191999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8191999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8191999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8191999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8192999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8192999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8192999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8192999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8192999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8193999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8193999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8193999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8193999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8193999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8194999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8194999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8194999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8194999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8194999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8195999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8195999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8195999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8195999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8195999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8196999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8196999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8196999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8196999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8196999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8197999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8197999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8197999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8197999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8197999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8198999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8198999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8198999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8198999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8198999999999868, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8199999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8199999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8199999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8199999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8199999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8200999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8200999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8200999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8200999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8200999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8201999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8201999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8201999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8201999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8201999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8202999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8202999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8202999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8202999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8202999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8203999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8203999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8203999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8203999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8203999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8204999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8204999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8204999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8204999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8204999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8205999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8205999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8205999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8205999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8205999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8206999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8206999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8206999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8206999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8206999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8207999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8207999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8207999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8207999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8207999999999867, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8208999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8208999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8208999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8208999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8208999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8209999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8209999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8209999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8209999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8209999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8210999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8210999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8210999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8210999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8210999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8210999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8211999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8211999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8211999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8211999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8212999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8212999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8212999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8212999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8212999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8213999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8213999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8213999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8213999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8213999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8214999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8214999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8214999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8214999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8214999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8215999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8215999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8215999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8215999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8215999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8216999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8216999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8216999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8216999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8216999999999866, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8217999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8217999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8217999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8217999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8217999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8218999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8218999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8218999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8218999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8218999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8219999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8219999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8219999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8219999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8219999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8220999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8220999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8220999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8220999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8220999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8221999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8221999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8221999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8221999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8221999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8222999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8222999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8222999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8222999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8222999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8223999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8223999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8223999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8223999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8223999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8224999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8224999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8224999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8224999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8224999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8225999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8225999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8225999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8225999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8225999999999865, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8226999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8226999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8226999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8226999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8226999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8227999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8227999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8227999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8227999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8227999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8228999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8228999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8228999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8228999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8228999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8229999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8229999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8229999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8229999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8229999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8230999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8230999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8230999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8230999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8230999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8231999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8231999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8231999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8231999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8231999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8232999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8232999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8232999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8232999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8232999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8233999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8233999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8233999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8233999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8233999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8234999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8234999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8234999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8234999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8234999999999864, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8235999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8235999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8235999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8235999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8235999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8236999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8236999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8236999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8236999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8236999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8237999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8237999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8237999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8237999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8238999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8238999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8238999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8238999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8238999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8239999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8239999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8239999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8239999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8239999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8240999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8240999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8240999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8240999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8240999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8241999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8241999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8241999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8241999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8241999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8242999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8242999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8242999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8242999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8242999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8243999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8243999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8243999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8243999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8243999999999863, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8244999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8244999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8244999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8244999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8244999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8245999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8245999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8245999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8245999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8245999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8246999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8246999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8246999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8246999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8246999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8247999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8247999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8247999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8247999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8247999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8248999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8248999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8248999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8248999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8248999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8249999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8249999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8249999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8249999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8249999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8250999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8250999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8250999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8250999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8250999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8251999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8251999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8251999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8251999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8251999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8252999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8252999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8252999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8252999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8252999999999862, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8253999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8253999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8253999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8253999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8253999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8254999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8254999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8254999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8254999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8254999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8255999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8255999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8255999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8255999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8255999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8256999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8256999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8256999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8256999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8256999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8257999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8257999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8257999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8257999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8257999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8258999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8258999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8258999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8258999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8258999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8259999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8259999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8259999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8259999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8260999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8260999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8260999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8260999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8260999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8261999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8261999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8261999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8261999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8261999999999861, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826299999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826299999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826299999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826299999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826299999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826399999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826399999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826399999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826399999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826399999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826499999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826499999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826499999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826499999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826499999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826599999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826599999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826599999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826599999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826599999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826699999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826699999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826699999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826699999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826699999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826799999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826799999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826799999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826799999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826799999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826899999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826899999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826899999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826899999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826899999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.826999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.826999999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.827099999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.827099999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.827099999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.827099999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.827099999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.827199999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.827199999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.827199999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.827199999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.827199999999986, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8272999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8272999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8272999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8272999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8272999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8273999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8273999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8273999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8273999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8273999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8274999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8274999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8274999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8274999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8274999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8275999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8275999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8275999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8275999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8275999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8276999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8276999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8276999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8276999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8276999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8277999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8277999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8277999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8277999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8277999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8278999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8278999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8278999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8278999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8278999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8279999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8279999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8279999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8279999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8279999999999859, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8280999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8280999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8280999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8280999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8280999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8281999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8281999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8281999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8281999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8281999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8282999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8282999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8282999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8282999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8282999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8283999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8283999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8283999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8283999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8283999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8284999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8284999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8284999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8284999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8284999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8285999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8285999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8285999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8285999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8285999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8286999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8286999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8286999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8286999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8286999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8287999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8287999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8287999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8287999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8287999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8288999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8288999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8288999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8288999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8288999999999858, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8289999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8289999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8289999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8289999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8289999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8290999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8290999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8290999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8290999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8290999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8291999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8291999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8291999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8291999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8291999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8292999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8292999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8292999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8292999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8292999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8293999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8293999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8293999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8293999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8293999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8294999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8294999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8294999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8294999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8295999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8295999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8295999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8295999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8295999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8296999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8296999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8296999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8296999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8296999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8297999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8297999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8297999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8297999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8297999999999857, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8298999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8298999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8298999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8298999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8298999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8299999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8299999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8299999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8299999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8299999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8300999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8300999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8300999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8300999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8300999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8301999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8301999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8301999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8301999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8301999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8302999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8302999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8302999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8302999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8302999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8303999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8303999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8303999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8303999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8303999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8304999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8304999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8304999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8304999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8304999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8305999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8305999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8305999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8305999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8305999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8306999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8306999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8306999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8306999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8306999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8307999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8307999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8307999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8307999999999856, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8308999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8308999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8308999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8308999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8308999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8309999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8309999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8309999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8309999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8309999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8310999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8310999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8310999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8310999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8310999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8311999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8311999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8311999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8311999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8311999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8312999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8312999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8312999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8312999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8312999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8313999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8313999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8313999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8313999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8313999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8314999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8314999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8314999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8314999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8314999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8315999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8315999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8315999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8315999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8315999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8316999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8316999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8316999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8316999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8316999999999855, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8317999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8317999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8317999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8317999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8317999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8318999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8318999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8318999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8318999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8318999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8319999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8319999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8319999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8319999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8319999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8320999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8320999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8320999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8320999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8320999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8321999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8321999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8321999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8321999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8321999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8322999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8322999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8322999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8322999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8322999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8323999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8323999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8323999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8323999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8323999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8324999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8324999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8324999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8324999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8324999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8325999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8325999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8325999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8325999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8325999999999854, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8326999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8326999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8326999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8326999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8326999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8327999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8327999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8327999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8327999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8327999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8328999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8328999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8328999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8328999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8328999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8329999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8329999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8329999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8329999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8329999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8330999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8330999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8330999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8330999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8330999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8331999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8331999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8331999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8331999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8331999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8332999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8332999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8332999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8332999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8332999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8333999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8333999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8333999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8333999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8334999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8334999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8334999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8334999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8334999999999853, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8335999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8335999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8335999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8335999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8335999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8336999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8336999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8336999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8336999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8336999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8337999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8337999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8337999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8337999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8337999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8338999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8338999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8338999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8338999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8338999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8339999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8339999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8339999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8339999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8339999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8340999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8340999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8340999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8340999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8340999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8341999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8341999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8341999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8341999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8341999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8342999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8342999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8342999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8342999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8342999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8343999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8343999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8343999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8343999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8343999999999852, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8344999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8344999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8344999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8344999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8344999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8345999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8345999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8345999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8345999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8345999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8346999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8346999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8346999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8346999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8346999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8347999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8347999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8347999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8347999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8347999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8348999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8348999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8348999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8348999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8348999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8349999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8349999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8349999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8349999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8349999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8350999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8350999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8350999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8350999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8350999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8351999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8351999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8351999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8351999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8351999999999851, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835299999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835299999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835299999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835299999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835299999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835399999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835399999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835399999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835399999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835399999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835499999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835499999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835499999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835499999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835499999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835599999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835599999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835599999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835599999999985, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.835699999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835699999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835699999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835699999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835699999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835799999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835799999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835799999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835799999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835799999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835899999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835899999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835899999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835899999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835899999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.835999999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.836099999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.836099999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.836099999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.836099999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.836099999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.836199999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.836199999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.836199999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.836199999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.836199999999985, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8362999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8362999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8362999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8362999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8362999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8363999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8363999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8363999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8363999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8363999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8364999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8364999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8364999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8364999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8364999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8365999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8365999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8365999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8365999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8365999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8366999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8366999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8366999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8366999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8366999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8367999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8367999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8367999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8367999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8367999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8368999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8368999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8368999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8368999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8368999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8369999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8369999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8369999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8369999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8369999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8370999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8370999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8370999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8370999999999849, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8371999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8371999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8371999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8371999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8371999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8372999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8372999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8372999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8372999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8372999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8373999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8373999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8373999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8373999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8373999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8374999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8374999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8374999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8374999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8374999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8375999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8375999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8375999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8375999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8375999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8376999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8376999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8376999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8376999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8376999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8377999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8377999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8377999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8377999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8377999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8378999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8378999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8378999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8378999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8378999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8379999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8379999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8379999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8379999999999848, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8380999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8380999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8380999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8380999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8380999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8381999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8381999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8381999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8381999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8382999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8382999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8382999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8382999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8382999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8383999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8383999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8383999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8383999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8383999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8384999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8384999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8384999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8384999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8384999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8385999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8385999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8385999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8385999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8385999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8386999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8386999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8386999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8386999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8386999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8387999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8387999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8387999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8387999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8387999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8388999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8388999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8388999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8388999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8388999999999847, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8389999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8389999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8389999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8389999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8389999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8390999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8390999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8390999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8390999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8390999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8391999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8391999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8391999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8391999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8391999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8392999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8392999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8392999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8392999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8392999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8393999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8393999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8393999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8393999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8393999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8394999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8394999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8394999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8394999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8394999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8395999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8395999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8395999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8395999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8395999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8396999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8396999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8396999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8396999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8396999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8397999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8397999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8397999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8397999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8397999999999846, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8398999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8398999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8398999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8398999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8398999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8399999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8399999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8399999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8399999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8399999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8400999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8400999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8400999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8400999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8400999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8401999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8401999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8401999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8401999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8401999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8402999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8402999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8402999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8402999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8402999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8403999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8403999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8403999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8403999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8404999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8404999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8404999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8404999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8404999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8405999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8405999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8405999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8405999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8405999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8406999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8406999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8406999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8406999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8406999999999845, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8407999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8407999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8407999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8407999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8407999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8408999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8408999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8408999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8408999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8408999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8409999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8409999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8409999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8409999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8409999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8410999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8410999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8410999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8410999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8410999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8411999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8411999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8411999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8411999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8411999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8412999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8412999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8412999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8412999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8412999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8413999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8413999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8413999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8413999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8413999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8414999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8414999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8414999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8414999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8414999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8415999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8415999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8415999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8415999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8415999999999844, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8416999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8416999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8416999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8416999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8416999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8417999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8417999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8417999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8417999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8417999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8418999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8418999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8418999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8418999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8418999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8419999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8419999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8419999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8419999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8419999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8419999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8420999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8420999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8420999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8420999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8420999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8421999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8421999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8421999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8421999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8421999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8422999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8422999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8422999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8422999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8422999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8423999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8423999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8423999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8423999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8423999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8424999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8424999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8424999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8424999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8424999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8425999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8425999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8425999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8425999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8425999999999843, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8426999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8426999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8426999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8426999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8426999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8427999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8427999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8427999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8427999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8427999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8428999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8428999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8428999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8428999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8428999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8429999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8429999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8429999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8429999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8430999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8430999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8430999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8430999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8430999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8431999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8431999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8431999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8431999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8431999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8432999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8432999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8432999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8432999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8432999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8433999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8433999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8433999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8433999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8433999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8434999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8434999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8434999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8434999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8434999999999842, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8435999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8435999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8435999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8435999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8435999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8436999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8436999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8436999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8436999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8436999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8437999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8437999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8437999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8437999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8437999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8438999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8438999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8438999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8438999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8438999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8439999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8439999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8439999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8439999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8439999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8440999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8440999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8440999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8440999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8440999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8441999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8441999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8441999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8441999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8441999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8442999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8442999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8442999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8442999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8442999999999841, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8443999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8443999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8443999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8443999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8443999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8444999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8444999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8444999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8444999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8444999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8445999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8445999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8445999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8445999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8445999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8446999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8446999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8446999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8446999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8446999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8447999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8447999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8447999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8447999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8447999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8448999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8448999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8448999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8448999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8448999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8449999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8449999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8449999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8449999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8449999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8450999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8450999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8450999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8450999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8450999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8451999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8451999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8451999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8451999999999984, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.845299999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.845299999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.845299999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.845299999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.845299999999984, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8453999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8453999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8453999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8453999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8453999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8454999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8454999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8454999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8454999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8454999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8455999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8455999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8455999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8455999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8455999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8456999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8456999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8456999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8456999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8456999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8457999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8457999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8457999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8457999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8457999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8458999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8458999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8458999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8458999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8458999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8459999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8459999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8459999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8459999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8459999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8460999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8460999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8460999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8460999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8460999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8461999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8461999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8461999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8461999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8461999999999839, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8462999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8462999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8462999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8462999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8462999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8463999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8463999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8463999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8463999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8463999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8464999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8464999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8464999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8464999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8464999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8465999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8465999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8465999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8465999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8465999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8466999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8466999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8466999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8466999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8466999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8467999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8467999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8467999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8467999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8467999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8468999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8468999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8468999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8468999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8468999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8469999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8469999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8469999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8469999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8469999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8470999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8470999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8470999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8470999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8470999999999838, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8471999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8471999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8471999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8471999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8471999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8472999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8472999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8472999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8472999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8472999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8473999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8473999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8473999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8473999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8473999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8474999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8474999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8474999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8474999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8474999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8475999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8475999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8475999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8475999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8475999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8476999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8476999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8476999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8476999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8476999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8477999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8477999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8477999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8477999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8477999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8478999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8478999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8478999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8478999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8478999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8479999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8479999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8479999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8479999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8479999999999837, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8480999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8480999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8480999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8480999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8480999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8481999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8481999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8481999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8481999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8481999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8482999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8482999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8482999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8482999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8482999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8483999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8483999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8483999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8483999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8483999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8484999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8484999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8484999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8484999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8484999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8485999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8485999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8485999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8485999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8485999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8486999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8486999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8486999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8486999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8486999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8487999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8487999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8487999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8487999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8487999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8488999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8488999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8488999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8488999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8488999999999836, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8489999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8489999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8489999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8489999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8489999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8490999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8490999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8490999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8490999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8490999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8491999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8491999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8491999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8491999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8491999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8492999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8492999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8492999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8492999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8492999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8493999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8493999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8493999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8493999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8493999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8494999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8494999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8494999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8494999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8494999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8495999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8495999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8495999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8495999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8495999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8496999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8496999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8496999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8496999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8496999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8497999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8497999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8497999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8497999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8497999999999835, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8498999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8498999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8498999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8498999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8498999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8499999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8499999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8499999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8499999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8500999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8500999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8500999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8500999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8500999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8501999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8501999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8501999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8501999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8501999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8502999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8502999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8502999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8502999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8502999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8503999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8503999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8503999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8503999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8503999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8504999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8504999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8504999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8504999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8504999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8505999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8505999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8505999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8505999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8505999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8506999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8506999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8506999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8506999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8506999999999834, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8507999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8507999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8507999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8507999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8507999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8508999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8508999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8508999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8508999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8508999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8509999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8509999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8509999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8509999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8509999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8510999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8510999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8510999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8510999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8510999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8511999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8511999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8511999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8511999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8511999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8512999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8512999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8512999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8512999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8512999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8513999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8513999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8513999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8513999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8513999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8514999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8514999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8514999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8514999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8514999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8515999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8515999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8515999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8515999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8515999999999833, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8516999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8516999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8516999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8516999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8516999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8517999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8517999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8517999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8517999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8517999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8518999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8518999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8518999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8518999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8518999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8519999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8519999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8519999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8519999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8519999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8520999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8520999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8520999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8520999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8520999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8521999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8521999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8521999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8521999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8521999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8522999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8522999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8522999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8522999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8522999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8523999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8523999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8523999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8523999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8523999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8524999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8524999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8524999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8524999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8524999999999832, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8525999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8525999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8525999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8525999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8525999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8526999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8526999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8526999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8526999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8526999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8527999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8527999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8527999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8527999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8527999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8528999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8528999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8528999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8528999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8528999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8529999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8529999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8529999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8529999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8529999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8530999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8530999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8530999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8530999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8530999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8531999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8531999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8531999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8531999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8531999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8532999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8532999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8532999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8532999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8532999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8533999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8533999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8533999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8533999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8533999999999831, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853499999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853499999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853499999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853499999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853499999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853599999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853599999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853599999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853599999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853599999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853699999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853699999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853699999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853699999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853699999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853799999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853799999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853799999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853799999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853799999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853899999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853899999999983, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.853899999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853899999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853899999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.853999999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.854099999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.854099999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.854099999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.854099999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.854099999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.854199999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.854199999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.854199999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.854199999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.854199999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.854299999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.854299999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.854299999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.854299999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.854299999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.854399999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.854399999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.854399999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.854399999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.854399999999983, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8544999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8544999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8544999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8544999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8544999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8545999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8545999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8545999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8545999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8545999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8546999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8546999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8546999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8546999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8546999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8546999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8547999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8547999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8547999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8547999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8548999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8548999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8548999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8548999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8548999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8549999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8549999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8549999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8549999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8549999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8550999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8550999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8550999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8550999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8550999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8551999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8551999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8551999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8551999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8551999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8552999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8552999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8552999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8552999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8552999999999829, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8553999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8553999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8553999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8553999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8553999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8554999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8554999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8554999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8554999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8554999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8555999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8555999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8555999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8555999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8555999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8556999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8556999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8556999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8556999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8556999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8557999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8557999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8557999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8557999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8557999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8558999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8558999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8558999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8558999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8558999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8559999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8559999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8559999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8559999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8559999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8560999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8560999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8560999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8560999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8560999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8561999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8561999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8561999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8561999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8561999999999828, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8562999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8562999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8562999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8562999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8562999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8563999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8563999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8563999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8563999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8563999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8564999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8564999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8564999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8564999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8564999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8565999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8565999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8565999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8565999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8565999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8566999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8566999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8566999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8566999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8566999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8567999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8567999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8567999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8567999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8567999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8568999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8568999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8568999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8568999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8568999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8569999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8569999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8569999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8569999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8569999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8570999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8570999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8570999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8570999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8570999999999827, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8571999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8571999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8571999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8571999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8571999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8572999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8572999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8572999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8572999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8572999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8573999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8573999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8573999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8573999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8573999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8574999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8574999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8574999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8574999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8574999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8575999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8575999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8575999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8575999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8575999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8576999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8576999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8576999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8576999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8576999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8577999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8577999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8577999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8577999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8577999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8578999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8578999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8578999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8578999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8578999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8579999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8579999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8579999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8579999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8579999999999826, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8580999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8580999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8580999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8580999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8580999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8581999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8581999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8581999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8581999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8581999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8582999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8582999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8582999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8582999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8582999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8583999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8583999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8583999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8583999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8583999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8584999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8584999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8584999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8584999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8584999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8585999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8585999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8585999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8585999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8585999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8586999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8586999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8586999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8586999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8586999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8587999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8587999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8587999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8587999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8587999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8588999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8588999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8588999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8588999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8588999999999825, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8589999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8589999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8589999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8589999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8589999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8590999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8590999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8590999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8590999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8590999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8591999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8591999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8591999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8591999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8591999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8592999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8592999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8592999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8592999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8592999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8593999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8593999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8593999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8593999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8593999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8594999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8594999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8594999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8594999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8594999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8595999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8595999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8595999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8595999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8596999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8596999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8596999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8596999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8596999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8597999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8597999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8597999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8597999999999824, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8598999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8598999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8598999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8598999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8598999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8599999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8599999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8599999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8599999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8600999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8600999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8600999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8600999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8600999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8601999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8601999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8601999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8601999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8601999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8602999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8602999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8602999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8602999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8602999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8603999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8603999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8603999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8603999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8603999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8604999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8604999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8604999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8604999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8604999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8605999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8605999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8605999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8605999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8605999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8606999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8606999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8606999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8606999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8606999999999823, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8607999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8607999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8607999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8607999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8607999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8608999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8608999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8608999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8608999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8608999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8609999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8609999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8609999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8609999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8609999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8610999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8610999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8610999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8610999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8611999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8611999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8611999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8611999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8611999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8612999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8612999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8612999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8612999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8612999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8613999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8613999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8613999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8613999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8613999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8614999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8614999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8614999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8614999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8614999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8615999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8615999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8615999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8615999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8615999999999822, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8616999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8616999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8616999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8616999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8616999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8617999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8617999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8617999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8617999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8617999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8618999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8618999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8618999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8618999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8618999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8619999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8619999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8619999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8619999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8619999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8620999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8620999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8620999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8620999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8620999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8621999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8621999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8621999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8621999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8621999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8622999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8622999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8622999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8622999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8622999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8623999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8623999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8623999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8623999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8623999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8624999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8624999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8624999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8624999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8624999999999821, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.862599999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.862599999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.862599999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.862599999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.862599999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.862699999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.862699999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.862699999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.862699999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.862699999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.862799999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.862799999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.862799999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.862799999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.862799999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.862899999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.862899999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.862899999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.862899999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.862899999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.862999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.862999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.862999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.862999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.862999999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.863099999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.863099999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.863099999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.863099999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.863099999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.863199999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.863199999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.863199999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.863199999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.863199999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.863299999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.863299999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.863299999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.863299999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.863299999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.863399999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.863399999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.863399999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.863399999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.863399999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.863499999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.863499999999982, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.863499999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.863499999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.863499999999982, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8635999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8635999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8635999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8635999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8635999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8636999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8636999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8636999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8636999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8636999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8637999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8637999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8637999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8637999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8637999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8638999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8638999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8638999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8638999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8638999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8639999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8639999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8639999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8639999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8639999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8640999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8640999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8640999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8640999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8640999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8641999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8641999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8641999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8641999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8641999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8642999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8642999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8642999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8642999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8642999999999819, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8643999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8643999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8643999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8643999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8644999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8644999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8644999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8644999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8644999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8645999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8645999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8645999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8645999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8645999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8646999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8646999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8646999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8646999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8646999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8647999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8647999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8647999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8647999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8647999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8648999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8648999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8648999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8648999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8648999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8649999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8649999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8649999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8649999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8649999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8650999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8650999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8650999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8650999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8650999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8651999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8651999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8651999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8651999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8651999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8652999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8652999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8652999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8652999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8652999999999818, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8653999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8653999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8653999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8653999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8653999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8654999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8654999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8654999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8654999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8654999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8655999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8655999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8655999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8655999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8655999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8656999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8656999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8656999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8656999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8656999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8657999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8657999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8657999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8657999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8657999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8658999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8658999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8658999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8658999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8658999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8659999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8659999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8659999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8659999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8659999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8660999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8660999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8660999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8660999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8660999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8661999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8661999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8661999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8661999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8661999999999817, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8662999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8662999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8662999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8662999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8662999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8663999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8663999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8663999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8663999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8663999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8664999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8664999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8664999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8664999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8664999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8665999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8665999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8665999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8665999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8665999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8666999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8666999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8666999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8666999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8666999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8667999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8667999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8667999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8667999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8667999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8668999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8668999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8668999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8668999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8668999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8669999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8669999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8669999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8669999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8669999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8670999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8670999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8670999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8670999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8670999999999816, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8671999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8671999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8671999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8671999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8671999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8672999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8672999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8672999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8672999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8672999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8673999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8673999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8673999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8673999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8673999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8674999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8674999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8674999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8674999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8674999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8675999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8675999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8675999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8675999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8675999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8676999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8676999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8676999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8676999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8676999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8677999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8677999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8677999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8677999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8677999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8678999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8678999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8678999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8678999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8678999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8679999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8679999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8679999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8679999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8679999999999815, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8680999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8680999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8680999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8680999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8680999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8681999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8681999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8681999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8681999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8681999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8682999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8682999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8682999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8682999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8682999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8683999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8683999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8683999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8683999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8683999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8684999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8684999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8684999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8684999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8684999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8685999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8685999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8685999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8685999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8685999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8686999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8686999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8686999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8686999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8686999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8687999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8687999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8687999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8687999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8687999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8688999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8688999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8688999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8688999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8688999999999814, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8689999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8689999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8689999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8689999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8689999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8690999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8690999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8690999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8690999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8690999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8690999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8691999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8691999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8691999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8691999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8691999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8692999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8692999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8692999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8692999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8692999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8693999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8693999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8693999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8693999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8693999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8694999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8694999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8694999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8694999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8694999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8695999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8695999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8695999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8695999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8695999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8696999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8696999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8696999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8696999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8696999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8697999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8697999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8697999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8697999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8697999999999813, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8698999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8698999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8698999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8698999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8698999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8699999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8699999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8699999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8699999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8699999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8700999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8700999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8700999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8700999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8700999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8701999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8701999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8701999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8701999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8701999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8702999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8702999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8702999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8702999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8702999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8703999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8703999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8703999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8703999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8703999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8704999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8704999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8704999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8704999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8704999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8705999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8705999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8705999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8705999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8705999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8706999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8706999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8706999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8706999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8706999999999812, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8707999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8707999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8707999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8707999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8707999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8708999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8708999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8708999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8708999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8708999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8709999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8709999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8709999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8709999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8709999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8710999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8710999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8710999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8710999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8710999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8711999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8711999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8711999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8711999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8711999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8712999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8712999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8712999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8712999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8712999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8713999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8713999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8713999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8713999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8713999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8714999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8714999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8714999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8714999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8714999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8715999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8715999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8715999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8715999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8715999999999811, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.871699999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.871699999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.871699999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.871699999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.871699999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.871799999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.871799999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.871799999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.871799999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.871799999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.871899999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.871899999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.871899999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.871899999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.871899999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.871999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.871999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.871999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.871999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.871999999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872099999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872099999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872099999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872099999999981, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.872099999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872199999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872199999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872199999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872199999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872199999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872299999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872299999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872299999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872299999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872299999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872399999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872399999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872399999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872399999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872399999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872499999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872499999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872499999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872499999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872499999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872599999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872599999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872599999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872599999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.872599999999981, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8726999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8726999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8726999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8726999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8726999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8727999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8727999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8727999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8727999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8727999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8728999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8728999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8728999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8728999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8728999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8729999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8729999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8729999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8729999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8729999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8730999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8730999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8730999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8730999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8730999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8731999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8731999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8731999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8731999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8731999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8732999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8732999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8732999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8732999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8732999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8733999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8733999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8733999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8733999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8733999999999809, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8734999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8734999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8734999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8734999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8734999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8735999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8735999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8735999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8735999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8735999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8736999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8736999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8736999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8736999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8736999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8737999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8737999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8737999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8737999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8737999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8738999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8738999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8738999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8738999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8738999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8739999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8739999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8739999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8739999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8740999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8740999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8740999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8740999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8740999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8741999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8741999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8741999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8741999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8741999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8742999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8742999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8742999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8742999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8742999999999808, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8743999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8743999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8743999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8743999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8743999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8744999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8744999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8744999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8744999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8744999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8745999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8745999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8745999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8745999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8745999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8746999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8746999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8746999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8746999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8746999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8747999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8747999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8747999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8747999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8747999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8748999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8748999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8748999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8748999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8748999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8749999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8749999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8749999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8749999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8749999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8750999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8750999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8750999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8750999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8750999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8751999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8751999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8751999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8751999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8751999999999807, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8752999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8752999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8752999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8752999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8752999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8753999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8753999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8753999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8753999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8753999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8754999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8754999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8754999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8754999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8754999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8755999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8755999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8755999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8755999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8755999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8756999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8756999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8756999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8756999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8756999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8757999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8757999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8757999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8757999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8757999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8758999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8758999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8758999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8758999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8758999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8759999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8759999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8759999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8759999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8759999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8760999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8760999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8760999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8760999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8760999999999806, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8761999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8761999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8761999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8761999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8761999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8762999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8762999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8762999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8762999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8762999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8763999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8763999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8763999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8763999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8763999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8764999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8764999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8764999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8764999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8764999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8765999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8765999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8765999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8765999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8765999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8766999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8766999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8766999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8766999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8766999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8767999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8767999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8767999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8767999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8767999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8768999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8768999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8768999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8768999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8768999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8769999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8769999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8769999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8769999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8769999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8770999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8770999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8770999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8770999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8770999999999805, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8771999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8771999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8771999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8771999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8771999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8772999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8772999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8772999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8772999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8772999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8773999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8773999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8773999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8773999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8773999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8774999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8774999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8774999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8774999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8774999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8775999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8775999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8775999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8775999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8775999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8776999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8776999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8776999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8776999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8776999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8777999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8777999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8777999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8777999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8777999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8778999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8778999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8778999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8778999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8778999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8779999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8779999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8779999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8779999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8779999999999804, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8780999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8780999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8780999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8780999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8780999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8781999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8781999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8781999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8781999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8781999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8782999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8782999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8782999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8782999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8782999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8783999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8783999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8783999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8783999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8783999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8784999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8784999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8784999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8784999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8784999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8785999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8785999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8785999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8785999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8785999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8786999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8786999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8786999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8786999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8786999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8787999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8787999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8787999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8787999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8788999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8788999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8788999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8788999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8788999999999803, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8789999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8789999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8789999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8789999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8789999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8790999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8790999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8790999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8790999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8790999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8791999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8791999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8791999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8791999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8791999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8792999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8792999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8792999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8792999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8792999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8793999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8793999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8793999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8793999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8793999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8794999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8794999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8794999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8794999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8794999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8795999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8795999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8795999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8795999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8795999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8796999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8796999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8796999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8796999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8796999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8797999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8797999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8797999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8797999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8797999999999802, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8798999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8798999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8798999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8798999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8798999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8799999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8799999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8799999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8799999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8799999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8800999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8800999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8800999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8800999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8800999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8801999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8801999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8801999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8801999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8801999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8802999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8802999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8802999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8802999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8802999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8803999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8803999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8803999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8803999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8803999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8804999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8804999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8804999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8804999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8804999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8805999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8805999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8805999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8805999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8805999999999801, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88069999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88069999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88069999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88069999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88069999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88079999999998, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.88079999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88079999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88079999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88079999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88089999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88089999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88089999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88089999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88089999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88099999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88099999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88099999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88099999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88099999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88109999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88109999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88109999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88109999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88109999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88119999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88119999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88119999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88119999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88119999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88129999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88129999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88129999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88129999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88129999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88139999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88139999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88139999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88139999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88139999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88149999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88149999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88149999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88149999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88149999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88159999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88159999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88159999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88159999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.88159999999998, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8816999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8816999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8816999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8816999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8816999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8817999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8817999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8817999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8817999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8817999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8818999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8818999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8818999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8818999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8818999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8819999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8819999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8819999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8819999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8819999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8820999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8820999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8820999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8820999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8820999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8821999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8821999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8821999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8821999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8821999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8822999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8822999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8822999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8822999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8822999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8823999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8823999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8823999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8823999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8823999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8824999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8824999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8824999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8824999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8824999999999799, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8825999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8825999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8825999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8825999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8825999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8826999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8826999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8826999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8826999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8826999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8827999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8827999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8827999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8827999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8827999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8828999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8828999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8828999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8828999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8828999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8829999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8829999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8829999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8829999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8829999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8830999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8830999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8830999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8830999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8830999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8831999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8831999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8831999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8831999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8831999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8832999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8832999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8832999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8832999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8832999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8833999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8833999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8833999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8833999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8833999999999798, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8834999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8834999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8834999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8834999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8834999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8835999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8835999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8835999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8835999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8836999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8836999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8836999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8836999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8836999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8837999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8837999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8837999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8837999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8837999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8838999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8838999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8838999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8838999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8838999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8839999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8839999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8839999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8839999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8839999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8840999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8840999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8840999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8840999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8840999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8841999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8841999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8841999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8841999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8841999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8842999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8842999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8842999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8842999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8842999999999797, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8843999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8843999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8843999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8843999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8843999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8844999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8844999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8844999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8844999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8844999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8845999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8845999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8845999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8845999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8845999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8846999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8846999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8846999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8846999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8846999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8847999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8847999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8847999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8847999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8847999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8848999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8848999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8848999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8848999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8848999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8849999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8849999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8849999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8849999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8849999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8850999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8850999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8850999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8850999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8850999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8851999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8851999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8851999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8851999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8851999999999796, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8852999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8852999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8852999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8852999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8852999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8853999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8853999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8853999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8853999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8853999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8854999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8854999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8854999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8854999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8854999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8855999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8855999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8855999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8855999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8855999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8856999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8856999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8856999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8856999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8856999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8857999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8857999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8857999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8857999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8857999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8858999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8858999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8858999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8858999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8858999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8858999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8859999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8859999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8859999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8859999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8860999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8860999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8860999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8860999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8860999999999795, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8861999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8861999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8861999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8861999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8861999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8862999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8862999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8862999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8862999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8862999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8863999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8863999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8863999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8863999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8863999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8864999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8864999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8864999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8864999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8864999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8865999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8865999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8865999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8865999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8865999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8866999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8866999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8866999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8866999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8866999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8867999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8867999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8867999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8867999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8867999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8868999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8868999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8868999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8868999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8868999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8869999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8869999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8869999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8869999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8869999999999794, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8870999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8870999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8870999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8870999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8870999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8871999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8871999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8871999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8871999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8871999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8872999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8872999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8872999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8872999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8872999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8873999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8873999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8873999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8873999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8873999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8874999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8874999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8874999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8874999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8874999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8875999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8875999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8875999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8875999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8875999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8876999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8876999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8876999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8876999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8876999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8877999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8877999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8877999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8877999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8877999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8878999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8878999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8878999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8878999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8878999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8879999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8879999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8879999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8879999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8879999999999793, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8880999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8880999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8880999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8880999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8880999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8881999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8881999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8881999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8881999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8881999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8882999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8882999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8882999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8882999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8882999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8883999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8883999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8883999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8883999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8884999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8884999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8884999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8884999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8884999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8885999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8885999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8885999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8885999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8885999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8886999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8886999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8886999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8886999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8886999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8887999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8887999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8887999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8887999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8887999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8888999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8888999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8888999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8888999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8888999999999792, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8889999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8889999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8889999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8889999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8889999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8890999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8890999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8890999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8890999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8890999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8891999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8891999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8891999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8891999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8891999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8892999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8892999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8892999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8892999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8892999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8893999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8893999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8893999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8893999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8893999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8894999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8894999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8894999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8894999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8894999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8895999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8895999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8895999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8895999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8895999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8896999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8896999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8896999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8896999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8896999999999791, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.889799999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.889799999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.889799999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.889799999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.889799999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.889899999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.889899999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.889899999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.889899999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.889999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.889999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.889999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.889999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.889999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.890099999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.890099999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.890099999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.890099999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.890099999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.890199999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.890199999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.890199999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.890199999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.890199999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.890299999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.890299999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.890299999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.890299999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.890299999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.890399999999979, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8903999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8903999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8903999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8903999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8904999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8904999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8904999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8904999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8904999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8905999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8905999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8905999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8905999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8905999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8906999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8906999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8906999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8906999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8906999999999979, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8907999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8907999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8907999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8907999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8907999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8908999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8908999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8908999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8908999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8908999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8909999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8909999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8909999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8909999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8909999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8910999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8910999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8910999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8910999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8910999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8911999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8911999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8911999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8911999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8911999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8912999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8912999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8912999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8912999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8912999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8913999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8913999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8913999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8913999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8913999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8914999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8914999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8914999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8914999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8914999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8915999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8915999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8915999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8915999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8915999999999789, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8916999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8916999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8916999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8916999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8916999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8917999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8917999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8917999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8917999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8917999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8918999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8918999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8918999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8918999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8919999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8919999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8919999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8919999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8919999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8920999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8920999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8920999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8920999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8920999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8921999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8921999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8921999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8921999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8921999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8922999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8922999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8922999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8922999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8922999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8923999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8923999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8923999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8923999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8923999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8924999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8924999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8924999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8924999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8924999999999788, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8925999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8925999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8925999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8925999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8925999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8926999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8926999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8926999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8926999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8926999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8927999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8927999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8927999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8927999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8927999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8928999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8928999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8928999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8928999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8928999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8929999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8929999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8929999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8929999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8929999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8930999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8930999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8930999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8930999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8930999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8931999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8931999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8931999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8931999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8932999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8932999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8932999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8932999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8932999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8933999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8933999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8933999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8933999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8933999999999787, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8934999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8934999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8934999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8934999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8934999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8935999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8935999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8935999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8935999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8935999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8936999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8936999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8936999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8936999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8936999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8937999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8937999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8937999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8937999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8937999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8938999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8938999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8938999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8938999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8938999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8939999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8939999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8939999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8939999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8939999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8940999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8940999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8940999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8940999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8940999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8941999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8941999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8941999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8941999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8941999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8942999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8942999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8942999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8942999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8942999999999786, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8943999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8943999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8943999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8943999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8943999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8944999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8944999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8944999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8944999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8944999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8945999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8945999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8945999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8945999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8945999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8946999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8946999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8946999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8946999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8946999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8947999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8947999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8947999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8947999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8947999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8948999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8948999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8948999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8948999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8948999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8949999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8949999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8949999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8949999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8949999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8950999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8950999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8950999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8950999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8950999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8951999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8951999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8951999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8951999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8951999999999785, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8952999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8952999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8952999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8952999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8952999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8953999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8953999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8953999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8953999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8953999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8954999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8954999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8954999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8954999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8954999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8955999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8955999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8955999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8955999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8955999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8956999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8956999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8956999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8956999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8956999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8957999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8957999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8957999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8957999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8957999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8958999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8958999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8958999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8958999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8958999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8959999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8959999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8959999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8959999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8959999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8960999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8960999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8960999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8960999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8960999999999784, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8961999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8961999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8961999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8961999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8961999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8962999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8962999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8962999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8962999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8962999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8963999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8963999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8963999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8963999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8963999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8964999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8964999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8964999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8964999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8964999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8965999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8965999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8965999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8965999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8965999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8966999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8966999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8966999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8966999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8966999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8967999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8967999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8967999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8967999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8967999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8968999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8968999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8968999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8968999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8968999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8969999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8969999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8969999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8969999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8969999999999783, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8970999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8970999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8970999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8970999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8970999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8971999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8971999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8971999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8971999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8971999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8972999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8972999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8972999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8972999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8972999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8973999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8973999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8973999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8973999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8973999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8974999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8974999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8974999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8974999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8974999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8975999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8975999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8975999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8975999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8975999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8976999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8976999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8976999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8976999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8976999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8977999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8977999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8977999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8977999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8977999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8978999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8978999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8978999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8978999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8978999999999782, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8979999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8979999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8979999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8979999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8979999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8980999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8980999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8980999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8980999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8980999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8981999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8981999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8981999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8981999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8981999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8982999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8982999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8982999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8982999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8982999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8983999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8983999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8983999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8983999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8984999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8984999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8984999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8984999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8984999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8985999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8985999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8985999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8985999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8985999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8986999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8986999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8986999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8986999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8986999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8987999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8987999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8987999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8987999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8987999999999781, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.898899999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.898899999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.898899999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.898899999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.898899999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.898999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.898999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.898999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.898999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.898999999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899099999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899099999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899099999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899099999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899099999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899199999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899199999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899199999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899199999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899199999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899299999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899299999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899299999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899299999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899299999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899399999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899399999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899399999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899399999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899399999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899499999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899499999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899499999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899499999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899499999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899599999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899599999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899599999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899599999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899599999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899699999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899699999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899699999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899699999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899699999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899799999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899799999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899799999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899799999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.899799999999978, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8998999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8998999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8998999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8998999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8998999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8999999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.8999999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8999999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8999999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.8999999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9000999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9000999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9000999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9000999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9000999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9001999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9001999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9001999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9001999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9001999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9002999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9002999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9002999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9002999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9002999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9003999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9003999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9003999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9003999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9003999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9004999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9004999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9004999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9004999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9004999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9005999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9005999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9005999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9005999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9005999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9006999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9006999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9006999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9006999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9006999999999779, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9007999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9007999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9007999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9007999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9007999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9008999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9008999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9008999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9008999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9008999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9009999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9009999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9009999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9009999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9009999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9010999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9010999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9010999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9010999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9010999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9011999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9011999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9011999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9011999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9011999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9012999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9012999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9012999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9012999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9012999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9013999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9013999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9013999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9013999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9013999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9014999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9014999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9014999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9014999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9014999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9015999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9015999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9015999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9015999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9015999999999778, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9016999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9016999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9016999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9016999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9016999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9017999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9017999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9017999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9017999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9017999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9018999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9018999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9018999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9018999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9018999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9019999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9019999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9019999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9019999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9019999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9020999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9020999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9020999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9020999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9020999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9021999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9021999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9021999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9021999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9021999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9022999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9022999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9022999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9022999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9022999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9023999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9023999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9023999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9023999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9023999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9024999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9024999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9024999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9024999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9024999999999777, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9025999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9025999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9025999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9025999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9025999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9026999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9026999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9026999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9026999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9026999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9027999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9027999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9027999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9027999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9028999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9028999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9028999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9028999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9028999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9029999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9029999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9029999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9029999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9029999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9030999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9030999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9030999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9030999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9030999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9031999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9031999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9031999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9031999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9031999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9032999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9032999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9032999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9032999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9032999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9033999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9033999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9033999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9033999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9033999999999776, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9034999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9034999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9034999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9034999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9034999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9035999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9035999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9035999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9035999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9035999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9036999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9036999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9036999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9036999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9036999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9037999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9037999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9037999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9037999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9037999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9038999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9038999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9038999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9038999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9038999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9039999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9039999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9039999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9039999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9039999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9040999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9040999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9040999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9040999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9040999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9041999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9041999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9041999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9041999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9041999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9042999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9042999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9042999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9042999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9042999999999775, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9043999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9043999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9043999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9043999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9043999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9044999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9044999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9044999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9044999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9044999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9045999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9045999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9045999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9045999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9045999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9046999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9046999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9046999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9046999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9046999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9047999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9047999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9047999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9047999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9047999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9048999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9048999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9048999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9048999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9048999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9049999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9049999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9049999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9049999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9049999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9050999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9050999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9050999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9050999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9050999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9051999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9051999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9051999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9051999999999774, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9052999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9052999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9052999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9052999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9052999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9053999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9053999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9053999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9053999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9054999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9054999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9054999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9054999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9054999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9055999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9055999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9055999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9055999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9055999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9056999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9056999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9056999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9056999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9056999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9057999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9057999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9057999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9057999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9057999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9058999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9058999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9058999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9058999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9058999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9059999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9059999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9059999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9059999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9059999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9060999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9060999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9060999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9060999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9060999999999773, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9061999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9061999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9061999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9061999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9061999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9062999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9062999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9062999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9062999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9062999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9063999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9063999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9063999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9063999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9063999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9064999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9064999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9064999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9064999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9064999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9065999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9065999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9065999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9065999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9065999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9066999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9066999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9066999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9066999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9066999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9067999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9067999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9067999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9067999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9067999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9068999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9068999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9068999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9068999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9068999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9069999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9069999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9069999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9069999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9069999999999772, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9070999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9070999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9070999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9070999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9070999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9071999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9071999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9071999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9071999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9071999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9072999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9072999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9072999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9072999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9072999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9073999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9073999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9073999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9073999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9073999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9074999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9074999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9074999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9074999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9074999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9075999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9075999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9075999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9075999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9076999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9076999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9076999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9076999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9076999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9077999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9077999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9077999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9077999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9077999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9078999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9078999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9078999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9078999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9078999999999771, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.907999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.907999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.907999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.907999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.907999999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908099999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908099999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908099999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908099999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908099999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908199999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908199999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908199999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908199999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908199999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908299999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908299999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908299999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908299999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908299999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908399999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908399999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908399999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908399999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908399999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908499999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908499999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908499999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908499999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908499999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908599999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908599999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908599999999977, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.908599999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908599999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908699999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908699999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908699999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908699999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908699999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908799999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908799999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908799999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908799999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908799999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908899999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908899999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908899999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908899999999977, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908999999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908999999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908999999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908999999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.908999999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.909099999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.909099999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.909099999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.909099999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.909199999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.909199999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.909199999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.909199999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.909199999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.909299999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.909299999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.909299999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.909299999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.909399999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.909399999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.909399999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.909399999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.909399999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.909499999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.909499999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.909499999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.909499999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.909499999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.909599999999769, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9095999999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9095999999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9095999999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9095999999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9096999999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9096999999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9096999999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9096999999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9096999999999769, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9097999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9097999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9097999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9097999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9097999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9098999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9098999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9098999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9098999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9098999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9099999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9099999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9099999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9099999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9099999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9100999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9100999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9100999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9100999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9100999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9101999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9101999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9101999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9101999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9102999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9102999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9102999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9102999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9102999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9103999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9103999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9103999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9103999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9103999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9104999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9104999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9104999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9104999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9104999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9105999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9105999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9105999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9105999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9105999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9106999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9106999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9106999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9106999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9106999999999768, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9107999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9107999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9107999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9107999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9107999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9108999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9108999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9108999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9108999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9108999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9109999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9109999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9109999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9109999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9109999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9110999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9110999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9110999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9110999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9110999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9111999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9111999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9111999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9111999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9111999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9112999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9112999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9112999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9112999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9112999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9113999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9113999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9113999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9113999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9113999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9114999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9114999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9114999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9114999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9114999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9115999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9115999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9115999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9115999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9115999999999767, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9116999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9116999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9116999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9116999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9116999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9117999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9117999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9117999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9117999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9117999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9118999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9118999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9118999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9118999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9118999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9119999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9119999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9119999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9119999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9119999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9120999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9120999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9120999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9120999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9120999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9121999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9121999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9121999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9121999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9121999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9122999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9122999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9122999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9122999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9122999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9123999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9123999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9123999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9123999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9124999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9124999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9124999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9124999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9124999999999766, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9125999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9125999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9125999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9125999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9125999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9126999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9126999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9126999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9126999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9126999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9127999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9127999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9127999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9127999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9127999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9128999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9128999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9128999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9128999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9128999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9129999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9129999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9129999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9129999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9129999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9130999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9130999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9130999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9130999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9130999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9131999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9131999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9131999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9131999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9131999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9132999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9132999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9132999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9132999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9132999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9133999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9133999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9133999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9133999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9133999999999765, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9134999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9134999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9134999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9134999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9134999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9135999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9135999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9135999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9135999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9135999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9136999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9136999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9136999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9136999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9136999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9137999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9137999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9137999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9137999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9137999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9138999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9138999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9138999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9138999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9138999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9139999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9139999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9139999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9139999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9139999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9139999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9140999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9140999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9140999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9140999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9140999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9141999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9141999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9141999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9141999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9141999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9142999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9142999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9142999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9142999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9142999999999764, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9143999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9143999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9143999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9143999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9143999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9144999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9144999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9144999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9144999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9144999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9145999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9145999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9145999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9145999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9145999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9146999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9146999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9146999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9146999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9146999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9147999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9147999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9147999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9147999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9148999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9148999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9148999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9148999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9148999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9149999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9149999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9149999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9149999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9150999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9150999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9150999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9150999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9150999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9151999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9151999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9151999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9151999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9151999999999763, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9152999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9152999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9152999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9152999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9152999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9153999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9153999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9153999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9153999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9153999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9154999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9154999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9154999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9154999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9154999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9155999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9155999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9155999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9155999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9155999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9156999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9156999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9156999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9156999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9156999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9157999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9157999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9157999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9157999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9157999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9158999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9158999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9158999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9158999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9158999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9159999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9159999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9159999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9159999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9159999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9160999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9160999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9160999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9160999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9160999999999762, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9161999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9161999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9161999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9161999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9161999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9162999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9162999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9162999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9162999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9162999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9163999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9163999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9163999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9163999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9163999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9164999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9164999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9164999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9164999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9164999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9165999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9165999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9165999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9165999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9165999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9166999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9166999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9166999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9166999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9166999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9167999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9167999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9167999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9167999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9167999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9168999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9168999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9168999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9168999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9168999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9169999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9169999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9169999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9169999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9169999999999761, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917099999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917099999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917099999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917099999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917099999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917199999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917199999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917199999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917199999999976, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.917299999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917299999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917299999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917299999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917299999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917399999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917399999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917399999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917399999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917399999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917499999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917499999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917499999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917499999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917499999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917599999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917599999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917599999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917599999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917599999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917699999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917699999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917699999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917699999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917699999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917799999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917799999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917799999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917799999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917799999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917899999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917899999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917899999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917899999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917899999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.917999999999976, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9180999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9180999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9180999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9180999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9180999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9181999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9181999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9181999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9181999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9181999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9182999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9182999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9182999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9182999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9182999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9183999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9183999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9183999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9183999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9183999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9184999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9184999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9184999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9184999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9184999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9185999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9185999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9185999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9185999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9185999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9186999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9186999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9186999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9186999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9186999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9187999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9187999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9187999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9187999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9187999999999759, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9188999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9188999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9188999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9188999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9188999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9189999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9189999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9189999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9189999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9189999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9190999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9190999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9190999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9190999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9190999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9191999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9191999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9191999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9191999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9191999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9192999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9192999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9192999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9192999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9192999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9193999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9193999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9193999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9193999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9193999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9194999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9194999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9194999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9194999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9194999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9195999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9195999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9195999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9195999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9195999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9196999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9196999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9196999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9196999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9196999999999758, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9197999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9197999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9197999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9197999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9197999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9198999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9198999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9198999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9198999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9198999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9198999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9199999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9199999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9199999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9199999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9199999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9200999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9200999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9200999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9200999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9200999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9201999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9201999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9201999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9201999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9201999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9202999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9202999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9202999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9202999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9202999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9203999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9203999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9203999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9203999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9203999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9204999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9204999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9204999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9204999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9204999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9205999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9205999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9205999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9205999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9205999999999757, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9206999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9206999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9206999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9206999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9206999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9207999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9207999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9207999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9207999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9207999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9208999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9208999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9208999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9208999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9208999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9209999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9209999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9209999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9209999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9209999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9210999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9210999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9210999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9210999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9210999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9211999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9211999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9211999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9211999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9211999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9212999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9212999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9212999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9212999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9212999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9213999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9213999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9213999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9213999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9213999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9214999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9214999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9214999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9214999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9214999999999756, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9215999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9215999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9215999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9215999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9215999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9216999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9216999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9216999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9216999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9216999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9217999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9217999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9217999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9217999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9217999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9218999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9218999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9218999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9218999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9218999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9218999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9219999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9219999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9219999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9219999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9220999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9220999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9220999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9220999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9220999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9221999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9221999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9221999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9221999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9221999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9222999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9222999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9222999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9222999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9222999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9223999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9223999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9223999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9223999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9223999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9224999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9224999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9224999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9224999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9224999999999755, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9225999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9225999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9225999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9225999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9225999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9226999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9226999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9226999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9226999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9226999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9227999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9227999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9227999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9227999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9227999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9228999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9228999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9228999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9228999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9228999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9229999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9229999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9229999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9229999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9229999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9230999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9230999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9230999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9230999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9230999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9231999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9231999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9231999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9231999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9231999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9232999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9232999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9232999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9232999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9232999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9233999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9233999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9233999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9233999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9233999999999754, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9234999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9234999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9234999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9234999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9234999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9235999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9235999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9235999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9235999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9235999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9236999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9236999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9236999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9236999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9236999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9237999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9237999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9237999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9237999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9237999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9238999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9238999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9238999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9238999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9238999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9239999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9239999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9239999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9239999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9239999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9240999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9240999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9240999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9240999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9240999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9241999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9241999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9241999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9241999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9241999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9242999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9242999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9242999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9242999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9242999999999753, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9243999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9243999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9243999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9243999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9243999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9244999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9244999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9244999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9244999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9244999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9245999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9245999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9245999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9245999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9245999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9246999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9246999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9246999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9246999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9246999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9247999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9247999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9247999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9247999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9247999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9248999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9248999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9248999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9248999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9248999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9249999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9249999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9249999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9249999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9249999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9250999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9250999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9250999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9250999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9250999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9251999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9251999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9251999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9251999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9251999999999752, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9252999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9252999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9252999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9252999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9252999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9253999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9253999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9253999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9253999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9253999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9254999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9254999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9254999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9254999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9254999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9255999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9255999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9255999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9255999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9255999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9256999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9256999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9256999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9256999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9256999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9257999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9257999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9257999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9257999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9257999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9258999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9258999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9258999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9258999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9258999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9259999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9259999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9259999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9259999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9259999999999751, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926099999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926099999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926099999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926099999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926099999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926199999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926199999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926199999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926199999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926199999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926299999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926299999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926299999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926299999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926299999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926399999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926399999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926399999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926399999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926399999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926499999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926499999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926499999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926499999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926499999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926599999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926599999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926599999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926599999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926599999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926699999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926699999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926699999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926699999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926699999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926799999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926799999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926799999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926799999999975, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.926899999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926899999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926899999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926899999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926899999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.926999999999975, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9270999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9270999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9270999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9270999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9270999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9271999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9271999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9271999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9271999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9271999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9272999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9272999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9272999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9272999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9272999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9273999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9273999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9273999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9273999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9273999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9274999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9274999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9274999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9274999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9274999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9275999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9275999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9275999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9275999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9275999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9276999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9276999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9276999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9276999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9276999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9277999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9277999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9277999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9277999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9277999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9278999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9278999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9278999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9278999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9278999999999749, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9279999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9279999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9279999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9279999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9279999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9280999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9280999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9280999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9280999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9280999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9281999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9281999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9281999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9281999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9281999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9282999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9282999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9282999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9282999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9282999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9283999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9283999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9283999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9283999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9283999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9284999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9284999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9284999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9284999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9284999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9285999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9285999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9285999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9285999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9285999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9286999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9286999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9286999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9286999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9286999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9287999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9287999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9287999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9287999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9287999999999748, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9288999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9288999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9288999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9288999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9288999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9289999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9289999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9289999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9289999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9289999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9290999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9290999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9290999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9290999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9290999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9291999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9291999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9291999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9291999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9291999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9292999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9292999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9292999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9292999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9292999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9293999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9293999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9293999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9293999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9293999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9294999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9294999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9294999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9294999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9294999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9295999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9295999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9295999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9295999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9295999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9296999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9296999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9296999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9296999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9296999999999747, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9297999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9297999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9297999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9297999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9297999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9298999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9298999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9298999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9298999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9298999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9299999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9299999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9299999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9299999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9299999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9300999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9300999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9300999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9300999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9300999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9301999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9301999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9301999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9301999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9301999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9302999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9302999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9302999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9302999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9303999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9303999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9303999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9303999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9303999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9304999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9304999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9304999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9304999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9304999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9305999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9305999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9305999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9305999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9305999999999746, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9306999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9306999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9306999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9306999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9306999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9307999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9307999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9307999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9307999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9307999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9308999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9308999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9308999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9308999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9308999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9309999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9309999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9309999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9309999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9309999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9310999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9310999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9310999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9310999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9310999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9311999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9311999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9311999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9311999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9311999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9312999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9312999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9312999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9312999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9312999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9313999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9313999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9313999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9313999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9313999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9314999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9314999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9314999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9314999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9314999999999745, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9315999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9315999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9315999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9315999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9315999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9316999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9316999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9316999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9316999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9316999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9317999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9317999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9317999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9317999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9317999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9318999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9318999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9318999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9318999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9318999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9319999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9319999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9319999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9319999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9319999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9320999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9320999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9320999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9320999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9320999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9321999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9321999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9321999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9321999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9321999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9322999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9322999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9322999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9322999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9322999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9323999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9323999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9323999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9323999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9323999999999744, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9324999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9324999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9324999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9324999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9324999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9325999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9325999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9325999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9325999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9325999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9326999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9326999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9326999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9326999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9326999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9327999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9327999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9327999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9327999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9327999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9328999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9328999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9328999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9328999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9328999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9329999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9329999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9329999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9329999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9330999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9330999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9330999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9330999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9330999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9331999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9331999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9331999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9331999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9331999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9332999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9332999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9332999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9332999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9332999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9333999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9333999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9333999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9333999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9333999999999743, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9334999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9334999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9334999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9334999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9334999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9335999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9335999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9335999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9335999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9335999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9336999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9336999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9336999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9336999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9336999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9337999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9337999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9337999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9337999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9337999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9338999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9338999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9338999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9338999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9338999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9339999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9339999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9339999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9339999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9339999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9340999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9340999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9340999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9340999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9340999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9341999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9341999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9341999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9341999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9341999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9342999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9342999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9342999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9342999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9342999999999742, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9343999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9343999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9343999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9343999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9343999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9344999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9344999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9344999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9344999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9344999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9345999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9345999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9345999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9345999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9345999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9346999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9346999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9346999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9346999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9346999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9347999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9347999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9347999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9347999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9347999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9348999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9348999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9348999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9348999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9348999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9349999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9349999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9349999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9349999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9349999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9350999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9350999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9350999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9350999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9350999999999741, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.935199999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.935199999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.935199999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.935199999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.935199999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.935299999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.935299999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.935299999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.935299999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.935299999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.935399999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.935399999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.935399999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.935399999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.935399999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.935499999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.935499999999974, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9354999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9354999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9354999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9355999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9355999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9355999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9355999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9355999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9356999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9356999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9356999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9356999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9357999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9357999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9357999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9357999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9357999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9358999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9358999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9358999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9358999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9358999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9359999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9359999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9359999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9359999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9359999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9360999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9360999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9360999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9360999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9360999999999974, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9361999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9361999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9361999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9361999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9361999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9362999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9362999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9362999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9362999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9362999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9363999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9363999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9363999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9363999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9364999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9364999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9364999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9364999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9364999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9365999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9365999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9365999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9365999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9365999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9366999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9366999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9366999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9366999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9366999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9367999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9367999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9367999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9367999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9367999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9368999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9368999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9368999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9368999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9368999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9369999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9369999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9369999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9369999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9369999999999739, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9370999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9370999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9370999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9370999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9370999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9371999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9371999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9371999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9371999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9371999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9372999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9372999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9372999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9372999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9372999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9373999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9373999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9373999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9373999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9373999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9374999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9374999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9374999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9374999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9374999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9375999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9375999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9375999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9375999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9375999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9376999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9376999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9376999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9376999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9376999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9377999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9377999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9377999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9377999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9377999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9378999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9378999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9378999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9378999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9378999999999738, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9379999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9379999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9379999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9379999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9379999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9380999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9380999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9380999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9380999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9380999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9381999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9381999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9381999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9381999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9381999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9382999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9382999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9382999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9382999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9382999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9383999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9383999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9383999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9383999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9383999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9384999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9384999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9384999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9384999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9384999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9385999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9385999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9385999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9385999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9385999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9386999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9386999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9386999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9386999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9386999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9387999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9387999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9387999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9387999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9387999999999737, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9388999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9388999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9388999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9388999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9388999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9389999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9389999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9389999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9389999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9389999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9390999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9390999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9390999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9390999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9390999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9391999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9391999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9391999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9391999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9391999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9392999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9392999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9392999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9392999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9392999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9393999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9393999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9393999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9393999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9393999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9394999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9394999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9394999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9394999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9394999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9395999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9395999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9395999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9395999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9395999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9396999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9396999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9396999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9396999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9396999999999736, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9397999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9397999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9397999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9397999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9397999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9398999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9398999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9398999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9398999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9398999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9399999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9399999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9399999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9399999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9399999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9400999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9400999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9400999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9400999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9400999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9401999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9401999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9401999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9401999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9401999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9402999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9402999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9402999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9402999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9402999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9403999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9403999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9403999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9403999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9403999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9404999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9404999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9404999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9404999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9404999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9405999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9405999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9405999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9405999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9405999999999735, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9406999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9406999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9406999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9406999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9406999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9407999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9407999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9407999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9407999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9407999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9408999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9408999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9408999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9408999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9408999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9409999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9409999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9409999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9409999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9409999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9410999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9410999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9410999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9410999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9410999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9411999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9411999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9411999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9411999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9412999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9412999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9412999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9412999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9412999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9413999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9413999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9413999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9413999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9413999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9414999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9414999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9414999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9414999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9414999999999734, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9415999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9415999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9415999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9415999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9415999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9416999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9416999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9416999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9416999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9416999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9417999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9417999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9417999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9417999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9417999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9418999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9418999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9418999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9418999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9418999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9419999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9419999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9419999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9419999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9419999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9420999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9420999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9420999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9420999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9420999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9421999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9421999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9421999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9421999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9421999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9422999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9422999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9422999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9422999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9422999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9423999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9423999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9423999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9423999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9423999999999733, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9424999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9424999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9424999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9424999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9424999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9425999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9425999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9425999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9425999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9425999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9426999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9426999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9426999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9426999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9426999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9427999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9427999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9427999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9427999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9427999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9428999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9428999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9428999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9428999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9428999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9429999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9429999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9429999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9429999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9429999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9430999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9430999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9430999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9430999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9430999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9431999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9431999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9431999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9431999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9431999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9432999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9432999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9432999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9432999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9432999999999732, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9433999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9433999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9433999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9433999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9433999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9434999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9434999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9434999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9434999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9434999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9435999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9435999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9435999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9435999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9435999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9436999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9436999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9436999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9436999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9436999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9437999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9437999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9437999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9437999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9437999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9438999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9438999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9438999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9438999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9438999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9439999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9439999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9439999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9439999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9439999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9440999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9440999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9440999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9440999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9440999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9441999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9441999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9441999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9441999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9441999999999731, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944299999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944299999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944299999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944299999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944299999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944399999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944399999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944399999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944399999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944399999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944499999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944499999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944499999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944499999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944499999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944599999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944599999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944599999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944599999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944599999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944699999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944699999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944699999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944699999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944699999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944799999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944799999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944799999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944799999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944799999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944899999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944899999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944899999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944899999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944899999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.944999999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.945099999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.945099999999973, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.945099999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.945099999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.945099999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.945199999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.945199999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.945199999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.945199999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.945199999999973, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9452999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9452999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9452999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9452999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9452999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9453999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9453999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9453999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9453999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9453999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9454999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9454999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9454999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9454999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9454999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9455999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9455999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9455999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9455999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9455999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9456999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9456999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9456999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9456999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9456999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9457999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9457999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9457999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9457999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9457999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9458999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9458999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9458999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9458999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9458999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9459999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9459999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9459999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9459999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9460999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9460999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9460999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9460999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9460999999999729, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9461999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9461999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9461999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9461999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9461999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9462999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9462999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9462999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9462999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9462999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9463999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9463999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9463999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9463999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9463999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9464999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9464999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9464999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9464999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9464999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9465999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9465999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9465999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9465999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9465999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9466999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9466999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9466999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9466999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9466999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9467999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9467999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9467999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9467999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9467999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9468999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9468999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9468999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9468999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9468999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9469999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9469999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9469999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9469999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9469999999999728, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9470999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9470999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9470999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9470999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9470999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9471999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9471999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9471999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9471999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9471999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9472999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9472999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9472999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9472999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9472999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9473999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9473999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9473999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9473999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9473999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9474999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9474999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9474999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9474999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9474999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9475999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9475999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9475999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9475999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9475999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9476999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9476999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9476999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9476999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9476999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9477999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9477999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9477999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9477999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9477999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9478999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9478999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9478999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9478999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9478999999999727, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9479999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9479999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9479999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9479999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9479999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9480999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9480999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9480999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9480999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9480999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9481999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9481999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9481999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9481999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9481999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9482999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9482999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9482999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9482999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9482999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9483999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9483999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9483999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9483999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9483999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9484999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9484999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9484999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9484999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9484999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9485999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9485999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9485999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9485999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9485999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9486999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9486999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9486999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9486999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9486999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9487999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9487999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9487999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9487999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9487999999999726, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9488999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9488999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9488999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9488999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9488999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9489999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9489999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9489999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9489999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9489999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9490999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9490999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9490999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9490999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9490999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9491999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9491999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9491999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9491999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9491999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9492999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9492999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9492999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9492999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9492999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9493999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9493999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9493999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9493999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9493999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9494999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9494999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9494999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9494999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9494999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9495999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9495999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9495999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9495999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9495999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9496999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9496999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9496999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9496999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9496999999999725, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9497999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9497999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9497999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9497999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9497999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9498999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9498999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9498999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9498999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9498999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9499999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9499999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9499999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9499999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9499999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9500999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9500999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9500999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9500999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9500999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9501999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9501999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9501999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9501999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9501999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9502999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9502999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9502999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9502999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9502999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9503999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9503999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9503999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9503999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9503999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9504999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9504999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9504999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9504999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9504999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9505999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9505999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9505999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9505999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9505999999999724, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9506999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9506999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9506999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9506999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9506999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9507999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9507999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9507999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9507999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9508999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9508999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9508999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9508999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9508999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9509999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9509999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9509999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9509999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9509999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9510999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9510999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9510999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9510999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9510999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9511999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9511999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9511999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9511999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9511999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9512999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9512999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9512999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9512999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9512999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9513999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9513999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9513999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9513999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9513999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9514999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9514999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9514999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9514999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9514999999999723, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9515999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9515999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9515999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9515999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9515999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9516999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9516999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9516999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9516999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9516999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9517999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9517999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9517999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9517999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9517999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9518999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9518999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9518999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9518999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9518999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9519999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9519999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9519999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9519999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9519999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9520999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9520999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9520999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9520999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9520999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9521999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9521999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9521999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9521999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9521999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9522999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9522999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9522999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9522999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9522999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9523999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9523999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9523999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9523999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9523999999999722, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9524999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9524999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9524999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9524999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9524999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9525999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9525999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9525999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9525999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9525999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9526999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9526999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9526999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9526999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9526999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9527999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9527999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9527999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9527999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9527999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9528999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9528999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9528999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9528999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9528999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9529999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9529999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9529999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9529999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9529999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9530999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9530999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9530999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9530999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9530999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9531999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9531999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9531999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9531999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9531999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9532999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9532999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9532999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9532999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9532999999999721, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953399999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953399999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953399999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953399999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953399999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953499999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953499999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953499999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953499999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953499999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953599999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953599999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953599999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953599999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953599999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953699999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953699999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953699999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953699999999972, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.953699999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953799999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953799999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953799999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953799999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953799999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953899999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953899999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953899999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953899999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953899999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.953999999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.954099999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.954099999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.954099999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.954099999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.954099999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.954199999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.954199999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.954199999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.954199999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.954199999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.954299999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.954299999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.954299999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.954299999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.954299999999972, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9543999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9543999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9543999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9543999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9543999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9544999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9544999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9544999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9544999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9544999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9545999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9545999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9545999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9545999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9546999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9546999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9546999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9546999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9546999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9547999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9547999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9547999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9547999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9547999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9548999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9548999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9548999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9548999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9548999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9549999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9549999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9549999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9549999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9549999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9550999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9550999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9550999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9550999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9550999999999719, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9551999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9551999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9551999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9551999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9551999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9552999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9552999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9552999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9552999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9552999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9553999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9553999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9553999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9553999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9553999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9554999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9554999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9554999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9554999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9554999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9555999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9555999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9555999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9555999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9556999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9556999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9556999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9556999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9556999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9557999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9557999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9557999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9557999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9557999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9558999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9558999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9558999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9558999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9558999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9559999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9559999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9559999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9559999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9559999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9560999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9560999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9560999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9560999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9560999999999718, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9561999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9561999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9561999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9561999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9561999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9562999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9562999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9562999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9562999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9562999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9563999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9563999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9563999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9563999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9563999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9564999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9564999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9564999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9564999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9564999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9565999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9565999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9565999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9565999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9565999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9566999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9566999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9566999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9566999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9566999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9567999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9567999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9567999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9567999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9567999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9568999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9568999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9568999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9568999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9569999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9569999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9569999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9569999999999717, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9570999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9570999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9570999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9570999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9570999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9571999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9571999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9571999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9571999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9571999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9572999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9572999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9572999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9572999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9572999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9573999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9573999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9573999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9573999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9573999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9574999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9574999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9574999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9574999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9574999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9575999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9575999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9575999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9575999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9575999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9576999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9576999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9576999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9576999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9576999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9577999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9577999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9577999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9577999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9577999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9578999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9578999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9578999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9578999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9578999999999716, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9579999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9579999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9579999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9579999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9579999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9580999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9580999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9580999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9580999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9580999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9581999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9581999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9581999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9581999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9582999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9582999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9582999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9582999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9582999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9583999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9583999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9583999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9583999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9583999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9584999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9584999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9584999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9584999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9584999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9585999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9585999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9585999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9585999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9585999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9586999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9586999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9586999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9586999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9586999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9587999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9587999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9587999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9587999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9587999999999715, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9588999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9588999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9588999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9588999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9588999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9589999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9589999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9589999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9589999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9589999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9590999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9590999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9590999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9590999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9590999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9591999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9591999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9591999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9591999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9591999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9592999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9592999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9592999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9592999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9592999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9593999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9593999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9593999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9593999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9593999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9594999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9594999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9594999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9594999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9594999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9595999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9595999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9595999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9595999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9595999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9596999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9596999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9596999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9596999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9596999999999714, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9597999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9597999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9597999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9597999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9597999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9598999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9598999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9598999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9598999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9598999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9599999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9599999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9599999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9599999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9599999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9600999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9600999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9600999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9600999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9600999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9601999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9601999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9601999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9601999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9601999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9602999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9602999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9602999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9602999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9602999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9603999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9603999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9603999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9603999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9604999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9604999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9604999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9604999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9604999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9605999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9605999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9605999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9605999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9605999999999713, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9606999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9606999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9606999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9606999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9606999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9607999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9607999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9607999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9607999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9607999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9608999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9608999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9608999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9608999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9608999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9609999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9609999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9609999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9609999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9609999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9610999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9610999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9610999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9610999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9610999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9611999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9611999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9611999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9611999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9611999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9612999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9612999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9612999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9612999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9612999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9613999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9613999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9613999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9613999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9613999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9614999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9614999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9614999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9614999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9614999999999712, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9615999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9615999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9615999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9615999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9615999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9616999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9616999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9616999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9616999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9616999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9617999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9617999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9617999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9617999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9617999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9618999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9618999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9618999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9618999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9618999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9619999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9619999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9619999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9619999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9619999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9620999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9620999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9620999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9620999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9620999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9621999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9621999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9621999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9621999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9621999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9622999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9622999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9622999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9622999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9622999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9623999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9623999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9623999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9623999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9623999999999711, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962499999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962499999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962499999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962499999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962499999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962599999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962599999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962599999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962599999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962599999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962699999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962699999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962699999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962699999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962699999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962799999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962799999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962799999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962799999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962799999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962899999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962899999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962899999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962899999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962899999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.962999999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.963099999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.963099999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.963099999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.963099999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.963099999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.963199999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.963199999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.963199999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.963199999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.963199999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.963299999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.963299999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.963299999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.963299999999971, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.963299999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.963399999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.963399999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.963399999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.963399999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.963399999999971, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9634999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9634999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9634999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9634999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9634999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9635999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9635999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9635999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9635999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9635999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9636999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9636999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9636999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9636999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9636999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9637999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9637999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9637999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9637999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9637999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9638999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9638999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9638999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9638999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9639999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9639999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9639999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9639999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9639999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9640999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9640999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9640999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9640999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9640999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9641999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9641999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9641999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9641999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9641999999999709, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9642999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9642999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9642999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9642999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9642999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9643999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9643999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9643999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9643999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9643999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9644999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9644999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9644999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9644999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9644999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9645999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9645999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9645999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9645999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9645999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9646999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9646999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9646999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9646999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9646999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9647999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9647999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9647999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9647999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9647999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9648999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9648999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9648999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9648999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9648999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9649999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9649999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9649999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9649999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9649999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9650999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9650999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9650999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9650999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9650999999999708, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9651999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9651999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9651999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9651999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9651999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9652999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9652999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9652999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9652999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9652999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9653999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9653999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9653999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9653999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9653999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9654999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9654999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9654999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9654999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9654999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9655999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9655999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9655999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9655999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9655999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9656999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9656999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9656999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9656999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9656999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9657999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9657999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9657999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9657999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9657999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9658999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9658999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9658999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9658999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9658999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9659999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9659999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9659999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9659999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9659999999999707, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9660999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9660999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9660999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9660999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9660999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9661999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9661999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9661999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9661999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9661999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9662999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9662999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9662999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9662999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9662999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9663999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9663999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9663999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9663999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9663999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9664999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9664999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9664999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9664999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9665999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9665999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9665999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9665999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9665999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9666999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9666999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9666999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9666999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9666999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9667999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9667999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9667999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9667999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9667999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9668999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9668999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9668999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9668999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9668999999999706, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9669999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9669999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9669999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9669999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9669999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9670999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9670999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9670999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9670999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9670999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9671999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9671999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9671999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9671999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9671999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9672999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9672999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9672999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9672999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9672999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9673999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9673999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9673999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9673999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9673999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9674999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9674999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9674999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9674999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9674999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9675999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9675999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9675999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9675999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9675999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9676999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9676999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9676999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9676999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9676999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9677999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9677999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9677999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9677999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9677999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9678999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9678999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9678999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9678999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9678999999999705, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9679999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9679999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9679999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9679999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9679999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9680999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9680999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9680999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9680999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9680999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9681999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9681999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9681999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9681999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9681999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9682999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9682999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9682999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9682999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9682999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9683999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9683999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9683999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9683999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9683999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9684999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9684999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9684999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9684999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9684999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9685999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9685999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9685999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9685999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9685999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9686999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9686999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9686999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9686999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9686999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9687999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9687999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9687999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9687999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9687999999999704, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9688999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9688999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9688999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9688999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9688999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9689999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9689999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9689999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9689999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9689999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9690999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9690999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9690999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9690999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9690999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9691999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9691999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9691999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9691999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9691999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9692999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9692999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9692999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9692999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9692999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9693999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9693999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9693999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9693999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9693999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9694999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9694999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9694999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9694999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9694999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9695999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9695999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9695999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9695999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9695999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9696999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9696999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9696999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9696999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9696999999999703, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9697999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9697999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9697999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9697999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9697999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9698999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9698999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9698999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9698999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9698999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9699999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9699999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9699999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9699999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9699999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9700999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9700999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9700999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9700999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9700999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9701999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9701999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9701999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9701999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9701999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9702999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9702999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9702999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9702999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9702999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9703999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9703999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9703999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9703999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9703999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9704999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9704999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9704999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9704999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9704999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9705999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9705999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9705999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9705999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9705999999999702, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9706999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9706999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9706999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9706999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9706999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9707999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9707999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9707999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9707999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9707999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9708999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9708999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9708999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9708999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9708999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9709999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9709999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9709999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9709999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9709999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9710999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9710999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9710999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9710999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9710999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9711999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9711999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9711999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9711999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9711999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9712999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9712999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9712999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9712999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9712999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9713999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9713999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9713999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9713999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9713999999999701, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97149999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97149999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97149999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97149999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97149999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97159999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97159999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97159999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97159999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97159999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97169999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97169999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97169999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97169999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97169999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97179999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97179999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97179999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97179999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97179999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97189999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97189999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97189999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97189999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97189999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97199999999997, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.97199999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97199999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97199999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97199999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97209999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97209999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97209999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97209999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97209999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97219999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97219999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97219999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97219999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97219999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97229999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97229999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97229999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97229999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97229999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97239999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97239999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97239999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97239999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.97239999999997, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9724999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9724999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9724999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9724999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9724999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9725999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9725999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9725999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9725999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9725999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9726999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9726999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9726999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9726999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9726999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9727999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9727999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9727999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9727999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9727999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9728999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9728999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9728999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9728999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9728999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9729999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9729999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9729999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9729999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9729999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9730999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9730999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9730999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9730999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9730999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9731999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9731999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9731999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9731999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9731999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9732999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9732999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9732999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9732999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9732999999999699, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9733999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9733999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9733999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9733999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9733999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9734999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9734999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9734999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9734999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9734999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9735999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9735999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9735999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9735999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9735999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9736999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9736999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9736999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9736999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9736999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9737999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9737999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9737999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9737999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9737999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9738999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9738999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9738999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9738999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9738999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9739999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9739999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9739999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9739999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9739999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9740999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9740999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9740999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9740999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9740999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9741999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9741999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9741999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9741999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9741999999999698, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9742999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9742999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9742999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9742999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9742999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9743999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9743999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9743999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9743999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9743999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9744999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9744999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9744999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9744999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9744999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9745999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9745999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9745999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9745999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9745999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9746999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9746999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9746999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9746999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9746999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9747999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9747999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9747999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9747999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9748999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9748999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9748999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9748999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9748999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9749999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9749999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9749999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9749999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9749999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9750999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9750999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9750999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9750999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9750999999999697, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9751999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9751999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9751999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9751999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9751999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9752999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9752999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9752999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9752999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9752999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9753999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9753999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9753999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9753999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9753999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9754999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9754999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9754999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9754999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9754999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9755999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9755999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9755999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9755999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9755999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9756999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9756999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9756999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9756999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9756999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9757999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9757999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9757999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9757999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9757999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9758999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9758999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9758999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9758999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9758999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9759999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9759999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9759999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9759999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9759999999999696, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9760999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9760999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9760999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9760999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9760999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9761999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9761999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9761999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9761999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9761999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9762999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9762999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9762999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9762999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9762999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9763999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9763999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9763999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9763999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9763999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9764999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9764999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9764999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9764999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9764999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9765999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9765999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9765999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9765999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9765999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9766999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9766999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9766999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9766999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9766999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9767999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9767999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9767999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9767999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9767999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9768999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9768999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9768999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9768999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9768999999999695, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9769999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9769999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9769999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9769999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9769999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9770999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9770999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9770999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9770999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9770999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9771999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9771999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9771999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9771999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9771999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9772999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9772999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9772999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9772999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9772999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9773999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9773999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9773999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9773999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9773999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9774999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9774999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9774999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9774999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9774999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9775999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9775999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9775999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9775999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9775999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9776999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9776999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9776999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9776999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9776999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9777999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9777999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9777999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9777999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9777999999999694, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9778999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9778999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9778999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9778999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9778999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9779999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9779999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9779999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9779999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9779999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9780999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9780999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9780999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9780999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9780999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9781999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9781999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9781999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9781999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9781999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9782999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9782999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9782999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9782999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9782999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9783999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9783999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9783999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9783999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9783999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9784999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9784999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9784999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9784999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9784999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9785999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9785999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9785999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9785999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9785999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9786999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9786999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9786999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9786999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9786999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9787999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9787999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9787999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9787999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9787999999999693, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9788999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9788999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9788999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9788999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9788999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9789999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9789999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9789999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9789999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9789999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9790999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9790999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9790999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9790999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9790999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9791999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9791999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9791999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9791999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9791999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9792999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9792999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9792999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9792999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9792999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9793999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9793999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9793999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9793999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9793999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9794999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9794999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9794999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9794999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9794999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9795999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9795999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9795999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9795999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9796999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9796999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9796999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9796999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9796999999999692, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9797999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9797999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9797999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9797999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9797999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9798999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9798999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9798999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9798999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9798999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9799999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9799999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9799999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9799999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9799999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9800999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9800999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9800999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9800999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9800999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9801999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9801999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9801999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9801999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9801999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9802999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9802999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9802999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9802999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9802999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9803999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9803999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9803999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9803999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9803999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9804999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9804999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9804999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9804999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9804999999999691, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.980599999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.980599999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.980599999999969, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9805999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9805999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9806999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9806999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9806999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9806999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9806999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9807999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9807999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9807999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9807999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9807999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9808999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9808999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9808999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9808999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9808999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9809999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9809999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9809999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9809999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9809999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9810999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9810999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9810999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9810999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9810999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9811999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9811999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9811999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9811999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9811999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9812999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9812999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9812999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9812999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9812999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9813999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9813999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9813999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9813999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9813999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9814999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9814999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9814999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9814999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9814999999999969, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9815999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9815999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9815999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9815999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9815999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9816999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9816999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9816999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9816999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9816999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9817999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9817999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9817999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9817999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9817999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9818999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9818999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9818999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9818999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9818999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9819999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9819999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9819999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9819999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9819999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9820999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9820999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9820999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9820999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9820999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9821999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9821999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9821999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9821999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9821999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9822999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9822999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9822999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9822999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9822999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9823999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9823999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9823999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9823999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9823999999999689, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9824999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9824999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9824999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9824999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9824999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9825999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9825999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9825999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9825999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9825999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9826999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9826999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9826999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9826999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9826999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9827999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9827999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9827999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9827999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9827999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9828999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9828999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9828999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9828999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9828999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9829999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9829999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9829999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9829999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9829999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9830999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9830999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9830999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9830999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9830999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9831999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9831999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9831999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9831999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9831999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9832999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9832999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9832999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9832999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9832999999999688, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9833999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9833999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9833999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9833999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9833999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9834999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9834999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9834999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9834999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9834999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9835999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9835999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9835999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9835999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9835999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9836999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9836999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9836999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9836999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9836999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9837999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9837999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9837999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9837999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9837999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9838999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9838999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9838999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9838999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9838999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9839999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9839999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9839999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9839999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9839999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9840999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9840999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9840999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9840999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9840999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9841999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9841999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9841999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9841999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9841999999999687, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9842999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9842999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9842999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9842999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9842999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9843999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9843999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9843999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9843999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9844999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9844999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9844999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9844999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9844999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9845999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9845999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9845999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9845999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9845999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9846999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9846999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9846999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9846999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9846999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9847999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9847999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9847999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9847999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9847999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9848999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9848999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9848999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9848999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9848999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9849999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9849999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9849999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9849999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9849999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9850999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9850999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9850999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9850999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9850999999999686, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9851999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9851999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9851999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9851999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9851999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9852999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9852999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9852999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9852999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9852999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9853999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9853999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9853999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9853999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9853999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9854999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9854999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9854999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9854999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9854999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9855999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9855999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9855999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9855999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9855999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9856999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9856999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9856999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9856999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9856999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9857999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9857999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9857999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9857999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9857999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9858999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9858999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9858999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9858999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9858999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9859999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9859999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9859999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9859999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9859999999999685, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9860999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9860999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9860999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9860999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9860999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9861999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9861999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9861999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9861999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9861999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9862999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9862999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9862999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9862999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9862999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9863999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9863999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9863999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9863999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9863999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9864999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9864999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9864999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9864999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9864999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9865999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9865999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9865999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9865999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9865999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9866999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9866999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9866999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9866999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9866999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9867999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9867999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9867999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9867999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9868999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9868999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9868999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9868999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9868999999999684, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9869999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9869999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9869999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9869999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9869999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9870999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9870999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9870999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9870999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9870999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9871999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9871999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9871999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9871999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9871999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9872999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9872999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9872999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9872999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9872999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9873999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9873999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9873999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9873999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9873999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9874999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9874999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9874999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9874999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9874999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9875999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9875999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9875999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9875999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9875999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9876999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9876999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9876999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9876999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9876999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9877999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9877999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9877999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9877999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9877999999999683, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9878999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9878999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9878999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9878999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9878999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9879999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9879999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9879999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9879999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9879999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9879999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9880999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9880999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9880999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9880999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9880999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9881999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9881999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9881999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9881999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9881999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9882999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9882999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9882999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9882999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9882999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9883999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9883999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9883999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9883999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9883999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9884999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9884999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9884999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9884999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9884999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9885999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9885999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9885999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9885999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9885999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9886999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9886999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9886999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9886999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9886999999999682, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9887999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9887999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9887999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9887999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9887999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9888999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9888999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9888999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9888999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9888999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9889999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9889999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9889999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9889999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9890999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9890999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9890999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9890999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9890999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9891999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9891999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9891999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9891999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9892999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9892999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9892999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9892999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9892999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9893999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9893999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9893999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9893999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9893999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9894999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9894999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9894999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9894999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9894999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9895999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9895999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9895999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9895999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9895999999999681, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.989699999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.989699999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.989699999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.989699999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.989699999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.989799999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.989799999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.989799999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.989799999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.989799999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.989899999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.989899999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.989899999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.989899999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.989899999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.989999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.989999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.989999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.989999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.989999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.990099999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.990099999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.990099999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.990099999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.990099999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.990199999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.990199999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.990199999999968, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9901999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9901999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9902999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9902999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9902999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9902999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9902999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9903999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9903999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9903999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9903999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9903999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9904999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9904999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9904999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9904999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9904999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9905999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9905999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9905999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9905999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9905999999999968, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9906999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9906999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9906999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9906999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9906999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9907999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9907999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9907999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9907999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9907999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9908999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9908999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9908999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9908999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9908999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9909999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9909999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9909999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9909999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9909999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9910999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9910999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9910999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9910999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9910999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9911999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9911999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9911999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9911999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9911999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9912999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9912999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9912999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9912999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9912999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9913999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9913999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9913999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9913999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9913999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9914999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9914999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9914999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9914999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9914999999999679, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9915999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9915999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9915999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9915999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9915999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9916999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9916999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9916999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9916999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9916999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9917999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9917999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9917999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9917999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9917999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9918999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9918999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9918999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9918999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9918999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9919999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9919999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9919999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9919999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9919999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9920999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9920999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9920999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9920999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9920999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9921999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9921999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9921999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9921999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9921999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9922999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9922999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9922999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9922999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9922999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9923999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9923999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9923999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9923999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9923999999999678, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9924999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9924999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9924999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9924999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9924999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9925999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9925999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9925999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9925999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9925999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9926999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9926999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9926999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9926999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9927999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9927999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9927999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9927999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9927999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9928999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9928999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9928999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9928999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9928999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9929999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9929999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9929999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9929999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9929999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9930999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9930999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9930999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9930999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9930999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9931999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9931999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9931999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9931999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9931999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9932999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9932999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9932999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9932999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9932999999999677, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9933999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9933999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9933999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9933999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9933999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9934999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9934999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9934999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9934999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9934999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9935999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9935999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9935999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9935999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9935999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9936999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9936999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9936999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9936999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9936999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9937999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9937999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9937999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9937999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9937999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9938999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9938999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9938999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9938999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9938999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9939999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9939999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9939999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9939999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9940999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9940999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9940999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9940999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9940999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9941999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9941999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9941999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9941999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9941999999999676, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9942999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9942999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9942999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9942999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9942999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9943999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9943999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9943999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9943999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9943999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9944999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9944999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9944999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9944999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9944999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9945999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9945999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9945999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9945999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9945999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9946999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9946999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9946999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9946999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9946999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9947999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9947999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9947999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9947999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9947999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9948999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9948999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9948999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9948999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9948999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9949999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9949999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9949999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9949999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9949999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9950999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9950999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9950999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9950999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9950999999999675, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9951999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9951999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9951999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9951999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9951999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9952999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9952999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9952999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9952999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9952999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9953999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9953999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9953999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9953999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9953999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9954999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9954999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9954999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9954999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9954999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9955999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9955999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9955999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9955999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9955999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9956999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9956999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9956999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9956999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9956999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9957999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9957999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9957999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9957999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9957999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9958999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9958999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9958999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9958999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9958999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9959999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9959999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9959999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9959999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9959999999999674, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9960999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9960999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9960999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9960999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9960999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9961999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9961999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9961999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9961999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9961999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9962999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9962999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9962999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9962999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9962999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9963999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9963999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9963999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9963999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9964999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9964999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9964999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9964999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9964999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9965999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9965999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9965999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9965999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9965999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9966999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9966999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9966999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9966999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9966999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9967999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9967999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9967999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9967999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9967999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9968999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9968999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9968999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9968999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9968999999999673, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9969999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9969999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9969999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9969999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9969999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9970999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9970999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9970999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9970999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9970999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9971999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9971999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9971999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9971999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9971999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9972999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9972999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9972999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9972999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9972999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9973999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9973999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9973999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9973999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9973999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9974999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9974999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9974999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9974999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9975999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9975999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9975999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9975999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9975999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9976999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9976999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9976999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9976999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9976999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9977999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9977999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9977999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9977999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9977999999999672, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9978999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9978999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9978999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9978999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9978999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9979999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9979999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9979999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9979999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9979999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9980999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9980999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9980999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9980999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9980999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9981999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9981999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9981999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9981999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9981999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9982999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9982999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9982999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9982999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9982999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9983999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9983999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9983999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9983999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9983999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9984999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9984999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9984999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9984999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9984999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9985999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9985999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9985999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9985999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9985999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9986999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9986999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9986999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9986999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9986999999999671, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.998799999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.998799999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.998799999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.998799999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.998799999999967, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.998899999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.998899999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.998899999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.998899999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.998899999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.998999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.998999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.998999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.998999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.998999999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999099999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999099999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999099999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999099999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999099999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999199999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999199999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999199999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999199999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999299999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999299999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999299999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999299999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999299999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999399999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999399999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999399999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999399999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999399999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999499999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999499999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999499999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999499999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999499999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999599999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999599999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999599999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999599999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999599999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999699999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999699999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999699999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999699999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.999699999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9997999999999669, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9997999999999669, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9997999999999669, penalty=l2, solver=lbfgs; total time=   0.0s
```

```
[CV] END ...C=0.9997999999999669, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9997999999999669, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9998999999999669, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9998999999999669, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9998999999999669, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9998999999999669, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9998999999999669, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9999999999999669, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9999999999999669, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9999999999999669, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9999999999999669, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=0.9999999999999669, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=1.000099999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=1.000099999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=1.000099999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=1.000099999999967, penalty=l2, solver=lbfgs; total time=   0.0s
[CV] END ...C=1.000099999999967, penalty=l2, solver=lbfgs; total time=   0.0s
Best Parameters: {'C': np.float64(0.9972999999999672), 'penalty': 'l2',
'solver': 'lbfgs'}
```

[8]:
```
from sklearn.model_selection import cross_validate
from sklearn.metrics import make_scorer, accuracy_score, recall_score,␣
 ↪precision_score, f1_score


scoring = {
    'accuracy': make_scorer(accuracy_score),
    'precision': make_scorer(precision_score),
    'recall': make_scorer(recall_score),
    'f1': make_scorer(f1_score)
}

# use cross_validate to do k_fold cross validation
cv_results = cross_validate(best_model, X_train, Y_train, cv=kf,␣
 ↪scoring=scoring, return_train_score=False)


print(f"Average Accuracy: {cv_results['test_accuracy'].mean():.4f}")
print(f"Average Precision: {cv_results['test_precision'].mean():.4f}")
print(f"Average Recall: {cv_results['test_recall'].mean():.4f}")
print(f"Average F1-Score: {cv_results['test_f1'].mean():.4f}")
```

```
Average Accuracy: 0.9687
Average Precision: 0.9760
Average Recall: 0.9855
Average F1-Score: 0.9807
```

[9]:
```
# Initialize lists for FPR and TPR
tprs = []
mean_fpr = np.linspace(0, 1, 100)
roc_auc_list = []

for train_index, test_index in kf.split(X):

    X_train2, X_test2 = X.iloc[train_index], X.iloc[test_index]
    y_train2, y_test2 = Y.iloc[train_index], Y.iloc[test_index]

    model1.fit(X_train2, y_train2)

    # Predict probabilities for ROC curve
    y_test_proba = best_model.predict_proba(X_test2)[:, 1]

    # Compute ROC curve and AUC
    fpr, tpr, thresholds = roc_curve(y_test2, y_test_proba)
    roc_auc = auc(fpr, tpr)
    roc_auc_list.append(roc_auc)

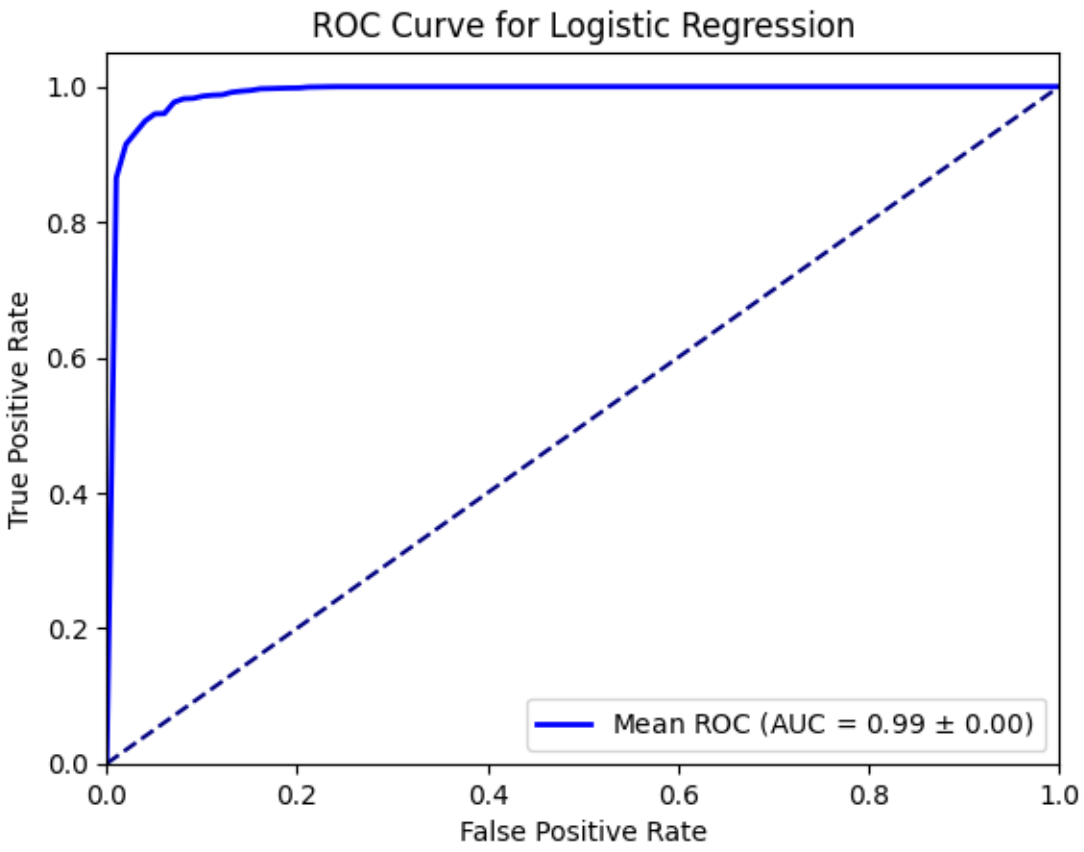
    # Interpolate TPR values for a common FPR range
    tprs.append(np.interp(mean_fpr, fpr, tpr))
    tprs[-1][0] = 0.0

# Calculate the mean TPR and its standard deviation
mean_tpr = np.mean(tprs, axis=0)
mean_tpr[-1] = 1.0
std_tpr = np.std(tprs, axis=0)

# Compute mean ROC AUC
mean_auc = np.mean(roc_auc_list)
std_auc = np.std(roc_auc_list)

# Plot the mean ROC curve
plt.figure()
plt.plot(mean_fpr, mean_tpr, color='b', label=r'Mean ROC (AUC = %0.2f $\pm$ %0.
 ↪2f)' % (mean_auc, std_auc), lw=2)
plt.plot([0, 1], [0, 1], color='navy', linestyle='--')

#plot
plt.xlim([0.0, 1.0])
plt.ylim([0.0, 1.05])
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curve for Logistic Regression')
plt.legend(loc='lower right')
plt.show()
```


ROC Curve for Logistic Regression
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Mean ROC (AUC = 0.99 ± 0.00)

DecisionTree(done)

December 7, 2024

1 Decision Tree

1.1 Data Loading

[ ]:
```
import numpy as np
import pandas as pd

from sklearn.model_selection import KFold
from sklearn.tree import DecisionTreeClassifier,plot_tree
from sklearn.metrics import accuracy_score,roc_curve,auc

import statsmodels.api as sm

import matplotlib.pyplot as plt
```

```
Matplotlib is building the font cache; this may take a moment.
```

[ ]:
```
df = pd.read_csv(r"C:\Users\13902\Downloads\Covid+Dataset.csv")
df1 = df.dropna()
df1.head()
```

[ ]:
```
  Breathing Problem Fever Dry Cough Sore throat Running Nose Asthma  \
0               Yes   Yes       Yes          Yes           Yes     No
1               Yes   Yes       Yes          Yes            No    Yes
2               Yes   Yes       Yes          Yes           Yes    Yes
3               Yes   Yes       Yes           No            No    Yes
4               Yes   Yes       Yes          Yes           Yes     No

  Chronic Lung Disease Headache Heart Disease Diabetes  ... Fatigue   \
0                   No       No            No      Yes  ...      Yes
1                  Yes      Yes            No       No  ...      Yes
2                  Yes      Yes            No      Yes  ...      Yes
3                   No       No           Yes      Yes  ...       No
4                  Yes      Yes           Yes      Yes  ...       No

  Gastrointestinal  Abroad travel Contact with COVID Patient  \
0              Yes            No                        Yes
1               No            No                         No
```

```
2               Yes           Yes                          No
3                No           Yes                          No
4               Yes            No                         Yes

  Attended Large Gathering Visited Public Exposed Places  \
0                       No                           Yes
1                      Yes                           Yes
2                       No                            No
3                      Yes                           Yes
4                       No                           Yes

  Family working in Public Exposed Places Wearing Masks  \
0                                     Yes            No
1                                      No            No
2                                      No            No
3                                      No            No
4                                      No            No

  Sanitization from Market COVID-19
0                       No      Yes
1                       No      Yes
2                       No      Yes
3                       No      Yes
4                       No      Yes

[5 rows x 21 columns]
```

1.2 Data Preparing

[ ]:
```
for i in df1.columns:
    df1[i] = df1[i].apply(lambda x: 1 if x == "Yes" else 0)
```

[ ]:
```
Y = df1["COVID-19"]
X = df1.drop("COVID-19", axis=1)
```

[ ]:
```
model = DecisionTreeClassifier()

kf = KFold(n_splits=5, shuffle=True, random_state=42)

accuracy_scores = []

for train_index, test_index in kf.split(X):

    X_train, X_test = X.iloc[train_index], X.iloc[test_index]
    Y_train, Y_test = Y.iloc[train_index], Y.iloc[test_index]

    model.fit(X_train, Y_train)
```

```
    predictions = model.predict(X_test)

    acc = accuracy_score(Y_test, predictions)
    accuracy_scores.append(acc)


print("Accuracy per fold:", accuracy_scores)

print("\nAverage accuracy:", np.mean(accuracy_scores))
```

```
Accuracy per fold: [0.984360625574977, 0.9825206991720331, 0.9862005519779209,
0.9797608095676172, 0.9769797421731123]

Average accuracy: 0.9819644856931321
```

[ ]:
```
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import KFold
from sklearn.metrics import accuracy_score
import numpy as np

model = DecisionTreeClassifier(
    criterion='gini',
    splitter='best',
    max_depth=None,
    min_samples_split=2,
    min_samples_leaf=1,
    min_weight_fraction_leaf=0.0,
    max_features=None,
    random_state=None,
    max_leaf_nodes=None,
    min_impurity_decrease=0.0,
    class_weight=None,
    ccp_alpha=0.0
)

kf = KFold(n_splits=5, shuffle=True, random_state=42)

accuracy_scores = []

for train_index, test_index in kf.split(X):

    X_train, X_test = X.iloc[train_index], X.iloc[test_index]
    Y_train, Y_test = Y.iloc[train_index], Y.iloc[test_index]


    model.fit(X_train, Y_train)
```

```
    predictions = model.predict(X_test)

    acc = accuracy_score(Y_test, predictions)
    accuracy_scores.append(acc)

print("Accuracy per fold:", accuracy_scores)
print("\nAverage accuracy:", np.mean(accuracy_scores))
```

```
Accuracy per fold: [0.984360625574977, 0.9825206991720331, 0.9862005519779209,
0.9797608095676172, 0.9769797421731123]

Average accuracy: 0.9819644856931321
```

1.3 Decision Tree

1.4 Optimizing Decision Tree Performance

[ ]:
```
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import GridSearchCV, KFold
from sklearn.metrics import accuracy_score
import numpy as np


model = DecisionTreeClassifier(criterion='entropy', random_state=42)


param_grid = {
    'min_samples_split': range(5, 20),
    'min_samples_leaf': range(5, 20),
    'max_depth': range(0,30),
    'max_features': ['auto', 'sqrt', 'log2'],
}

grid_search = GridSearchCV(estimator=model, param_grid=param_grid, cv=5,␣
 ↪scoring='accuracy')

grid_search.fit(X, Y)


best_params = grid_search.best_params_
best_score = grid_search.best_score_

print("Best Parameters:", best_params)
print("Best Cross-validation Accuracy:", best_score)

best_model = grid_search.best_estimator_
```

```
accuracy_scores = []

kf = KFold(n_splits=5, shuffle=True, random_state=42)
for train_index, test_index in kf.split(X):
    X_train, X_test = X.iloc[train_index], X.iloc[test_index]
    Y_train, Y_test = Y.iloc[train_index], Y.iloc[test_index]

    best_model.fit(X_train, Y_train)
    predictions = best_model.predict(X_test)

    acc = accuracy_score(Y_test, predictions)
    accuracy_scores.append(acc)

print("\nAccuracy per fold:", accuracy_scores)
print("\nAverage accuracy:", np.mean(accuracy_scores))
```

```
c:\Users\13902\AppData\Local\Programs\Python\Python312\Lib\site-
packages\sklearn\model_selection\_validation.py:540: FitFailedWarning:
36000 fits failed out of a total of 101250.
The score on these train-test partitions for these parameters will be set to
nan.
If these failures are not expected, you can try to debug them by setting
error_score='raise'.

Below are more details about the failures:
--------------------------------------------------------------------------------
3375 fits failed with the following error:
Traceback (most recent call last):
  File "c:\Users\13902\AppData\Local\Programs\Python\Python312\Lib\site-
packages\sklearn\model_selection\_validation.py", line 888, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
  File "c:\Users\13902\AppData\Local\Programs\Python\Python312\Lib\site-
packages\sklearn\base.py", line 1466, in wrapper
    estimator._validate_params()
  File "c:\Users\13902\AppData\Local\Programs\Python\Python312\Lib\site-
packages\sklearn\base.py", line 666, in _validate_params
    validate_parameter_constraints(
  File "c:\Users\13902\AppData\Local\Programs\Python\Python312\Lib\site-
packages\sklearn\utils\_param_validation.py", line 95, in
validate_parameter_constraints
    raise InvalidParameterError(
sklearn.utils._param_validation.InvalidParameterError: The 'max_depth' parameter
of DecisionTreeClassifier must be an int in the range [1, inf) or None. Got 0
instead.

--------------------------------------------------------------------------------
32625 fits failed with the following error:
```

```
Traceback (most recent call last):
  File "c:\Users\13902\AppData\Local\Programs\Python\Python312\Lib\site-
packages\sklearn\model_selection\_validation.py", line 888, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
  File "c:\Users\13902\AppData\Local\Programs\Python\Python312\Lib\site-
packages\sklearn\base.py", line 1466, in wrapper
    estimator._validate_params()
  File "c:\Users\13902\AppData\Local\Programs\Python\Python312\Lib\site-
packages\sklearn\base.py", line 666, in _validate_params
    validate_parameter_constraints(
  File "c:\Users\13902\AppData\Local\Programs\Python\Python312\Lib\site-
packages\sklearn\utils\_param_validation.py", line 95, in
validate_parameter_constraints
    raise InvalidParameterError(
sklearn.utils._param_validation.InvalidParameterError: The 'max_features'
parameter of DecisionTreeClassifier must be an int in the range [1, inf), a
float in the range (0.0, 1.0], a str among {'sqrt', 'log2'} or None. Got 'auto'
instead.

  warnings.warn(some_fits_failed_message, FitFailedWarning)
c:\Users\13902\AppData\Local\Programs\Python\Python312\Lib\site-
packages\numpy\ma\core.py:2881: RuntimeWarning: invalid value encountered in
cast
  _data = np.array(data, dtype=dtype, copy=copy,

Best Parameters: {'max_depth': 12, 'max_features': 'sqrt', 'min_samples_leaf':
7, 'min_samples_split': 5}
Best Cross-validation Accuracy: 0.905580432399647

Accuracy per fold: [0.9576816927322908, 0.9613615455381784, 0.9696412143514259,
0.9494020239190433, 0.9511970534069981]

Average accuracy: 0.9578567059895873

c:\Users\13902\AppData\Local\Programs\Python\Python312\Lib\site-
packages\sklearn\model_selection\_search.py:1103: UserWarning: One or more of
the test scores are non-finite: [       nan        nan        nan ... 0.85037349
0.85037349 0.85037349]
  warnings.warn(
```

[ ]:
```
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import GridSearchCV


clf = DecisionTreeClassifier(random_state=42)


param_grid = {
```

```
    'criterion': ['gini', 'entropy'],
    'max_depth': range(3, 20),
    'min_samples_split': range(2, 10),
    'min_samples_leaf': range(1, 5),
}


grid_search = GridSearchCV(
    estimator=clf,
    param_grid=param_grid,
    cv=5,
    scoring='f1',
    n_jobs=-1,
    verbose=1
)
grid_search.fit(X, Y)


print("optimum parameter", grid_search.best_params_)
print("optimum score:", grid_search.best_score_)
```

```
Fitting 5 folds for each of 1088 candidates, totalling 5440 fits
 {'criterion': 'entropy', 'max_depth': 9, 'min_samples_leaf': 1,
'min_samples_split': 7}
 0.8956699342097773
```

[ ]:
```
from sklearn.metrics import accuracy_score, f1_score, precision_score,␣
 ↪recall_score


best_params = grid_search.best_params_
model2 = DecisionTreeClassifier(**best_params, random_state=42)


kf = KFold(n_splits=5, shuffle=True, random_state=42)


accuracy_scores = []
f1_scores = []
precision_scores = []
recall_scores = []


for fold, (train_index, test_index) in enumerate(kf.split(X), start=1):

    X_train, X_test = X.iloc[train_index], X.iloc[test_index]
    Y_train, Y_test = Y.iloc[train_index], Y.iloc[test_index]
```

```
    model2.fit(X_train, Y_train)

    predictions = model2.predict(X_test)


    acc = accuracy_score(Y_test, predictions)
    f1 = f1_score(Y_test, predictions, average='weighted')
    precision = precision_score(Y_test, predictions, average='weighted')
    recall = recall_score(Y_test, predictions, average='weighted')


    accuracy_scores.append(acc)
    f1_scores.append(f1)
    precision_scores.append(precision)
    recall_scores.append(recall)

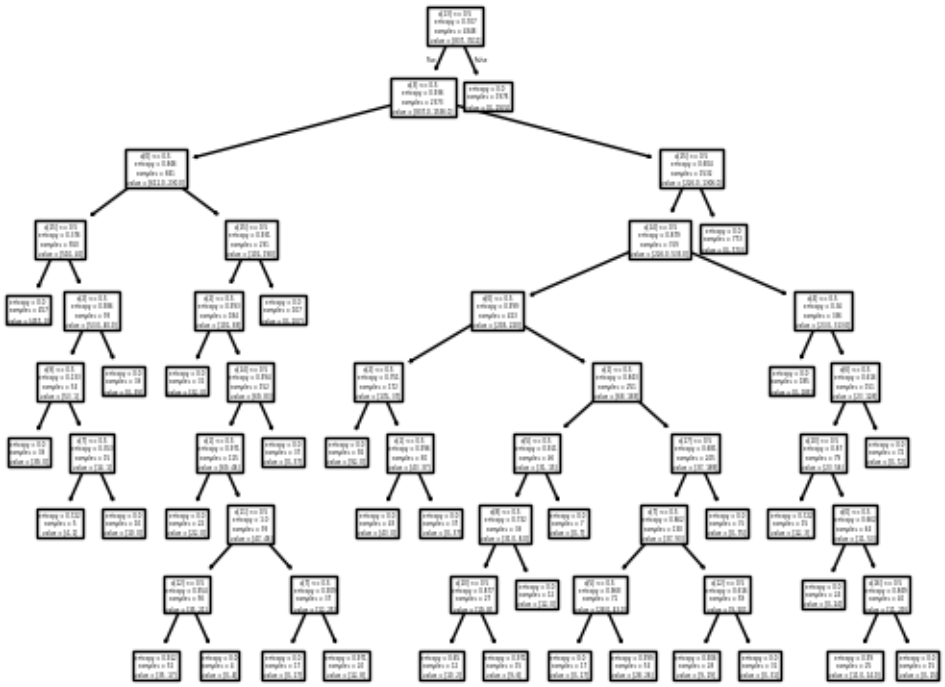

    print(f"Fold {fold}: Accuracy={acc:.4f}, F1={f1:.4f}, Precision={precision:.
↪4f}, Recall={recall:.4f}")


print("\nAverage Metrics:")
print(f"Accuracy: {np.mean(accuracy_scores):.4f}")
print(f"F1 Score: {np.mean(f1_scores):.4f}")
print(f"Precision: {np.mean(precision_scores):.4f}")
print(f"Recall: {np.mean(recall_scores):.4f}")
```

```
Fold 1: Accuracy=0.9724, F1=0.9726, Precision=0.9730, Recall=0.9724
Fold 2: Accuracy=0.9742, F1=0.9744, Precision=0.9747, Recall=0.9742
Fold 3: Accuracy=0.9807, F1=0.9809, Precision=0.9819, Recall=0.9807
Fold 4: Accuracy=0.9798, F1=0.9801, Precision=0.9810, Recall=0.9798
Fold 5: Accuracy=0.9715, F1=0.9719, Precision=0.9730, Recall=0.9715

Average Metrics:
Accuracy: 0.9757
F1 Score: 0.9760
Precision: 0.9767
Recall: 0.9757
```

[ ]:
```
plot_tree(model2)
plt.show()
```



[ ]:
```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import roc_curve, auc
from sklearn.model_selection import KFold


model2 = DecisionTreeClassifier()


kf = KFold(n_splits=5, shuffle=True, random_state=42)


tprs = []
mean_fpr = np.linspace(0, 1, 100)


for train_index, test_index in kf.split(X):
    X_train, X_test = X.iloc[train_index], X.iloc[test_index]
    y_train, y_test = Y.iloc[train_index], Y.iloc[test_index]


    model2.fit(X_train, y_train)
```

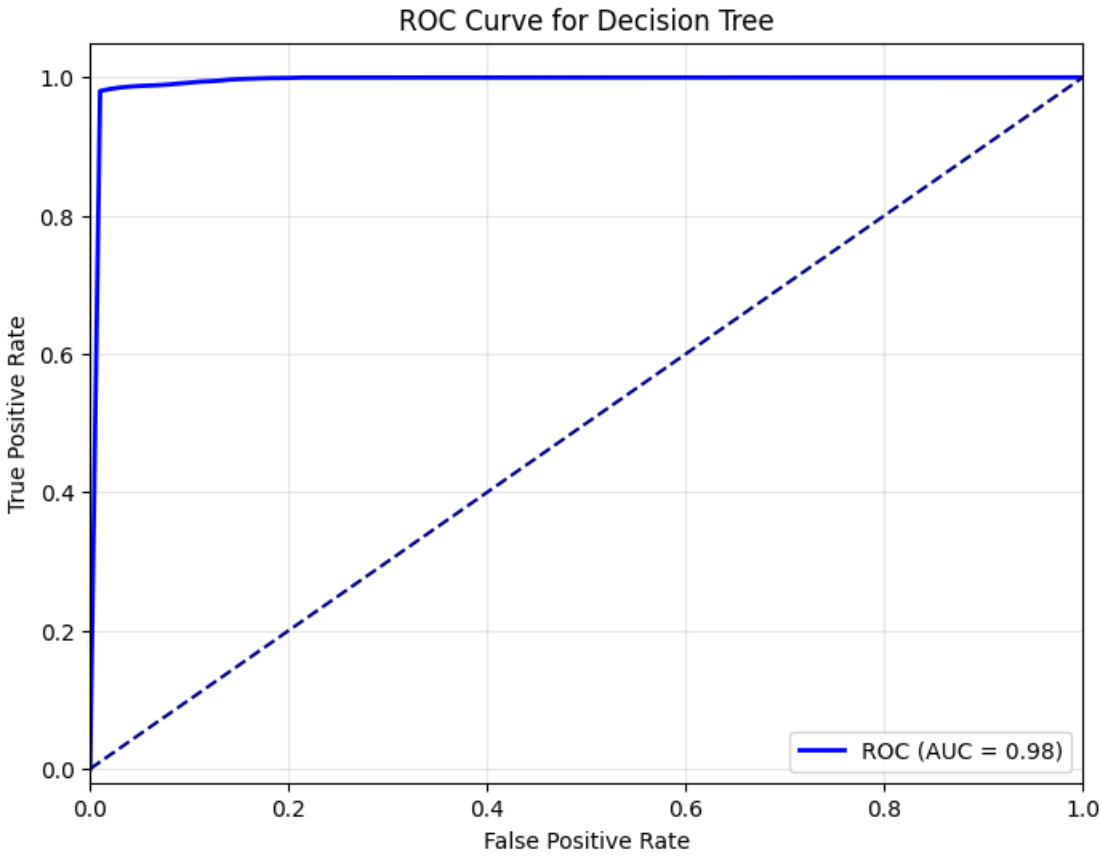
```
    y_proba = model2.predict_proba(X_test)[:, 1]


    fpr, tpr, _ = roc_curve(y_test, y_proba)
    tprs.append(np.interp(mean_fpr, fpr, tpr))
    tprs[-1][0] = 0.0

mean_tpr = np.mean(tprs, axis=0)
mean_tpr[-1] = 1.0


plt.figure(figsize=(8, 6))
plt.plot(mean_fpr, mean_tpr, color='b', label='ROC (AUC = 0.98)', lw=2)  # AUC␣
↪= 0.98
plt.plot([0, 1], [0, 1], color='navy', linestyle='--')


plt.xlim([0.0, 1.0])
plt.ylim([-0.02, 1.05])
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curve for Decision Tree')
plt.legend(loc='lower right')
plt.grid(alpha=0.3)
plt.show()
```


ROC Curve for Decision Tree
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
ROC (AUC = 0.98)

SVM(done)

December 7, 2024

0.1 Support Vector Machine

[1]:
```
import numpy as np
import pandas as pd

from sklearn import svm
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score,␣
↪recall_score, f1_score, roc_curve, auc
import matplotlib.pyplot as plt

from sklearn.model_selection import cross_val_score

from sklearn.model_selection import KFold,GridSearchCV
```

0.1.1 Data Preparing

[2]:
```
df = pd.read_csv('Covid Dataset.csv')
df1 = df.dropna()
```

[3]:
```
for i in df1.columns:
    df1[i] = df1[i].apply(lambda x: 1 if x == 'Yes' else 0)
df1
```

[3]:

	Breathing Problem	Fever	Dry Cough	Sore throat	Running Nose	Asthma	\
0	1	1	1	1	1	0	
1	1	1	1	1	0	1	
2	1	1	1	1	1	1	
3	1	1	1	0	0	1	
4	1	1	1	1	1	0	
...	...	...	...	...	...	...	
5429	1	1	0	1	1	1	
5430	1	1	1	0	1	1	
5431	1	1	1	0	0	0	
5432	1	1	1	0	1	0	
5433	1	1	1	0	1	1	

	Chronic Lung Disease	Headache	Heart Disease	Diabetes	...	Fatigue	\

```
0                       0         0              0         1  ...         1
1                       1         1              0         0  ...         1
2                       1         1              0         1  ...         1
3                       0         0              1         1  ...         0
4                       1         1              1         1  ...         0
...                   ...       ...            ...       ...  ...       ...
5429                    1         0              0         0  ...         1
5430                    0         1              0         1  ...         1
5431                    0         0              1         0  ...         0
5432                    0         1              1         0  ...         0
5433                    0         1              0         1  ...         1

      Gastrointestinal   Abroad travel  Contact with COVID Patient  \
0                     1              0                           1
1                     0              0                           0
2                     1              1                           0
3                     0              1                           0
4                     1              0                           1
...                 ...            ...                         ...
5429                  1              0                           0
5430                  0              0                           0
5431                  0              0                           0
5432                  0              0                           0
5433                  0              0                           0

      Attended Large Gathering  Visited Public Exposed Places  \
0                            0                              1
1                            1                              1
2                            0                              0
3                            1                              1
4                            0                              1
...                        ...                            ...
5429                         0                              0
5430                         0                              0
5431                         0                              0
5432                         0                              0
5433                         0                              0

      Family working in Public Exposed Places  Wearing Masks  \
0                                           1              0
1                                           0              0
2                                           0              0
3                                           0              0
4                                           0              0
...                                       ...            ...
5429                                        0              0
5430                                        0              0
```

```
5431                                    0              0
5432                                    0              0
5433                                    0              0

      Sanitization from Market  COVID-19
0                            0         1
1                            0         1
2                            0         1
3                            0         1
4                            0         1
...                        ...       ...
5429                         0         1
5430                         0         1
5431                         0         0
5432                         0         0
5433                         0         0

[5434 rows x 21 columns]
```

[4]:
```
Y = df1['COVID-19']
X = df1.drop('COVID-19', axis=1)
```

0.1.2 Modeling

[5]:
```
model1 = svm.SVC(kernel='linear')
```

Evaluation Metrics

[6]:
```
kf = KFold(n_splits=5, shuffle=True, random_state=42)

accuracy_scores = []
precision_scores = []
recall_scores = []
f1_scores = []

for train_index, test_index in kf.split(X):

    X_train1, X_test1 = X.iloc[train_index], X.iloc[test_index]
    Y_train1, Y_test1 = Y.iloc[train_index], Y.iloc[test_index]

    model1.fit(X_train1, Y_train1)
    predictions = model1.predict(X_test1)

    acc = accuracy_score(Y_test1, predictions)
    precision = precision_score(Y_test1, predictions)
    recall = recall_score(Y_test1, predictions)
    f1 = f1_score(Y_test1, predictions)
```

```
    accuracy_scores.append(acc)
    precision_scores.append(precision)
    recall_scores.append(recall)
    f1_scores.append(f1)

print("Accuracy per fold:", accuracy_scores)
print("Precision per fold:", precision_scores)
print("Recall per fold:", recall_scores)
print("F1-Score per fold:", f1_scores)

print("\nAverage Accuracy:", np.mean(accuracy_scores))
print("Average Precision:", np.mean(precision_scores))
print("Average Recall:", np.mean(recall_scores))
print("Average F1-Score:", np.mean(f1_scores))
```

```
Accuracy per fold: [0.9733210671573137, 0.9558417663293468, 0.9760809567617296,
0.9724011039558418, 0.9594843462246777]
Precision per fold: [np.float64(0.968236582694414),
np.float64(0.9486338797814208), np.float64(0.9849710982658959),
np.float64(0.9778516057585825), np.float64(0.9519650655021834)]
Recall per fold: [np.float64(1.0), np.float64(0.998849252013809),
np.float64(0.9849710982658959), np.float64(0.9888017917133258), np.float64(1.0)]
F1-Score per fold: [np.float64(0.9838619922092376),
np.float64(0.9730941704035875), np.float64(0.9849710982658959),
np.float64(0.9832962138084632), np.float64(0.9753914988814317)]

Average Accuracy: 0.9674258480857819
Average Precision: 0.9663316464004993
Average Recall: 0.9945244283986062
Average F1-Score: 0.9801229947137232
```

0.1.3 Improving Model

[7]:
```
from sklearn.model_selection import GridSearchCV


param_grid = {
    'C': np.arange(0.1, 10.1, 0.1),  # C controls the trade-off between smooth␣
 ↪decision boundary and classifying training points correctly. Smaller values␣
 ↪allow more margin violations.
    'gamma': [1, 0.1, 0.01],        # Gamma defines the influence of a single␣
 ↪training example. A higher value leads to overfitting, while a smaller value␣
 ↪increases generalization.
    'kernel': ['rbf', 'linear']     # Kernel specifies the type of decision␣
 ↪boundary: 'rbf' for non-linear and 'linear' for a straight-line boundary.
}
```

```
kf = KFold(n_splits=5, shuffle=True, random_state=42)


grid = GridSearchCV(svm.SVC(), param_grid, cv=kf, verbose=2)

grid.fit(X_train1, Y_train1)


print('Optimal parameters:', grid.best_params_)
```

```
Fitting 5 folds for each of 600 candidates, totalling 3000 fits
[CV] END ...C=0.1, gamma=1, kernel=rbf; total time=   0.2s
[CV] END ...C=0.1, gamma=1, kernel=rbf; total time=   0.1s
[CV] END ...C=0.1, gamma=1, kernel=rbf; total time=   0.1s
[CV] END ...C=0.1, gamma=1, kernel=rbf; total time=   0.1s
[CV] END ...C=0.1, gamma=1, kernel=rbf; total time=   0.1s
[CV] END ...C=0.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.1, gamma=0.1, kernel=rbf; total time=   0.1s
[CV] END ...C=0.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.1, gamma=0.1, kernel=rbf; total time=   0.1s
[CV] END ...C=0.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.1, gamma=0.01, kernel=rbf; total time=   0.1s
[CV] END ...C=0.1, gamma=0.01, kernel=rbf; total time=   0.1s
[CV] END ...C=0.1, gamma=0.01, kernel=rbf; total time=   0.1s
[CV] END ...C=0.1, gamma=0.01, kernel=rbf; total time=   0.1s
[CV] END ...C=0.1, gamma=0.01, kernel=rbf; total time=   0.1s
[CV] END ...C=0.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.2, gamma=1, kernel=rbf; total time=   0.1s
[CV] END ...C=0.2, gamma=1, kernel=rbf; total time=   0.1s
[CV] END ...C=0.2, gamma=1, kernel=rbf; total time=   0.1s
[CV] END ...C=0.2, gamma=1, kernel=rbf; total time=   0.1s
[CV] END ...C=0.2, gamma=1, kernel=rbf; total time=   0.1s
[CV] END ...C=0.2, gamma=1, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=0.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.2, gamma=0.01, kernel=rbf; total time=   0.1s
[CV] END ...C=0.2, gamma=0.01, kernel=rbf; total time=   0.1s
[CV] END ...C=0.2, gamma=0.01, kernel=rbf; total time=   0.1s
[CV] END ...C=0.2, gamma=0.01, kernel=rbf; total time=   0.1s
[CV] END ...C=0.2, gamma=0.01, kernel=rbf; total time=   0.1s
[CV] END ...C=0.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.30000000000000004, gamma=1, kernel=rbf; total time=   0.1s
[CV] END ...C=0.30000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.30000000000000004, gamma=1, kernel=rbf; total time=   0.1s
[CV] END ...C=0.30000000000000004, gamma=1, kernel=rbf; total time=   0.1s
[CV] END ...C=0.30000000000000004, gamma=1, kernel=rbf; total time=   0.1s
[CV] END ...C=0.30000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.30000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.30000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.30000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.30000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.30000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.30000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.30000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.30000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.30000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.30000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.30000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.30000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.30000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.30000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.30000000000000004, gamma=0.01, kernel=rbf; total time=   0.1s
[CV] END ...C=0.30000000000000004, gamma=0.01, kernel=rbf; total time=   0.1s
[CV] END ...C=0.30000000000000004, gamma=0.01, kernel=rbf; total time=   0.1s
[CV] END ...C=0.30000000000000004, gamma=0.01, kernel=rbf; total time=   0.1s
```

```
[CV] END ...C=0.30000000000000004, gamma=0.01, kernel=rbf; total time=   0.1s
[CV] END ...C=0.30000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.30000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.30000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.30000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.30000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.4, gamma=0.01, kernel=rbf; total time=   0.1s
[CV] END ...C=0.4, gamma=0.01, kernel=rbf; total time=   0.1s
[CV] END ...C=0.4, gamma=0.01, kernel=rbf; total time=   0.1s
[CV] END ...C=0.4, gamma=0.01, kernel=rbf; total time=   0.1s
[CV] END ...C=0.4, gamma=0.01, kernel=rbf; total time=   0.1s
[CV] END ...C=0.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.5, gamma=0.1, kernel=rbf; total time=   0.0s
```

```
[CV] END ...C=0.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.5, gamma=0.01, kernel=rbf; total time=   0.1s
[CV] END ...C=0.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=0.5, gamma=0.01, kernel=rbf; total time=   0.1s
[CV] END ...C=0.5, gamma=0.01, kernel=rbf; total time=   0.1s
[CV] END ...C=0.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=0.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.6, gamma=0.01, kernel=rbf; total time=   0.1s
[CV] END ...C=0.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=0.6, gamma=0.01, kernel=rbf; total time=   0.1s
[CV] END ...C=0.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=0.6, gamma=0.01, kernel=rbf; total time=   0.1s
[CV] END ...C=0.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.6, gamma=0.01, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=0.7000000000000001, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=0.01, kernel=rbf; total time=   0.1s
[CV] END ...C=0.7000000000000001, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.7000000000000001, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.8, gamma=0.1, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=0.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=0.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=0.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=0.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=0.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=0.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=0.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=0.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=0.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=0.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=0.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=0.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=0.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=0.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=0.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.0, gamma=1, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=1.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.1, gamma=0.01, kernel=rbf; total time=   0.0s
```

```
[CV] END ...C=1.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.2000000000000002, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
```

```
[CV] END ...C=1.3000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.3000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.4000000000000001, gamma=0.01, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=1.5000000000000002, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.5000000000000002, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.6, gamma=0.1, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=1.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.7000000000000002, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=1, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=1.8000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.8000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=0.01, kernel=rbf; total time=   0.0s
```

```
[CV] END ...C=1.9000000000000001, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=1.9000000000000001, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.1, gamma=0.1, kernel=rbf; total time=   0.0s
```

```
[CV] END ...C=2.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.2, gamma=0.01, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=2.3000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.3000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=2.4000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.4000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.5000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.6, gamma=1, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=2.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.7, gamma=0.01, kernel=rbf; total time=   0.0s
```

```
[CV] END ...C=2.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.8000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
```

```
[CV] END ...C=2.9000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=2.9000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.0000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=3.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.2, gamma=0.1, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=3.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.3000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=1, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=3.4000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.4000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
```

```
[CV] END ...C=3.5000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.5000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.7, gamma=0.1, kernel=rbf; total time=   0.0s
```

```
[CV] END ...C=3.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.8000000000000003, gamma=0.01, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=3.9000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=3.9000000000000004, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.0, gamma=0.1, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=4.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.2, gamma=1, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=4.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3, gamma=0.01, kernel=rbf; total time=   0.0s
```

```
[CV] END ...C=4.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.3999999999999995, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.5, gamma=0.1, kernel=rbf; total time=   0.0s
```

```
[CV] END ...C=4.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.6, gamma=0.01, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=4.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.8, gamma=0.1, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=4.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=4.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=4.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=4.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=4.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=4.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.0, gamma=1, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=5.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.1, gamma=0.01, kernel=rbf; total time=   0.0s
```

```
[CV] END ...C=5.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.3, gamma=0.1, kernel=rbf; total time=   0.0s
```

```
[CV] END ...C=5.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.4, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.4, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.4, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.4, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.4, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.4, gamma=0.01, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=5.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.6, gamma=0.1, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=5.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.8, gamma=1, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=5.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=5.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=5.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=5.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.9, gamma=0.01, kernel=rbf; total time=   0.0s
```

```
[CV] END ...C=5.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=5.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=5.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.1, gamma=0.1, kernel=rbf; total time=   0.0s
```

```
[CV] END ...C=6.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.2, gamma=0.01, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=6.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.4, gamma=0.1, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=6.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.4, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.4, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.4, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.4, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.4, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.6, gamma=1, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=6.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.7, gamma=0.01, kernel=rbf; total time=   0.0s
```

```
[CV] END ...C=6.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=6.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.9, gamma=0.1, kernel=rbf; total time=   0.0s
```

```
[CV] END ...C=6.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=6.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=6.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=6.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=6.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.0, gamma=0.01, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=7.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.2, gamma=0.1, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=7.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.4, gamma=1, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=7.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.4, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.4, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.4, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.4, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.4, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.5, gamma=0.01, kernel=rbf; total time=   0.0s
```

```
[CV] END ...C=7.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.7, gamma=0.1, kernel=rbf; total time=   0.0s
```

```
[CV] END ...C=7.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.8, gamma=0.01, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=7.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=7.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=7.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=7.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=7.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=7.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.0, gamma=0.1, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=8.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.2, gamma=1, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=8.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.3, gamma=0.01, kernel=rbf; total time=   0.0s
```

```
[CV] END ...C=8.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.4, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.4, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.4, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.4, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.4, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.5, gamma=0.1, kernel=rbf; total time=   0.0s
```

```
[CV] END ...C=8.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.6, gamma=0.01, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=8.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.7, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.7, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.7, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.7, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.7, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.7, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.8, gamma=0.1, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=8.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=8.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=8.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=8.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=8.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=8.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.0, gamma=1, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=9.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.1, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.1, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.1, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.1, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.1, gamma=0.01, kernel=rbf; total time=   0.0s
```

```
[CV] END ...C=9.1, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.1, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.2, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.2, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.2, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.2, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.2, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.2, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.3, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.3, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.3, gamma=0.1, kernel=rbf; total time=   0.0s
```

```
[CV] END ...C=9.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.3, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.3, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.3, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.3, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.4, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.4, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.4, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.4, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.4, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.4, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.4, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.4, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.4, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.4, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.4, gamma=0.01, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=9.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.5, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.5, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.5, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.5, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.5, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.5, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.6, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.6, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.6, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.6, gamma=0.1, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=9.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.6, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.6, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.6, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.700000000000001, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.8, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.8, gamma=1, kernel=linear; total time=   0.0s
```

```
[CV] END ...C=9.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.8, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.8, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.8, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.8, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.8, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.9, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.9, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=9.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.9, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=9.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.9, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=9.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.9, gamma=0.01, kernel=rbf; total time=   0.0s
```

```
[CV] END ...C=9.9, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=9.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=9.9, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=10.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=10.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=10.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=10.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=10.0, gamma=1, kernel=rbf; total time=   0.0s
[CV] END ...C=10.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=10.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=10.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=10.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=10.0, gamma=1, kernel=linear; total time=   0.0s
[CV] END ...C=10.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=10.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=10.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=10.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=10.0, gamma=0.1, kernel=rbf; total time=   0.0s
[CV] END ...C=10.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=10.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=10.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=10.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=10.0, gamma=0.1, kernel=linear; total time=   0.0s
[CV] END ...C=10.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=10.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=10.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=10.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=10.0, gamma=0.01, kernel=rbf; total time=   0.0s
[CV] END ...C=10.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=10.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=10.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=10.0, gamma=0.01, kernel=linear; total time=   0.0s
[CV] END ...C=10.0, gamma=0.01, kernel=linear; total time=   0.0s
Optimal parameters: {'C': np.float64(5.6), 'gamma': 0.1, 'kernel': 'rbf'}
```

[8]:
```
model2 = svm.SVC(kernel='rbf', gamma=0.1, C=5.6)
```

Evaluation Metrics

[9]:
```
kf = KFold(n_splits=5, shuffle=True, random_state=42)

accuracy_scores = []
precision_scores = []
recall_scores = []
f1_scores = []
```

```
for train_index, test_index in kf.split(X):

    X_train2, X_test2 = X.iloc[train_index], X.iloc[test_index]
    Y_train2, Y_test2 = Y.iloc[train_index], Y.iloc[test_index]

    model2.fit(X_train2, Y_train2)
    predictions = model2.predict(X_test2)

    acc = accuracy_score(Y_test2, predictions)
    precision = precision_score(Y_test2, predictions)
    recall = recall_score(Y_test2, predictions)
    f1 = f1_score(Y_test2, predictions)

    accuracy_scores.append(acc)
    precision_scores.append(precision)
    recall_scores.append(recall)
    f1_scores.append(f1)

print("Accuracy per fold:", accuracy_scores)
print("Precision per fold:", precision_scores)
print("Recall per fold:", recall_scores)
print("F1-Score per fold:", f1_scores)

print("\nAverage Accuracy:", np.mean(accuracy_scores))
print("Average Precision:", np.mean(precision_scores))
print("Average Recall:", np.mean(recall_scores))
print("Average F1-Score:", np.mean(f1_scores))
```

```
Accuracy per fold: [0.984360625574977, 0.9724011039558418, 0.9880404783808647,
0.9816007359705612, 0.9769797421731123]
Precision per fold: [np.float64(0.9942987457240593),
np.float64(0.9740112994350283), np.float64(0.994199535962877),
np.float64(0.98989898989899), np.float64(0.9895953757225433)]
Recall per fold: [np.float64(0.9864253393665159),
np.float64(0.9919447640966629), np.float64(0.9907514450867052),
np.float64(0.9876819708846585), np.float64(0.981651376146789)]
F1-Score per fold: [np.float64(0.9903463940942646),
np.float64(0.9828962371721779), np.float64(0.9924724956572091),
np.float64(0.9887892376681614), np.float64(0.9856073690270581)]

Average Accuracy: 0.9806765372110714
Average Precision: 0.9884007893486995
Average Recall: 0.9876909791162662
Average F1-Score: 0.9880223467237741
```

ROC curve

[10]:
```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import KFold
from sklearn.svm import SVC
from sklearn.metrics import roc_curve, auc


X = df1.drop('COVID-19', axis=1).values
Y = df1['COVID-19'].values


kf = KFold(n_splits=5, shuffle=True, random_state=42)


model = SVC(kernel='rbf', gamma=1, C=0.8, probability=True)

fpr_all = []
tpr_all = []
roc_auc_all = []

for train_index, test_index in kf.split(X):
    X_train, X_test = X[train_index], X[test_index]
    y_train, y_test = Y[train_index], Y[test_index]


    model.fit(X_train, y_train)


    y_score = model.predict_proba(X_test)[:, 1]


    fpr, tpr, _ = roc_curve(y_test, y_score)
    roc_auc = auc(fpr, tpr)

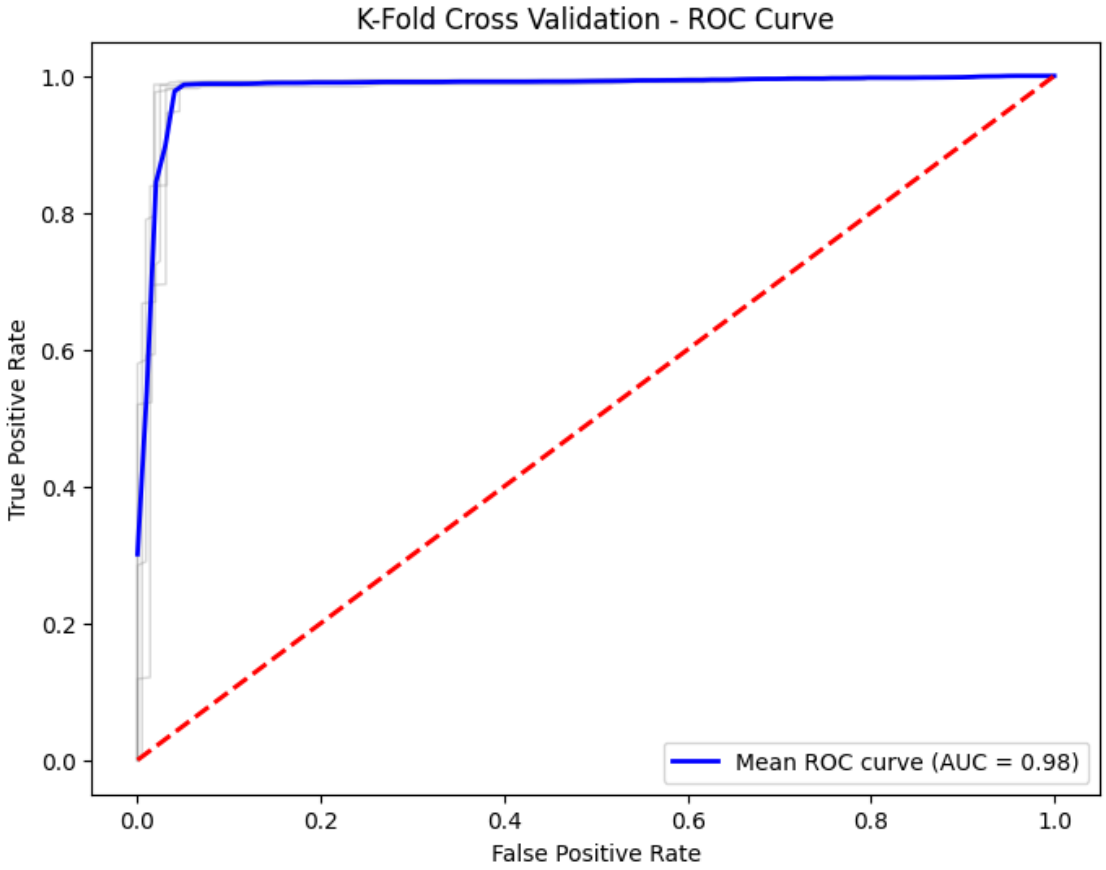
    fpr_all.append(fpr)
    tpr_all.append(tpr)
    roc_auc_all.append(roc_auc)


mean_fpr = np.linspace(0, 1, 100)
mean_tpr = np.mean([np.interp(mean_fpr, fpr, tpr) for fpr, tpr in zip(fpr_all,␣
 ↪tpr_all)], axis=0)
mean_roc_auc = np.mean(roc_auc_all)


plt.figure(figsize=(8, 6))
for i in range(len(fpr_all)):
```

```
    plt.plot(fpr_all[i], tpr_all[i], lw=1, alpha=0.3, color='gray')

plt.plot(mean_fpr, mean_tpr, color='b', label=f'Mean ROC curve (AUC =␣
 ↪{mean_roc_auc:.2f})', lw=2)
plt.plot([0, 1], [0, 1], linestyle='--', color='red', lw=2)
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('K-Fold Cross Validation - ROC Curve')
plt.legend(loc='lower right')
plt.show()
```



ahp

December 7, 2024

1 AHP_Threshold

[1]:
```
import numpy as np
import pandas as pd
from sklearn.naive_bayes import BernoulliNB
from sklearn.tree import DecisionTreeClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, precision_score, recall_score,␣
↪f1_score, roc_curve, auc
import matplotlib.pyplot as plt
```

[2]:
```
df = pd.read_csv("Covid Dataset.csv")
df1 = df.dropna()
df1.head()
```

[2]:

	Breathing Problem	Fever	Dry Cough	Sore throat	Running Nose	Asthma	\
0	Yes	Yes	Yes	Yes	Yes	No	
1	Yes	Yes	Yes	Yes	No	Yes	
2	Yes	Yes	Yes	Yes	Yes	Yes	
3	Yes	Yes	Yes	No	No	Yes	
4	Yes	Yes	Yes	Yes	Yes	No	

	Chronic Lung Disease	Headache	Heart Disease	Diabetes	...	Fatigue	\
0	No	No	No	Yes	...	Yes	
1	Yes	Yes	No	No	...	Yes	
2	Yes	Yes	No	Yes	...	Yes	
3	No	No	Yes	Yes	...	No	
4	Yes	Yes	Yes	Yes	...	No	

	Gastrointestinal	Abroad travel	Contact with COVID Patient	\
0	Yes	No	Yes	
1	No	No	No	
2	Yes	Yes	No	
3	No	Yes	No	
4	Yes	No	Yes	

```
  Attended Large Gathering Visited Public Exposed Places  \
0                       No                           Yes
1                      Yes                           Yes
2                       No                            No
3                      Yes                           Yes
4                       No                           Yes

  Family working in Public Exposed Places Wearing Masks  \
0                                     Yes            No
1                                      No            No
2                                      No            No
3                                      No            No
4                                      No            No

  Sanitization from Market COVID-19
0                       No      Yes
1                       No      Yes
2                       No      Yes
3                       No      Yes
4                       No      Yes

[5 rows x 21 columns]
```

[3]:
```
for i in df1.columns:
    df1[i] = df1[i].apply(lambda x: 1 if x == "Yes" else 0)

df1.head()
```

[3]:
```
   Breathing Problem  Fever  Dry Cough  Sore throat  Running Nose  Asthma  \
0                  1      1          1            1             1       0
1                  1      1          1            1             0       1
2                  1      1          1            1             1       1
3                  1      1          1            0             0       1
4                  1      1          1            1             1       0

   Chronic Lung Disease  Headache  Heart Disease  Diabetes  ...  Fatigue   \
0                     0         0              0         1  ...         1
1                     1         1              0         0  ...         1
2                     1         1              0         1  ...         1
3                     0         0              1         1  ...         0
4                     1         1              1         1  ...         0

   Gastrointestinal   Abroad travel  Contact with COVID Patient  \
0                  1              0                           1
1                  0              0                           0
2                  1              1                           0
```

```
3                   0              1                            0
4                   1              0                            1

   Attended Large Gathering  Visited Public Exposed Places  \
0                         0                              1
1                         1                              1
2                         0                              0
3                         1                              1
4                         0                              1

   Family working in Public Exposed Places  Wearing Masks  \
0                                        1              0
1                                        0              0
2                                        0              0
3                                        0              0
4                                        0              0

   Sanitization from Market  COVID-19
0                         0         1
1                         0         1
2                         0         1
3                         0         1
4                         0         1

[5 rows x 21 columns]
```

[4]:
```
Y = df1["COVID-19"]
X = df1.drop("COVID-19", axis=1)
```

[5]:
```
X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size=0.2,␣
↪random_state=42)
```

[6]:
```
model1 = BernoulliNB()
model2 = DecisionTreeClassifier()
model3 = LogisticRegression(C=0.9973, penalty='l2', solver='lbfgs',␣
↪max_iter=1000)
model4 = SVC(kernel='rbf', gamma=1, C=0.8, probability=True)

model1.fit(X_train, y_train)
model2.fit(X_train, y_train)
model3.fit(X_train, y_train)
model4.fit(X_train, y_train)
```

[6]:
```
SVC(C=0.8, gamma=1, probability=True)
```

[7]:
```
prob1 = model1.predict_proba(X_test)[:, 1]
prob2 = model2.predict_proba(X_test)[:, 1]
```

```
prob3 = model3.predict_proba(X_test)[:, 1]
prob4 = model4.predict_proba(X_test)[:, 1]
```

[8]:
```
weights = np.array([0.0550, 0.1178, 0.2634, 0.5638])

weighted_probs = (
    weights[0] * prob1 +
    weights[1] * prob2 +
    weights[2] * prob3 +
    weights[3] * prob4
)
```

[9]:
```
thresholds = np.arange(0, 1.01, 0.01)
best_threshold = 0
best_f1 = 0
f1_scores = []

for t in thresholds:

    final_predictions = (weighted_probs >= t).astype(int)

    f1 = f1_score(y_test, final_predictions)
    f1_scores.append(f1)

    if f1 > best_f1:
        best_f1 = f1
        best_threshold = t

print(f"Best threshold: {best_threshold}")
print(f"Corresponding F1 Score: {best_f1:.4f}")
```

```
Best threshold: 0.31
Corresponding F1 Score: 0.9903
```

[10]:
```
final_predictions = (weighted_probs >= best_threshold).astype(int)
```

[11]:
```
accuracy = accuracy_score(y_test, final_predictions)
precision = precision_score(y_test, final_predictions)
recall = recall_score(y_test, final_predictions)

print(f"Best threshold: {best_threshold:.2f}")
print(f"Accuracy: {accuracy:.4f}")
print(f"Precision: {precision:.4f}")
print(f"Recall: {recall:.4f}")
print(f"F1 Score: {best_f1:.4f}")
```

```
Best threshold: 0.31
Accuracy: 0.9844
```

```
Precision: 0.9943
Recall: 0.9864
F1 Score: 0.9903
```

cnn_code

December 7, 2024

System info - OS: Microsoft Windows 10 Enterprise LTSC 10.0.17763 N/A Build 17763 - CPU: Intel(R) Core(TM) i7-10875H @ 2.30G - GPU: NVIDIA GeForce RTX 2060 - RAM: 16.0 GB

0. Environment Version

```
torch.__version__ = 2.5.1+cu118
```

```
numpy.__version__ = '2.0.2'
```

```
matplotlib.__version__ = '3.9.2'
```

```
PIL.__version__ = '11.0.0'
```

1. Determine CUDA availability.

[1]:
```
import torch

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
device
```

[1]:
```
device(type='cuda')
```

2. Define data enhancement and pre-precess operations.

[2]:
```
from torchvision import transforms

transform = transforms.Compose([
    transforms.RandomHorizontalFlip(),  # rand_flip
    transforms.RandomRotation(10),      # rand_rotation
    transforms.ColorJitter(brightness=0.2, contrast=0.2),  # adj_img
    transforms.Resize((224, 224)),      # adj_size
    transforms.ToTensor(),              # to_Tensor
    transforms.Normalize(
        mean=[0.485, 0.456, 0.406],     # normalize
        std=[0.229, 0.224, 0.225]
    )
])
```

3. Load and pre-precess data

[3]:
```
import os
from PIL import Image
```

```
# file_path
normal_path = "data/NORMAL"
covid_path = "data/COVID19"


# func: load_image
def load_images_from_folder(folder, transform):
    images = []
    for filename in os.listdir(folder):
        if filename.endswith(".jpeg") or filename.endswith(".jpg"):
            img_path = os.path.join(folder, filename)
            img = Image.open(img_path).convert("RGB")  # to_RGB
            img = transform(img)  # pre-process
            images.append(img)
    return images


# load_image
normal_images = load_images_from_folder(normal_path, transform)
covid_images = load_images_from_folder(covid_path, transform)
```

4. Create label and combine data

[4]:
```
# create_label
normal_labels = [0] * len(normal_images)  # 0: normal
covid_labels = [1] * len(covid_images)  # 1: covid

# combine_image&label
all_images = normal_images + covid_images
all_labels = normal_labels + covid_labels

all_images[0], all_labels[0]
```

[4]:
```
(tensor([[[-1.8439, -1.8439, -1.8439,  ..., -1.8439, -1.8439, -1.8439],
          [-1.8439, -1.8439, -1.8439,  ..., -1.8439, -1.8439, -1.8439],
          [-1.8439, -1.8439, -1.8439,  ..., -1.8439, -1.8439, -1.8439],
          ...,
          [-1.8439, -1.8439, -1.8439,  ..., -1.8439, -1.8439, -1.8439],
          [-1.8439, -1.8439, -1.8439,  ..., -1.8439, -1.8439, -1.8439],
          [-1.8439, -1.8439, -1.8439,  ..., -1.8439, -1.8439, -1.8439]],

         [[-1.7556, -1.7556, -1.7556,  ..., -1.7556, -1.7556, -1.7556],
          [-1.7556, -1.7556, -1.7556,  ..., -1.7556, -1.7556, -1.7556],
          [-1.7556, -1.7556, -1.7556,  ..., -1.7556, -1.7556, -1.7556],
          ...,
          [-1.7556, -1.7556, -1.7556,  ..., -1.7556, -1.7556, -1.7556],
          [-1.7556, -1.7556, -1.7556,  ..., -1.7556, -1.7556, -1.7556],
```

```
          [-1.7556, -1.7556, -1.7556,  ..., -1.7556, -1.7556, -1.7556]],

         [[-1.5256, -1.5256, -1.5256,  ..., -1.5256, -1.5256, -1.5256],
          [-1.5256, -1.5256, -1.5256,  ..., -1.5256, -1.5256, -1.5256],
          [-1.5256, -1.5256, -1.5256,  ..., -1.5256, -1.5256, -1.5256],
          ...,
          [-1.5256, -1.5256, -1.5256,  ..., -1.5256, -1.5256, -1.5256],
          [-1.5256, -1.5256, -1.5256,  ..., -1.5256, -1.5256, -1.5256],
          [-1.5256, -1.5256, -1.5256,  ..., -1.5256, -1.5256, -1.5256]]]),
 0)
```

5. Prepare Dataset

[5]:
```
from torch.utils.data import TensorDataset
from sklearn.model_selection import KFold

# to_TensorDataset
dataset = TensorDataset(torch.stack(all_images), torch.tensor(all_labels))

# setup_KFold
k_folds = 5
kfold = KFold(n_splits=k_folds, shuffle=True, random_state=2330036105)
```

6. Create CNN (LeNet-5 Arch)

[6]:
```
import torch.nn as nn
import torch.nn.functional as F

class LeNet5(nn.Module):
    def __init__(self, num_classes=2):
        super(LeNet5, self).__init__()
        self.conv1 = nn.Conv2d(3, 6, kernel_size=5, stride=1, padding=2)  # Input: 3x224x224
        self.conv2 = nn.Conv2d(6, 16, kernel_size=5, stride=1)            # Input: 6x112x112
        self.pool = nn.MaxPool2d(kernel_size=2, stride=2)

        # Dynamically calculate the input size to the first fully connected layer
        dummy_input = torch.zeros(1, 3, 224, 224)  # Batch size 1, 3 channels, 224x224
        out = self.pool(F.relu(self.conv1(dummy_input)))
        out = self.pool(F.relu(self.conv2(out)))
        self.fc1_input_features = out.numel()

        self.fc1 = nn.Linear(self.fc1_input_features, 120)
        self.fc2 = nn.Linear(120, 84)
        self.fc3 = nn.Linear(84, num_classes)
```

```
    def forward(self, x):
        x = F.relu(self.conv1(x))  # Conv1 + ReLU
        x = F.max_pool2d(x, kernel_size=2, stride=2)  # Pooling1
        x = F.relu(self.conv2(x))  # Conv2 + ReLU
        x = F.max_pool2d(x, kernel_size=2, stride=2)  # Pooling2
        x = x.view(x.size(0), -1)  # Flatten
        x = F.relu(self.fc1(x))  # Fully Connected1 + ReLU
        x = F.relu(self.fc2(x))  # Fully Connected2 + ReLU
        x = self.fc3(x)  # Fully Connected3 (Output)
        return x
```

7. Draw the loss plot.

[7]:
```
import matplotlib.pyplot as plt
from torch.utils.data import DataLoader, TensorDataset, Subset

# fold
fold_train_losses = []
fold_val_losses = []

results = []
epoch_step = 10

for fold, (train_idx, val_idx) in enumerate(kfold.split(dataset)):
    print(f"FOLD {fold}")
    print("-" * 30)

    # Subset data for train and validation
    train_subset = Subset(dataset, train_idx)
    val_subset = Subset(dataset, val_idx)

    # Create DataLoaders
    train_loader = DataLoader(train_subset, batch_size=32, shuffle=True)
    val_loader = DataLoader(val_subset, batch_size=32, shuffle=False)

    # Initialize model, loss, and optimizer
    model = LeNet5(num_classes=2).to(device)
    criterion = nn.CrossEntropyLoss()
    optimizer = torch.optim.Adam(model.parameters(), lr=0.001)

    train_losses = []
    val_losses = []

    # Training loop
    for epoch in range(epoch_step):  # Train for 5 epochs per fold
        model.train()
        train_loss = 0.0
```

```
        for inputs, labels in train_loader:
            inputs, labels = inputs.to(device), labels.to(device)

            # Zero gradients
            optimizer.zero_grad()

            # Forward + backward + optimize
            outputs = model(inputs)
            loss = criterion(outputs, labels)
            loss.backward()
            optimizer.step()

            train_loss += loss.item()
        avg_train_loss = train_loss / len(train_loader)
        train_losses.append(avg_train_loss)
        print(f"Epoch {epoch+1}, Training Loss: {avg_train_loss}")

    # Validation loop
    model.eval()
    for epoch in range(epoch_step):
        val_loss = 0.0
        correct = 0
        total = 0
        with torch.no_grad():
            for inputs, labels in val_loader:
                inputs, labels = inputs.to(device), labels.to(device)
                outputs = model(inputs)
                loss = criterion(outputs, labels)
                val_loss += loss.item()

                _, predicted = torch.max(outputs, 1)
                total += labels.size(0)
                correct += (predicted == labels).sum().item()

        avg_val_loss = val_loss / len(val_loader)
        val_losses.append(avg_val_loss)
        val_accuracy = 100 * correct / total

    # store each fold val_loss
    fold_val_losses.append(val_losses)

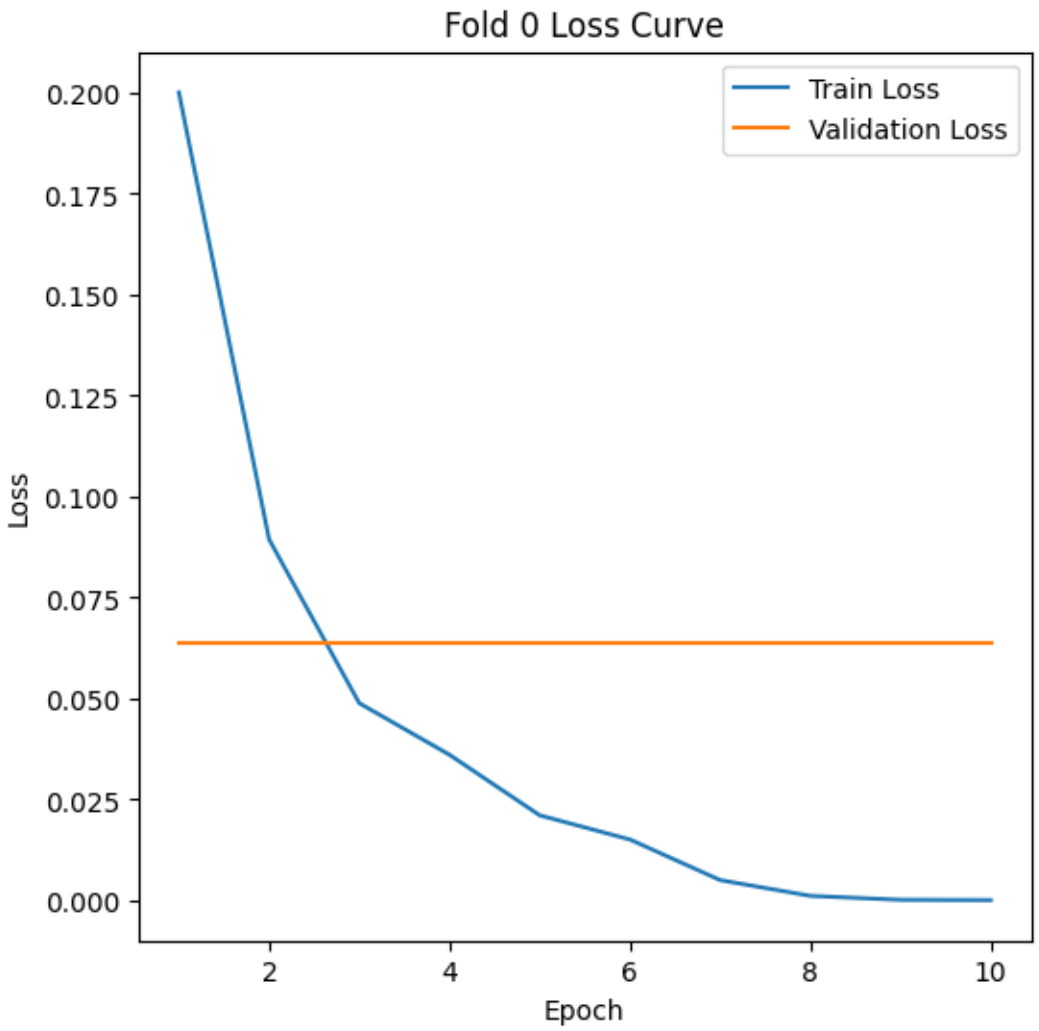

    # store the loss the this flod
    fold_train_losses.append(train_losses)
    fold_val_losses.append(val_losses)

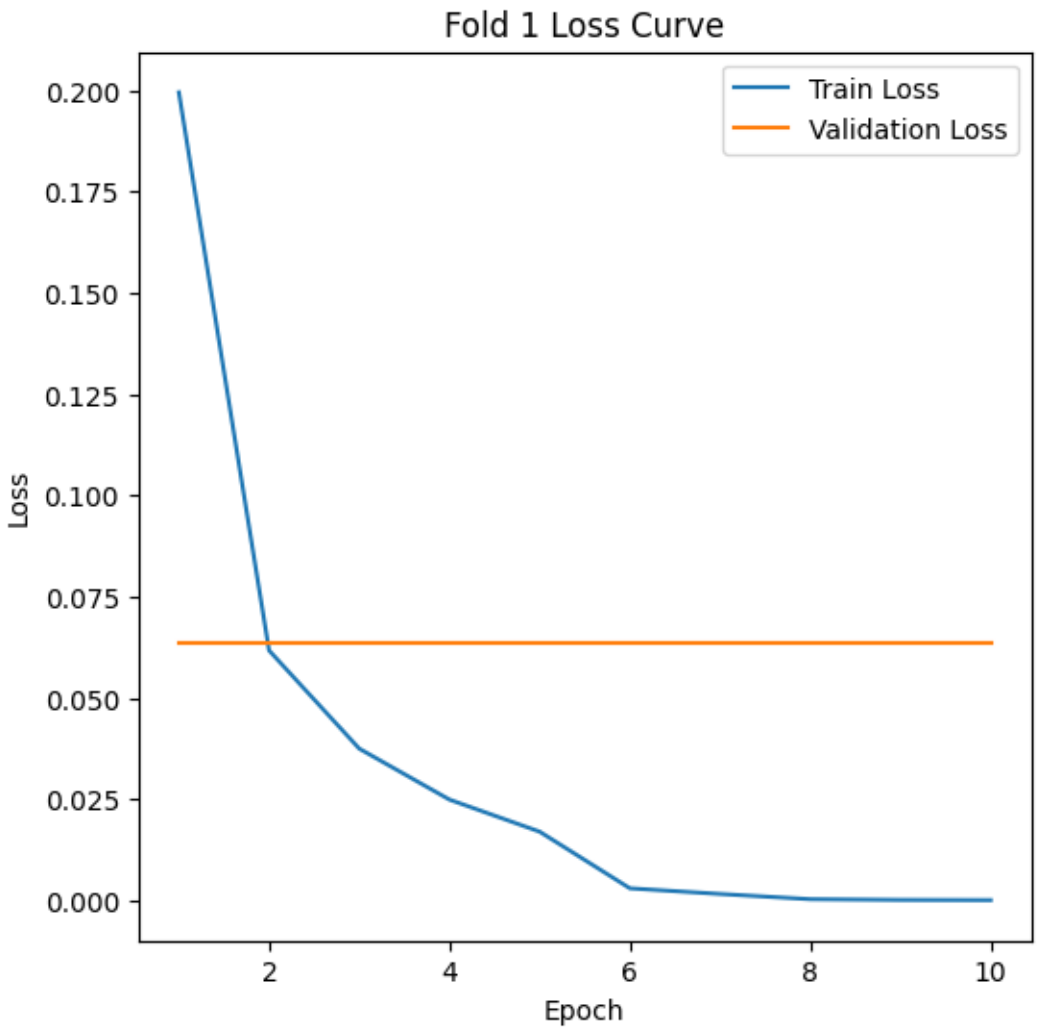
# draw plot
```

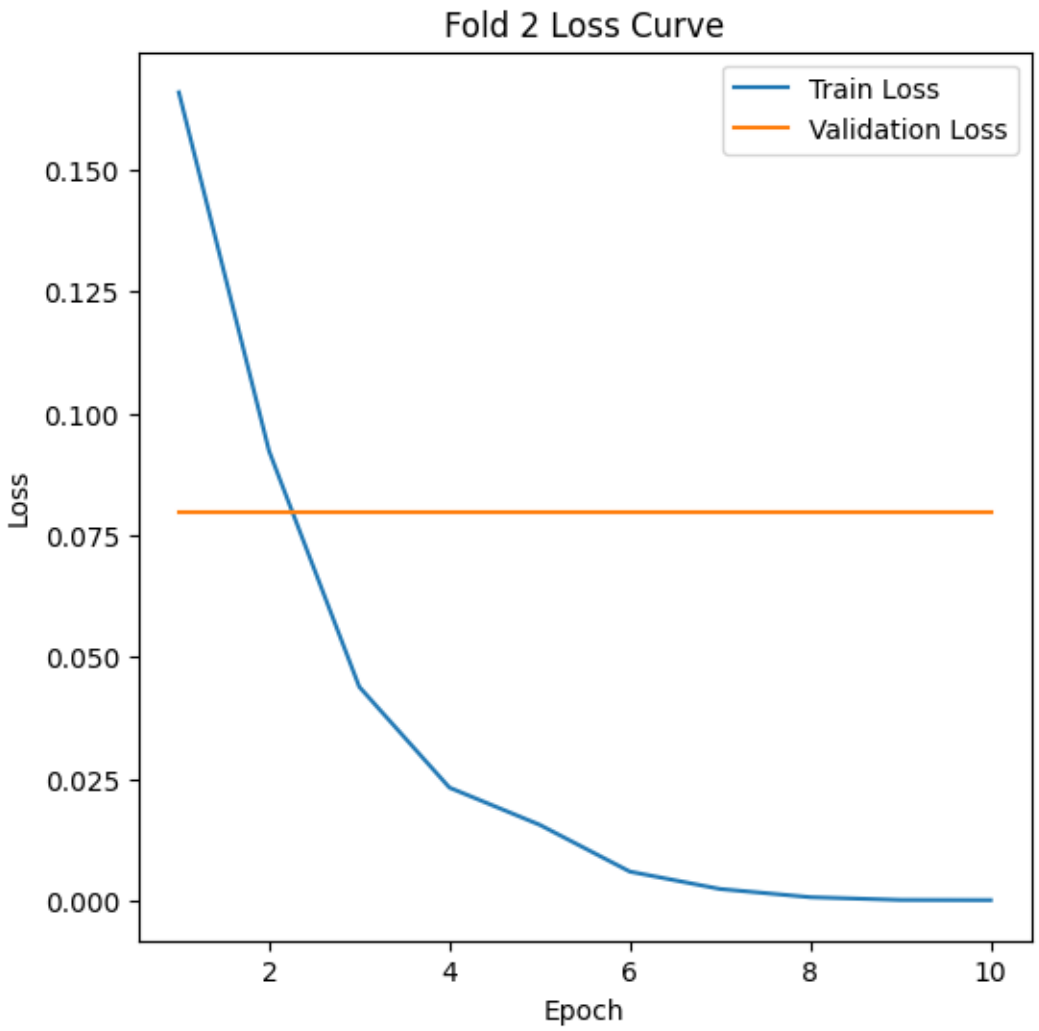
```
for fold in range(len(fold_train_losses)):
    plt.figure(figsize=(6, 6))
    epochs = len(fold_train_losses[fold])
    plt.plot(range(1, epochs + 1), fold_train_losses[fold], label="Train Loss")
    plt.plot(range(1, epochs + 1), fold_val_losses[fold], label="Validation␣
 ↪Loss")
    plt.title(f"Fold {fold} Loss Curve")
    plt.xlabel("Epoch")
    plt.ylabel("Loss")
    plt.legend()
    plt.savefig(f"Epoch_{fold}.pdf")
```

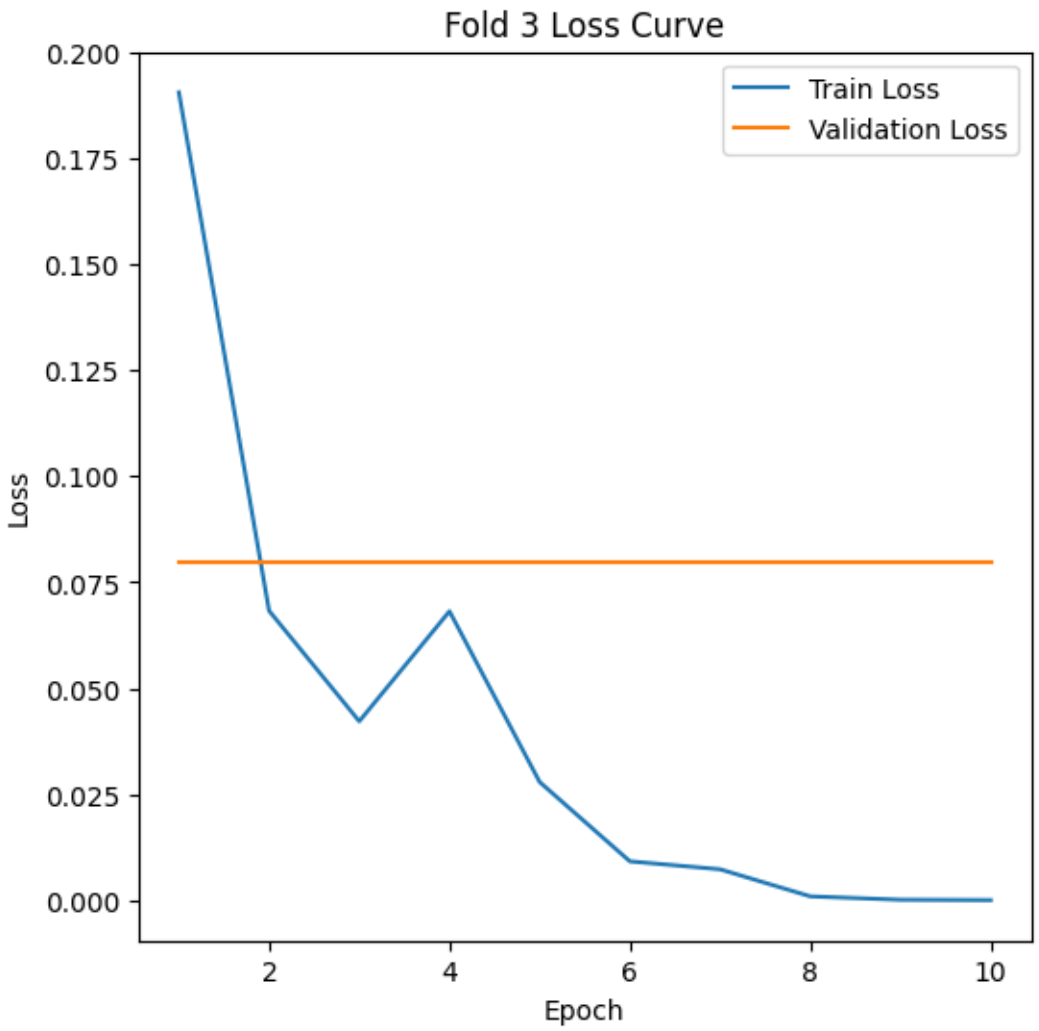
```
FOLD 0
------------------------------
Epoch 1, Training Loss: 0.20012495282140594
Epoch 2, Training Loss: 0.08944252435186947
Epoch 3, Training Loss: 0.04887148730954828
Epoch 4, Training Loss: 0.0360938016437339
Epoch 5, Training Loss: 0.02109737974953734
Epoch 6, Training Loss: 0.015111716234026028
Epoch 7, Training Loss: 0.005055969418730523
Epoch 8, Training Loss: 0.0011495215950600578
Epoch 9, Training Loss: 0.0001685494151102089
Epoch 10, Training Loss: 9.095180826921865e-05
FOLD 1
------------------------------
Epoch 1, Training Loss: 0.1994859877007979
Epoch 2, Training Loss: 0.06172237366748353
Epoch 3, Training Loss: 0.037483199023107
Epoch 4, Training Loss: 0.02490204432804064
Epoch 5, Training Loss: 0.016984422731588386
Epoch 6, Training Loss: 0.0030054471547676562
Epoch 7, Training Loss: 0.001609753061470824
Epoch 8, Training Loss: 0.00031907313575208754
Epoch 9, Training Loss: 0.00013915966571457998
Epoch 10, Training Loss: 8.570705077625428e-05
FOLD 2
------------------------------
Epoch 1, Training Loss: 0.16585468080032756
Epoch 2, Training Loss: 0.09224118550170075
Epoch 3, Training Loss: 0.043840883325578436
Epoch 4, Training Loss: 0.02315176557048862
Epoch 5, Training Loss: 0.015521704835043271
Epoch 6, Training Loss: 0.00591260446176252
Epoch 7, Training Loss: 0.00234501472767045
Epoch 8, Training Loss: 0.0006586071838901935
Epoch 9, Training Loss: 9.612204396742325e-05
Epoch 10, Training Loss: 5.662442657010329e-05
```

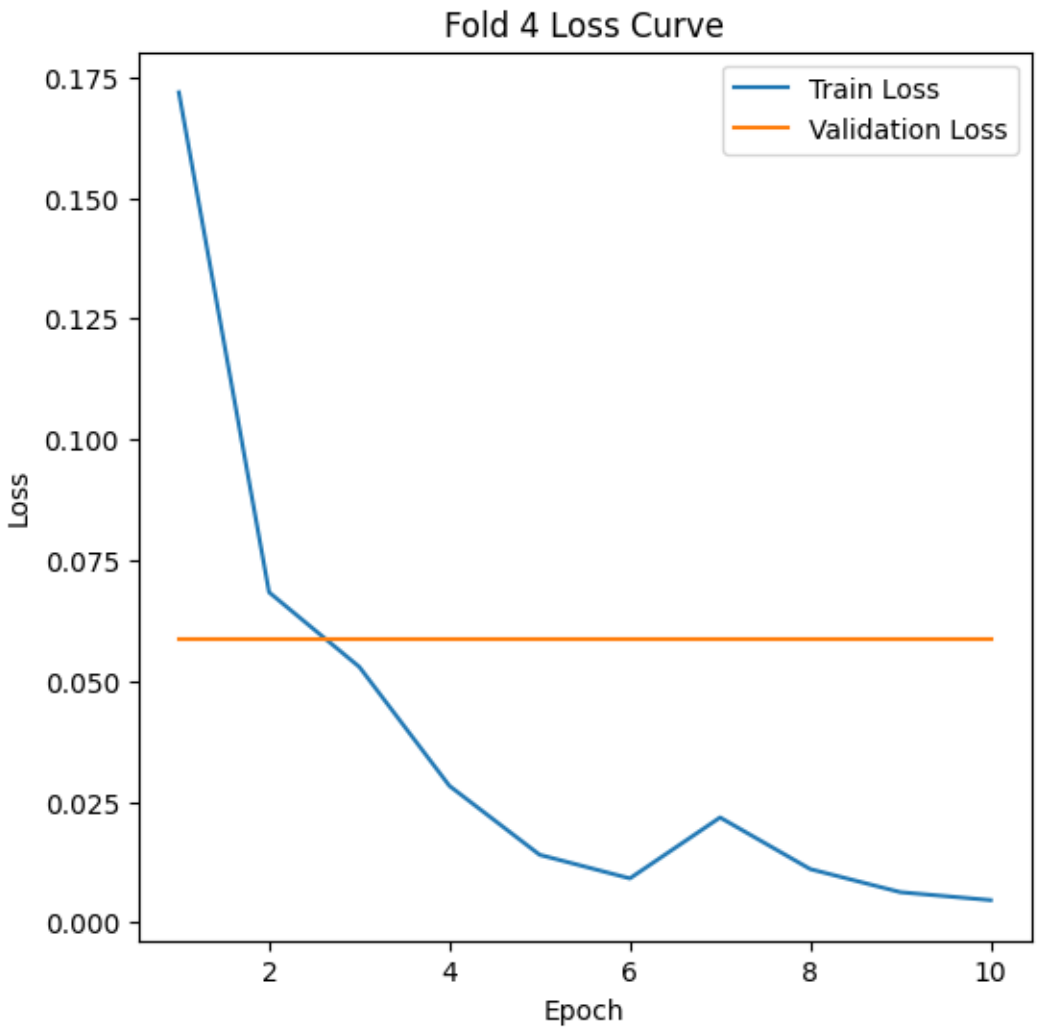
```
FOLD 3
------------------------------
Epoch 1, Training Loss: 0.1905515785501511
Epoch 2, Training Loss: 0.06829844080601577
Epoch 3, Training Loss: 0.04222189968762299
Epoch 4, Training Loss: 0.06817414228916513
Epoch 5, Training Loss: 0.027895306392801338
Epoch 6, Training Loss: 0.009191990701462297
Epoch 7, Training Loss: 0.007311866285346224
Epoch 8, Training Loss: 0.0009287134860482251
Epoch 9, Training Loss: 0.00012857985209983007
Epoch 10, Training Loss: 3.8993656292834406e-05
FOLD 4
------------------------------
Epoch 1, Training Loss: 0.17181157672570813
Epoch 2, Training Loss: 0.06834194742798529
Epoch 3, Training Loss: 0.052916959508567944
Epoch 4, Training Loss: 0.028266193039059914
Epoch 5, Training Loss: 0.014005916158297353
Epoch 6, Training Loss: 0.009124416479608044
Epoch 7, Training Loss: 0.021730033151335967
Epoch 8, Training Loss: 0.011045629712633079
Epoch 9, Training Loss: 0.006266337797064302
Epoch 10, Training Loss: 0.004624040154491619
```


Fold 0 Loss Curve
Train Loss
Validation Loss
Loss
Epoch
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
2
4
6
8
10


Fold 1 Loss Curve
Train Loss
Validation Loss
Loss
Epoch
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
2
4
6
8
10


Fold 2 Loss Curve
Train Loss
Validation Loss
0.150
0.125
0.100
0.075
0.050
0.025
0.000
Loss
2
4
6
8
10
Epoch


Fold 3 Loss Curve
Train Loss
Validation Loss
Loss
Epoch
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
2
4
6
8
10



Since the average epoch of train making loss increase (Fold) is 5, we choose 5 as the `max_epoch`.

8. Train and result

[8]:
```
from torch.utils.data import DataLoader, TensorDataset, Subset

results = []

for fold, (train_idx, val_idx) in enumerate(kfold.split(dataset)):
    print(f"FOLD {fold}")
    print("-" * 30)

    # Subset data for train and validation
    train_subset = Subset(dataset, train_idx)
    val_subset = Subset(dataset, val_idx)
```

```
    # Create DataLoaders
    train_loader = DataLoader(train_subset, batch_size=32, shuffle=True)
    val_loader = DataLoader(val_subset, batch_size=32, shuffle=False)

    # Initialize model, loss, and optimizer
    model = LeNet5(num_classes=2).to(device)
    criterion = nn.CrossEntropyLoss()
    optimizer = torch.optim.Adam(model.parameters(), lr=0.001)

    # Training loop
    for epoch in range(5):  # Train for 4 epochs per fold
        model.train()
        train_loss = 0.0
        for inputs, labels in train_loader:
            inputs, labels = inputs.to(device), labels.to(device)

            # Zero gradients
            optimizer.zero_grad()

            # Forward + backward + optimize
            outputs = model(inputs)
            loss = criterion(outputs, labels)
            loss.backward()
            optimizer.step()

            train_loss += loss.item()
        print(f"Epoch {epoch+1}, Training Loss: {train_loss/len(train_loader)}")

    # Validation loop
    model.eval()
    val_loss = 0.0
    correct = 0
    total = 0
    with torch.no_grad():
        for inputs, labels in val_loader:
            inputs, labels = inputs.to(device), labels.to(device)
            outputs = model(inputs)
            loss = criterion(outputs, labels)
            val_loss += loss.item()

            _, predicted = torch.max(outputs, 1)
            total += labels.size(0)
            correct += (predicted == labels).sum().item()

    val_accuracy = 100 * correct / total
    print(f"Validation Loss: {val_loss/len(val_loader)}")
    print(f"Validation Accuracy: {val_accuracy}%")
```

```
    results.append({"fold": fold, "val_loss": val_loss/len(val_loader),␣
 ↪"val_accuracy": val_accuracy})

# Print Results
print("Results per fold:")
for result in results:
    print(f"Fold {result['fold']} - Loss: {result['val_loss']:.4f}, Accuracy:␣
 ↪{result['val_accuracy']:.2f}%")

# Average results
avg_accuracy = sum([r["val_accuracy"] for r in results]) / k_folds
print(f"Average Validation Accuracy: {avg_accuracy:.2f}%")
```

```
FOLD 0
------------------------------
Epoch 1, Training Loss: 0.16034568115827386
Epoch 2, Training Loss: 0.07291057705879211
Epoch 3, Training Loss: 0.030889871944155958
Epoch 4, Training Loss: 0.00496997184570689
Epoch 5, Training Loss: 0.0012420150054847706
Validation Loss: 0.0774984132222869
Validation Accuracy: 98.61111111111111%
FOLD 1
------------------------------
Epoch 1, Training Loss: 0.20397578531669247
Epoch 2, Training Loss: 0.07869687122810218
Epoch 3, Training Loss: 0.03840740164149018
Epoch 4, Training Loss: 0.02490861793221147
Epoch 5, Training Loss: 0.011611136093523345
Validation Loss: 0.0728559670928267
Validation Accuracy: 98.37962962962963%
FOLD 2
------------------------------
Epoch 1, Training Loss: 0.18318944431289477
Epoch 2, Training Loss: 0.08788924005003301
Epoch 3, Training Loss: 0.05504274893664168
Epoch 4, Training Loss: 0.02862022963866669
Epoch 5, Training Loss: 0.014232347173405017
Validation Loss: 0.0764019488693951
Validation Accuracy: 98.14814814814815%
FOLD 3
------------------------------
Epoch 1, Training Loss: 0.18860488326323252
Epoch 2, Training Loss: 0.06797932615262214
Epoch 3, Training Loss: 0.04834213945152307
Epoch 4, Training Loss: 0.04122462990387619
Epoch 5, Training Loss: 0.017685895481716014
```

```
Validation Loss: 0.03289019786045953
Validation Accuracy: 98.37962962962963%
FOLD 4
------------------------------
Epoch 1, Training Loss: 0.19677320931589715
Epoch 2, Training Loss: 0.0514154281484446
Epoch 3, Training Loss: 0.043386625600288865
Epoch 4, Training Loss: 0.015869536421464377
Epoch 5, Training Loss: 0.005992273599582
Validation Loss: 0.1071749727221751
Validation Accuracy: 97.4477958236659%
Results per fold:
Fold 0 - Loss: 0.0775, Accuracy: 98.61%
Fold 1 - Loss: 0.0729, Accuracy: 98.38%
Fold 2 - Loss: 0.0764, Accuracy: 98.15%
Fold 3 - Loss: 0.0329, Accuracy: 98.38%
Fold 4 - Loss: 0.1072, Accuracy: 97.45%
Average Validation Accuracy: 98.19%
```

Appendix A: environment.yml (conda environment) name: C:39 channels: - defaults dependencies: - ca-certificates=2024.9.24=haa95532$_0$ − $openssl$ = 3.0.15 = $h827c3e9_0$ − pip = 24.2 = $py39haa95532_0$ − $python$ = 3.9.20 = $h8205438_1$ − $setuptools$ = 75.1.0 = $py39haa95532_0$ − $sqlite$ = 3.45.3 = $h2bbff1b_0$ − vc = 14.40 = $h2eaa2aa_1$ − $vs2015_runtime$ = 14.40.33807 = $h98bb1dd_1$ − $wheel$ = 0.44.0 = $py39haa95532_0$ − pip : −$absl$ − py == 2.1.0 − $anyio$ == 4.6.2.$post1$ − $argon2$ − $cffi$ == 23.1.0 − $argon2$ − $cffi$ − $bindings$ == 21.2.0 − $arrow$ == 1.3.0 − $asttokens$ == 2.4.1 − $async$ − lru == 2.0.4 − $attrs$ == 24.2.0 − $babel$ == 2.16.0 − $beautifulsoup4$ == 4.12.3 − $bleach$ == 6.2.0 − $certifi$ == 2024.8.30 − $cffi$ == 1.17.1 − $charset$ − $normalizer$ == 3.4.0−$colorama$ == 0.4.6−$comm$ == 0.2.2−$contourpy$ == 1.3.0−$cycler$ == 0.12.1−$debugpy$ == 1.8.8 − $decorator$ == 5.1.1 − $defusedxml$ == 0.7.1 − $exceptiongroup$ == 1.2.2 − $executing$ == 2.1.0 − $fastjsonschema$ == 2.20.0 − $filelock$ == 3.13.1 − $fonttools$ == 4.55.0 − $fqdn$ == 1.5.1 − $fsspec$ == 2024.2.0 − $h11$ == 0.14.0 − $h5py$ == 3.12.1 − $httpcore$ == 1.0.7 − $httpx$ == 0.27.2 − $idna$ == 3.10 − $importlib$ − $metadata$ == 8.5.0 − $importlib$ − $resources$ == 6.4.5 − $ipykernel$ == 6.29.5 − $ipython$ == 8.18.1 − $ipywidgets$ == 8.1.5 − $isoduration$ == 20.11.0 − $jedi$ == 0.19.2 − $jinja2$ == 3.1.4 − $joblib$ == 1.4.2 − $json5$ == 0.9.28 − $jsonpointer$ == 3.0.0 − $jsonschema$ == 4.23.0 − $jsonschema$ − $specifications$ == 2024.10.1 − $jupyter$ == 1.1.1 − $jupyter$ − $client$ == 8.6.3 − $jupyter$ − $console$ == 6.6.3 − $jupyter$ − $core$ == 5.7.2 − $jupyter$ − $events$ == 0.10.0 − $jupyter$ − lsp == 2.2.5 − $jupyter$ − $server$ == 2.14.2 − $jupyter$ − $server$ − $terminals$ == 0.5.3 − $jupyterlab$ == 4.2.6 − $jupyterlab$ − $pygments$ == 0.3.0 − $jupyterlab$ − $server$ == 2.27.3 − $jupyterlab$ − $widgets$ == 3.0.13 − $kaggle$ == 1.6.17 − $kagglehub$ == 0.3.4 − $keras$ == 3.6.0−$kiwisolver$ == 1.4.7−$markdown$−it−py == 3.0.0−$markupsafe$ == 3.0.2−$matplotlib$ == 3.9.2−$matplotlib$−$inline$ == 0.1.7−$mdurl$ == 0.1.2−$mistune$ == 3.0.2−ml−$dtypes$ == 0.5.0−$mpmath$ == 1.3.0−$namex$ == 0.0.8−$nbclient$ == 0.10.0−$nbconvert$ == 7.16.4−$nbformat$ == 5.10.4−$nest$−$asyncio$ == 1.6.0−$networkx$ == 3.2.1−$notebook$ == 7.2.2−$notebook$−$shim$ == 0.2.4−$numpy$ == 2.0.2−$optree$ == 0.13.1−$overrides$ == 7.7.0−$packaging$ == 24.2−$pandas$ == 2.2.3 − $pandocfilters$ == 1.5.1 − $parso$ == 0.8.4 − $pillow$ == 11.0.0 − $platformdirs$ == 4.3.6 − $prometheus$ − $client$ == 0.21.0 − $prompt$ − $toolkit$ == 3.0.48 − $psutil$ == 6.1.0 − $pure$ − $eval$ == 0.2.3 − $pycparser$ == 2.22 − $pygments$ == 2.18.0 − $pyparsing$ == 3.2.0 − $python$ − $dateutil$ == 2.9.0.$post0$ − $python$ − $json$ − $logger$ == 2.0.7 − $python$ − $slugify$ == 8.0.4 − $pytz$ == 2024.2 −

$pywin32 == 308 - pywinpty == 2.0.14 - pyyaml == 6.0.2 - pyzmq == 26.2.0 - referencing == 0.35.1 - requests == 2.32.3 - rfc3339 - validator == 0.1.4 - rfc3986 - validator == 0.1.1 - rich == 13.9.4 - rpds - py == 0.21.0 - scikit - learn == 1.5.2 - scipy == 1.13.1 - send2trash == 1.8.3 - six == 1.16.0 - sniffio == 1.3.1 - soupsieve == 2.6 - stack - data == 0.6.3 - sympy == 1.13.1 - terminado == 0.18.1 - text - unidecode == 1.3 - threadpoolctl == 3.5.0 - tinycss2 == 1.4.0 - tomli == 2.1.0 - torch == 2.5.1 + cu118 - torchaudio == 2.5.1 + cu118 - torchvision == 0.20.1 + cu118 - tornado == 6.4.1 - tqdm == 4.67.0 - traitlets == 5.14.3 - types - python - dateutil == 2.9.0.20241003 - typing - extensions == 4.12.2 - tzdata == 2024.2 - uri - template == 1.3.0 - urllib3 == 2.2.3 - wcwidth == 0.2.13 - webcolors == 24.11.1 - webencodings == 0.5.1 - websocket - client == 1.8.0 - wget == 3.2 - widgetsnbextension == 4.0.13 - zipp == 3.21.0 prefix : C : 39 AppendixB : Exmapleimage(NORMAL/COVID19)$

[9]:
```
import matplotlib.pyplot as plt
fig, ax = plt.subplots(1,2)
ax[0].imshow(Image.open("data/NORMAL/NORMAL(9).jpg"))
ax[0].set_title("data/NORMAL/NORMAL(9).jpg")
ax[0].axis("off")
ax[1].imshow(Image.open("data/COVID19/COVID19(10).jpg"), cmap = "gray")
ax[1].set_title("data/COVID19/COVID19(1).jpg")
ax[1].axis("off")
```

[9]:
```
(np.float64(-0.5), np.float64(1037.5), np.float64(1030.5), np.float64(-0.5))
```

